SARTH: Three Stage Hindi Songs Classification based on Genre, Lyrics & Lyricist

A Thesis submitted to Gujarat Technological University

for the Award of

Doctor of Philosophy

in

Computer/IT Engineering

by
Pathak Darshita Suryakantbhai
199999913551

Under Supervision of
Dr.Tejas P. Patalia



GUJARAT TECHNOLOGICAL UNIVERSITY
AHMEDABAD

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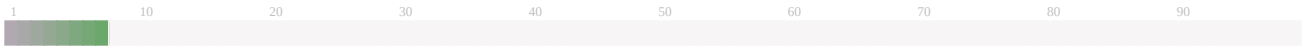
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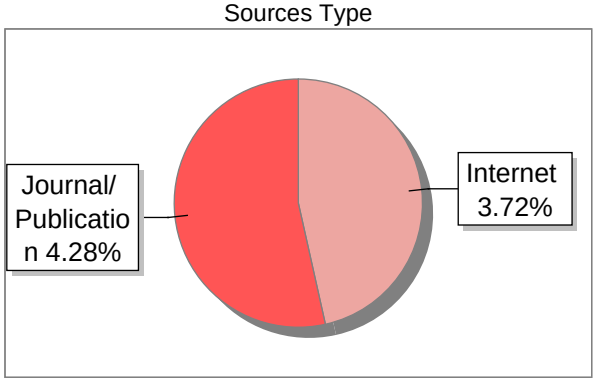
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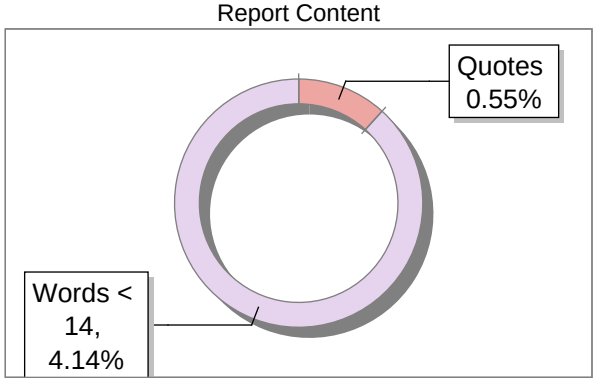
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Abstract

The classification of musical genres has received a lot of attention in recent years due to the several uses it can serve in music information retrieval. Additionally, since genres frequently co-occur in music tracks in Indian Film Songs, it is desirable to record and model the genre correlations in order to enhance the effectiveness of multi-label music genre, lyrics and lyricist classification. The thesis presents a novel multi-modal strategy that uses a music spectral feature-based classification and content based analysis method to address these problems. Names of given genres are classified in relation to Indian music by using abbreviation like Indian flavor. In addition, a genre correlations extraction module is offered to identify and model probable genre correlations based on the characteristics of the proposed approach. Machine Learning is one of the techniques for classifying songs in light of topics/subjects as has been finished in past exploration involving verses as the object of research.

Combining these many characteristics enhances the categorization process and allows for a more sophisticated comprehension of music. The method improves its accuracy in classifying music by taking into account the genre as well as the lyrics and lyricist's traits. Moreover, this approach captures nuanced characteristics of music that transcend conventional genre borders, enabling individualized music recommendation systems.

Test results show how well the suggested approach works to correctly categorize various musical genres while taking authorial and lyrical qualities into consideration. The results highlight how diverse music categorization schemes may improve digital platform music discovery, organization, and recommendation.

In this review, a programmed order framework based regarding the matter utilizing genre, lyrics and lyricist information sources was made. Thus, it is found that the framework execution with the expansion of classification of data in contrasted with utilizing verses as it were. This concentrate likewise attempts to apply the three phases in the classification and looks at it to the single level order strategy carried out in past research. The results indicate that the order of three stage classification with improved random forest algorithm improvises results than other machine learning algorithms.

Dedication

I dedicate my thesis to my life my ребенок *Mr. Chitrang Vyas, whose unwavering love, support, and understanding have been the cornerstone of my journey throughout the completion of this thesis. His encouragement and belief in my abilities sustained me through the challenges and triumphs of this endeavor.*

Acknowledgements

I am profoundly fortunate to have my universe my parents, **Mr. Suryakant Pathak and Mrs. Rita Pathak**, my father-in-law **Mr. Kamendu Vyas** and mother-in-law **Mrs. Saroj Vyas** with me in my academic journey whose confidence in me and unwavering commitment have shaped my life.

I would like to extend my heartily gratitude to my thesis advisor, **Dr. Tejas P. Patalia Head & Professor, Computer Engineering Department, V.V.P. Engineering College** whose guidance, expertise, and unwavering support have been instrumental throughout this research journey. He provided invaluable direction, encouragement, and constructive feedback at every stage of this project. His mentorship not only shaped the trajectory of this thesis but also significantly contributed to my personal and academic growth.

The achievement of this research work would not have been possible without the Doctorate Progress Committee (DPC) members, **Dr. Chirag Patel, Assistant Professor, LE Morbi and Dr. Sohil Pandya, Assistant Professor, CHARUSAT**. I am heartily thankful for their sincere considerations and valuable suggestions through my research. They have made my work a rich, exciting and created environment in which I did research in this field. Their insightful and constructive feedback has helped me to improve all of the work I have done here.

I am grateful to the **GTU community**. Their caring, openness, and support has made my tenure here a pleasant one. They have always been there to help make things possible and have bent over backward to smoothen my path. They have always been reasonable, and often more than reasonable.

Finally, I express my deepest gratitude and appreciation especially to my **AVPTI Computer Engineering Staff Members**, **My AVPTI Parivar**, **My family members** and *"कुदरत शक्ति"*.

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List of Abbreviations

Sr No.	Abbreviations	Full Form
1	MIR	Music Information Retrieval
2	MIDI	Musical Instrument Digital Interface
3	MFCC	Mel-Frequency Cepstral Coefficient
4	CNN	Convolution Neural Network
5	KNN	K-nearest Neighbour
6	SVM	Support Vector Machine
7	RNN	Recurrent neural network
8	MIREX	Music Information Retrieval Evaluation eXchange
9	IDE	Integrated Development Environment
10	ID3	Iterative Dichotomiser 3
11	GUI	Graphical User Interface
12	LRC	Lyrics
13	API	Application programming interface
14	TF-IDF	Term Frequency - Inverse Document Frequency
15	WAV	Waveform Audio File Format
16	DOM	Document Object Model
17	XLS	Excel spreadsheet format
18	CSS	Cascading Style Sheet
19	URL	Uniform Resource Locator
20	HTML	Hyper Text Markup Language
21	POM	Project Object Model
22	BSD	Berkeley Software Distribution
23	MP3	MPEG Audio Layer 3
24	MySQL	My Structured Query Language
25	LPC	Linear predictive coding
26	FFT	Fast Fourier Transform
27	RMS	Root mean square
28	OSCC	Octave Scale Cepstral Co-efficient
29	DWCH	Daubechies Wavelet Coefficient Histograms
30	DCT	Discrete Cosine Transformation

31	CD	Compact Disc
32	ZCR	Zero Crossing Rate
33	LDA	Latent Dirichlet Allocation
34	NLP	Natural Language Processing
35	RF	Random Forest
36	NB	Naïve Bayes
37	OGG	Ogging

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CHAPTER-1

Introduction

Thesis Motivation

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CHAPTER 1

Introduction

In this introductory section, the broader context of the music information retrieval is outlined. It begins with a general overview of the information retrieval, gradually focuses on thesis motivation which later leads to scope and objectives of research. These objectives are directly related to addressing the identified research problem which provide a roadmap for the rest of the study and guide the research process.

1.1 Introduction

Information retrieval used to be an activity employed by only a few professionals like reference librarians, paralegals and similar professional searchers. Today's world is different; hundreds of millions of individuals utilize web search engines or email searches to retrieve information on a daily basis. Information retrieval is quickly replacing traditional database-style searching as the most popular method of accessing information.[1]

Retrieving all documents from a database that includes parts or aspects comparable to a particular data fragment is a generic scenario for multimedia information retrieval that has earned a lot of interest, based on the query-by-example paradigm. Yet, multimedia items often exhibit significant temporal or geographical differences, even while possessing structural or semantic similarities. This makes content-based multimedia retrieval a challenging field with many unanswered questions. [45]

The interdisciplinary study of information retrieval from music is known as multimedia information retrieval (MIR) or multimedia objects. MIR is a narrow but rapidly expanding topic of study with a wide range of practical uses. A background in signal processing, machine learning, academic music studies, musicology, psychology or any combination of these fields may be possessed by anyone working in MIR. According to Stephen Downie, in an attempt to make the enormous library of music accessible to everyone, Music Information Retrieval

(MIR) is "a multidisciplinary research endeavor that strives to develop novel interfaces, innovative content-based searching schemes, and growing interconnected methods of delivery". Furthermore, the domains of music content, music similarity, and music psychology are the focus of this MIR. [46]

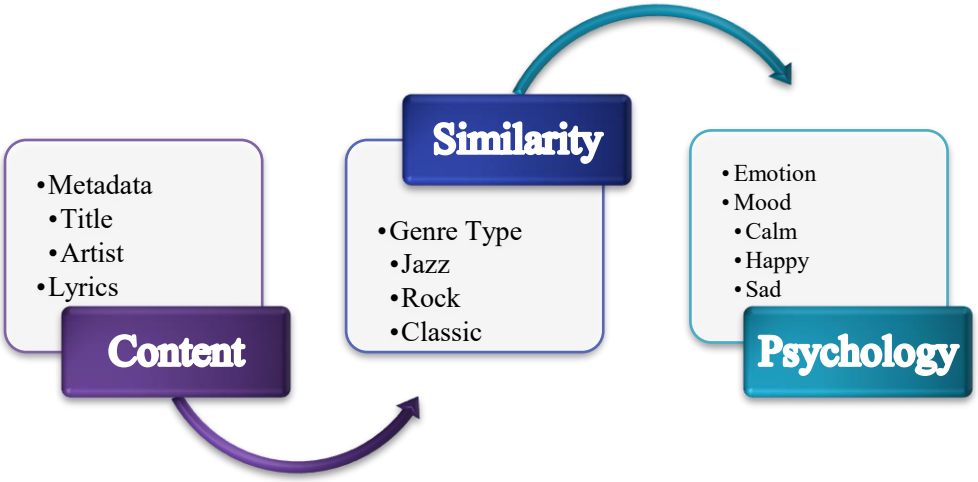


Figure 1.1 Areas of Music Information Retrieval

The quantity of songs, particularly in India, ranges in number and assortment. This is clear from the quantity of singles as well as collections sent off by a gathering of music/vocalists consistently. Subsequently it needs a framework that can sort songs to help the music audience while looking for specific tunes. In a web-based study that has been finished in the past exploration, while looking for songs 33.4% and 17.9% of 427 music audience respondents will generally choose tunes in light of topic/primary subject and storyline separately.[47]

Normally, numerous music applications can arrange songs by genres, artists, and albums. This is reasonable in light of the fact that in the ‘sound record’ there is data about the artists, genres and albums so that music applications can automatically create playlists. However, to sort tunes in view of subject(s) requires a cycle to figure out the topic/subject of the song. In the event that the order of tunes in view of topics/subjects is led manually, it is extremely wasteful on the grounds that it requires exertion and an investment consuming considering the quantity of songs on the world are various and endless. It is because on the grounds that in classifying the tune in light of the theme subject/topic, playlist producer (sorter) should pay attention to the songs individually and know the significance of the songs.

1.2 Thesis Motivation

Songs are an integral part of movies. Songs represent, enhance and complement emotions in the movie. The Indian Film industry, widely known as 'Bollywood', has great importance of movie songs. The Indian Film Industry music forms a noteworthy part of everyday life in India and has grown rapidly with Industrial collaboration all over the world. The versatility in the Hindi music industry ranges widely from Dance to Orchestra, from Ghazals to Thumris, from Hip-Hops to Devotional, the industry serves you multitude of genre at its epitome. Lyricists, Producers, Singers and Artists are involved with this entertainment industry. Whether person is travelling in a car, bus or auto-rickshaw or listening to the radio, or attending a wedding or sangeet ceremony or in funeral, the songs play vital role to set the appropriate background. Classification of songs into different parts based on music spectral features involves music information retrieval algorithm for automatic categorization of songs.

To address the problem, consider an example where user simply wants to listen to a particular kind of song and thus searches for that kind of a song. But such normal search procedure doesn't have rich database which contains it's Track/Genre type in repository or doesn't have the ability to search inside metadata, and thus problem exists. For instance, the user searches "Folk" kind of song, but relevant result is not displayed. Also some smart mobile music players are nowadays providing track classification of songs, but it's actually done by online library update system and the same doesn't have accuracy of even 20% for Hindi songs.

Subsequently, one method for sorting tunes naturally in the light of fundamental topic/subject melodies can be done utilizing machine learning algorithms.[1] According to Harimurti Kridalaksana, the subject is part of the clause that marks what is spoken by the speaker. [22] In this study, the subject of the song is the central theme conveyed by the singer in a song, such as love, friendship & family.

1.3 Research Objectives

The scope of Music Information Retrieval (MIR) encompasses various interdisciplinary areas, combining aspects of music, computer science, signal processing, and information retrieval. Following is a summary of the research scope within MIR [48]:

- **Music Recommendation Systems:** MIR research has contributed to the development of sophisticated music recommendation systems, such as those used by streaming services such as Pandora, Apple Music and Spotify. These systems utilize algorithms that analyze user preferences, music features, and contextual information to provide personalized recommendations.
- **Content-Based Music Retrieval:** MIR techniques enable users to search for music based on its acoustic features, such as melody, rhythm, timbre and harmony. This has applications in music search engines, digital music libraries, and audio fingerprinting systems.
- **Music Transcription and Analysis:** MIR research has resulted in improvements in automatic transcription of music, a process that converts audio information into symbolic representations like MIDI files or sheet music. Digital archives, music production and music education are all impacted by this.
- **Music Emotion Recognition:** Analyzing music's emotional content is a crucial component of MIR research. Techniques for automatic music emotion recognition have applications in mood-based playlist generation, affective computing and music therapy.
- **Music Genre Classification and Tagging:** Identification of a variety of genres of music or assignment of descriptive tags based on its audio content can be done by MIR algorithms. This is useful for organizing music collections, improving music recommendation systems and analyzing trends.
- **Live Music Performance Analysis:** MIR techniques are used to analyze live music performances, including aspects such as tempo estimation, beat tracking and expressive timing. This has uses in music education and music production and live event broadcasting.

- **Musicological Research:** MIR research contributes to musicological studies by providing tools for analyzing large corpora of music recordings, identifying patterns and trends in musical styles and exploring cultural and historical influences on music.
- **Cross-Disciplinary Collaboration**: MIR research involves researchers from different fields working together, including computer science, music theory, psychology and audio engineering. This interdisciplinary approach fosters innovation and leads to novel perspectives on the nature and perception of music.

1.4 Scope of the work

Research in Music Information Retrieval (MIR) encompasses a broad range of subjects, demonstrating the multidisciplinary character of the field of study methods for measuring similarity between music tracks and clustering them into meaningful groups. Research in this area involves exploring different feature representations of music, such as audio features, metadata and user-generated tags, as well as developing similarity metrics and clustering algorithms.

Research involves studying techniques for automatically categorizing music into genre categories. This includes creating classifiers that are able to differentiate between different musical styles, such as classical, rock, jazz and electronic music, based on audio features, lyrics or other metadata. Exploring methods for recognizing and analyzing the emotional content of music involves creating models with automated detection emotions expressed in music, such as happiness, sadness, excitement, or relaxation and investigating how these emotional cues can be utilized in various applications, such as mood-based playlist generation or affective computing.

1) SARTH can be used to search songs based on their "Lyrics".
2) Since the music alone will "classify" tracks into distinct genre types, finding specific "Genre/Track type" songs will be easier.
3) Will increase the effectiveness of "Searching Song" because it is entirely emotion-based, making it easier to find the song you're looking for.

1.5 Objectives of Research

The primary objective of this thesis is centred around classification of Hindi songs based on genre, lyrics and lyricist.

Classifying music songs can be done based on various objectives based on the situation and goal. These are a few typical thesis goals, depending on the extent of the effort:

1) **Genre Classification**: This is perhaps the most common form of classification, where songs are categorized into genres such as rock, pop, hip-hop, classical, jazz, electronic, etc. This classification can be based on musical features like rhythm, instrumentation, vocal style, etc.
2) **Mood or Emotion Classification**: Songs can be classified based on the emotions they evoke, such as happy, sad, energetic, relaxing, etc. This can involve analyzing musical elements like tempo, key, chord progressions and lyrical content.
3) **Lyric-based Classification**: Classifying songs based on their lyrical content, such as love songs, protest songs, storytelling songs, etc. Lyrics can be analyzed and categorized using natural language processing methods.
4) **Artist or Band Classification**: Categorizing songs according to the artist or band performing them. This can involve building models to recognize the unique style or characteristics of different artists.

Each of these objectives will be implemented on Hindi songs' data sources and then apply feature extraction techniques and machine learning techniques for efficient categorization.

1.6 Original Contribution by the Thesis

Studies have explored the emotional effects of music and how different musical elements (example: melody, rhythm, harmony) elicit specific emotional responses. Original research has expanded our knowledge of the mechanisms underlying music-induced emotions and their impact on mood regulation, stress reduction and social bonding. Classifying Hindi songs into their label category will lead song to define in specific emotional class. Proposed three stage classification approach of

implementation focuses on preliminary emotion model of Plutchik Dyads which helps to illustrate how emotions can blend and influence each other, highlighting the nuanced ways in which individuals experience and express their feelings. By examining the through studies of the dynamics of emotion pairs, academics and professionals can better comprehend the complexity of emotions and social relationships. However, specific to music emotional label, arousal and valence play important factors while categorizing songs to particular class, analyzing the Circumplex Model allows for the dynamic representation of emotional states over time, as individuals may experience shifts in valence and arousal depending on internal and external factors. The arrangement of emotions in the Circumplex Model forms a circular structure, with neighboring emotions sharing similarities in both valence and arousal. This structure implies that feelings are arranged in a predictable and methodical way, allowing for the prediction of emotional experiences based on their position within the model. Thus, as per this structure, few of the classes which fall in same quadrant are combined together among 125 genres and target to classify most popular 10 genres of Hindi film industry.

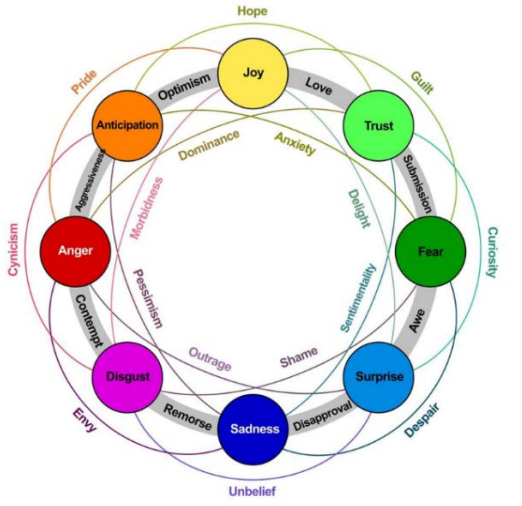


Figure 1.2(a) Plutchik Dyads Model

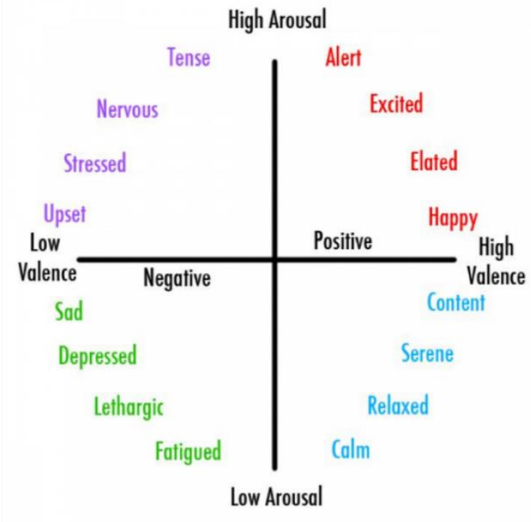


Figure 1.2(b) Russel's Circumplex Model

1.7 Work Plan

In the initial stage of my doctorate of philosophy in Computer Engineering, I started with my course work of research. The course work had two subjects, Research Methodology and Research Publication and Ethics along with course work as per the curriculum of Gujarat Technological University.

• After that, I did the literature survey of my research from different reputed journals, books and many web references.

• I did the comparative analysis of the literature and found the research gap for the research from whatever I found from the literature survey.

• Based on this research gap and my objective, I had started implementation work of my proposed algorithm.

• After implementation of proposed algorithm, whatever result I found, I compared that result with the previously available result to find the success of the proposed work.

• After getting the proper result of my proposed work, I started thesis writing, and subsequently I completed my entire Ph.D. work.

1.8 Structure of the thesis

Chapter 1 is Introduction. It describes the subject in brief, its significance and relevant literature. It clearly states the research questions that thesis aims to address and the objectives research intends to achieve. It defines the scope of study and any limitations or constraints and identifies gaps or unresolved issues in the literature that this research seeks to address.

Chapter 2 of literature work summarize previous research relevant to the topic, including theoretical frameworks, key concepts and empirical studies. It discusses relevant theories and concepts that inform music research.

Chapter 3 describes the overall approach and design and explains techniques of data collection (e.g., crawling, retrieval, surveys, analysis of musical compositions) which outlines data analysis procedures and techniques.

Chapter 4 describes Training Model Structure. Training a methodology for categorizing music using machine learning which involves several steps of Data

Collection that gathers a dataset of music songs coupled with the labels or classification-relevant features that correspond to them.

This dataset should be diverse and representative of the classification task that one wants to perform and feature Extraction where techniques of audio signal processing will take the relevant data out of the audio signal or metadata of the music recordings. This can contain rhythmic elements like beat and tempo as well as other musical characteristics like key and mode, as well as spectral information like MFCCs and spectrograms. Next, data preprocessing is performed to prepare the retrieved features. To ensure that the input data are suitable for training the model, this may involve managing missing values, scaling, dimensionality reduction (e.g.: PCA), normalization and other preparatory steps to improve the quality of the data. Later, choosing the machine learning model that best fits the need for classification.

Popular models for categorizing music include decision trees, random forests, support vector machines (SVM), k-nearest neighbors (KNN), neural networks (e.g., CNNs, RNNs) and ensemble approaches. The size of the dataset, the computational resources available, the task difficulty and other factors all play a role in the model selection process. The dataset is then divided into training and validation sets. Utilizing the validation set to adjust hyperparameters and the training data, optimize the performance of the chosen model. This could entail methods such as cross-validation and to determine the ideal hyperparameter values, use grid search.

Finally, use the validation set or an independent test set to evaluate the trained model's performance. For classification tasks, accuracy, precision, recall, and F1-score are frequently used evaluation metrics.

Chapter 5 is a presentation of findings. It presents the findings of research in a clear and organized manner, interprets and discusses the results, addressing each research question or objective and provides examples or excerpts from musical compositions or information to back up conclusions. It compares the results with earlier studies and the ideas covered in the literature review, and talks about the ramifications of the findings and how they add to the body of knowledge already known in the field.

It includes acknowledgement of any limitations of study and suggestions of areas for future research and summarization of the main findings and their significance, restating the research questions and reflecting on the broader implications of work.

Chapter of conclusion recaps the main findings and contributions of the research and practical implications which discusses the practical implications of the findings for music practice, education or policy and highlights any theoretical contributions or advancements resulting from the research. It Includes some closing thoughts and observations, as well as suggestions for future research directions based on the study's shortcomings on the overall significance of proposed work.

Last chapter of References lists all the sources cited in thesis following the appropriate citation style.

CHAPTER-2

Music Information Retrieval

Music Information Retrieval Methods

Issues Of MIR

How Musical are Humans?

Impact of Music in Human Life

Indian Music Film Industry

Western v/s Indian Music

Lyrics & Genre

Evolutionary Change in Lyric Content based Analysis

Literature Review

CHAPTER 2

Literature Review

This chapter provides a detailed background study of the basic theory used in this thesis. To minimize the conciseness and to avoid inconsequential discussions, we are not giving a thorough explanation of the underlying idea over here. Here, a brief introduction that highlights the basic concepts and contributions to the understanding of this thesis is provided. The basic concepts and definitions are used in subsequent chapters of this thesis.

The major consideration of this chapter is defined as follows.

2.1 Music Information Retrieval

2.1.1 Introduction

- **What is Music Information Retrieval?**

The interdisciplinary study of information retrieval from music is called music information retrieval (MIR). MIR is a narrow but rapidly expanding topic of study with a wide range of practical uses. MIR participants may have backgrounds in signal processing, machine learning, academic music studies, psychology, musicology or any combination of these fields. [50]

Music information retrieval systems seem to have reached an impasse. A significant number of attempts at creating efficient and effective systems to search through large databases of music have brought about almost the same number of solutions or approaches. Businesses and academics utilize MIR to classify, modify, and even produce music. [50]

- **Is Music Information? [1]**

In the context of Information Retrieval, the word ‘information’ can be taken to mean ‘structured data that is suited to enhance a person’s knowledge of the world’. Information in this sense is best exemplified by functional prose that has ‘aboutness’. Scholarly papers, newspaper articles, computer manuals and travel guides clearly fall in this category; they are about some aspects of the world. Outside the domain of

textual documents, diagrams, news broadcasts and documentary films can be said to be about something. On the other hand, the aboutness of literary writing, poetry for example, is problematic, because the factual subject matter is usually not the most important reason why one is interested in the work in question. Therefore, one cannot separate the content of the work from the way it is expressed without losing some of its essence. By extension this lack of aboutness applies to other art forms as well, including music.

Yet considering music and specifically music notation, as information [51] makes much sense as a professional view of music. Digitally encoded music notation can be queried for such things as recurrent melodies or chords or processed to make statements about its structure or authorship. In this way it can surely contribute to one's knowledge of the musical world. However, it is hard to imagine how musical content and expression could be separated. It is, for example, unclear how one could embed the information from one piece in another other than by quotation.

In addition, musical information tells us very little about the world outside music that is not subjective. Music possesses only a weak aboutness [51] and is thus rather 'pseudo information'. Generally, meaning in music is like a very problematic concept; it does however merit further exploration, while the time has shown very clearly that music is very meaningful to so many people.

- **How Musical Are Humans? [1]**

One reason why MIR may concentrate on professional approaches to music is that historically the focus of attention in both musicology and music psychology used to be on the production of music by composers and performers. Only recently the study of listeners' 'passive' competence has begun to receive similar attention. Petres [23] gives an interesting overview of passive musical competence and the problems of testing it without presupposing professional skills. In her view, 'the ordinary adult listener is a musical expert, although s/he may be unaware of it.' Most people acquire this expertise by being exposed to music, but a surprising percentage of the population (around 50% in the UK and USA) have received some musical instruction as well. Genuinely, a musicality (tone-deafness, the inability to distinguish between pitches) is genetically determined and occurs in about 4% of the population. It is not the result of

lack of motivation or training. Non-trained listeners generally possess the following musical abilities, among other things:

• They are able to distinguish subtle stylistic differences;

• They are equally good at learning songs as professionals;

• They can identify out-of-key notes;

• They recognize patterns of implication and realization [53];

• They are able move to music (as in tapping the beat of music).

2.1.2 Major Elements Song

Main Element of any Hindi song is having standard by its fixed format that is contained by ID tag. The ID3 metadata container is typically utilized in association with the MP3 audio file format. It permits the storage of metadata in the file itself, including the track number, artist, album, title and other details.

Field	**Length**	**Description**
header	3	"TAG"
title	30	30 characters of the title
artist	30	30 characters of the artist's name
album	30	30 characters of the album name
year	4	A four-digit year
comment	28 or 30	The comment.
zero-byte	1	If a track number is stored, this byte contains a binary 0.
track	1	The number of the track on the album, or 0. Invalid, if previous byte is not a binary 0.
genre	1	Index in a list of genres, or 255

Table 2.1 Metadata of Song (Mp3) [20]

2.1.3 Architecture of Music Information Retrieval system

The Architecture of typical music information retrieval system is having following major components shown in figure.

Generally, User interacts with two primary elements that provide basic GUI utility with Music Player and Music Query Interface.

To match and display relevant results based on proposed algorithm applied to system results are matched with music objects and query result is to be displayed.

The interaction of Music Objects is dealt with Music Database while for various musicology features music index repository ought to be used.

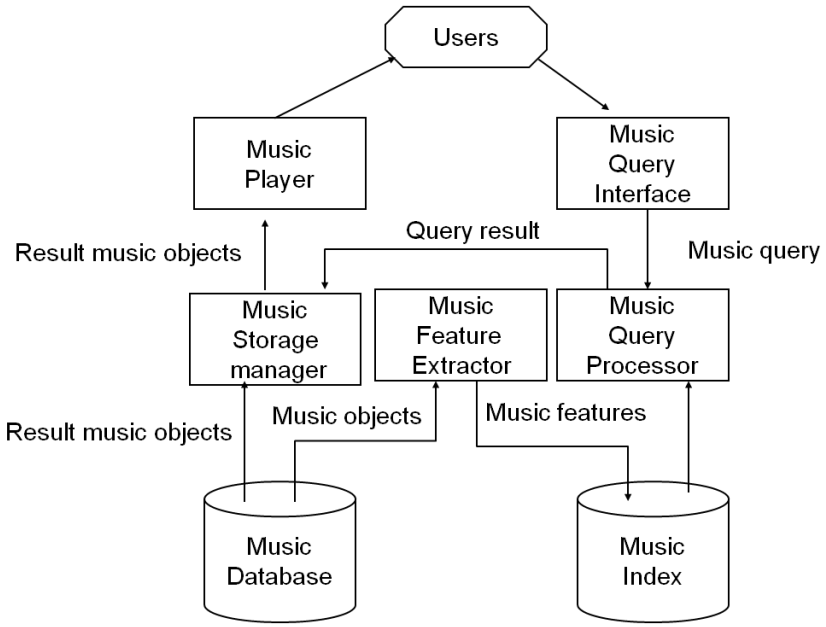


Figure 2.1 Architecture Of MIR

2.2 Music Information Retrieval Methods [50]

2.2.1 Data Source

While scores provide a rational and understandable explanation of the music, it is frequently impractical to have access to sheet music, digital or otherwise. Similar purposes have also been assigned to MIDI music; however, unless the music was composed specifically with the MIDI standards in mind—which is not often the

case—some information is lost during the conversion from any other format to MIDI. When audio is a component of the study, digital audio formats like WAV, mp3, and Ogg are used. Although lossy file types like mp3 and ogg are compatible with the human ear, they could be deficient in important research data. Furthermore, certain encodings produce artifacts that any computerized analyzer could find misleading. Despite this, the mp3's widespread use has meant that these are the primary sources for a great deal of the field's study. In order to provide a more comprehensive knowledge of the music within its cultural context, MIR is increasingly using metadata mined from the web. Most recently, this has included analysis of social tags for music.

2.2.2 Feature Representation

Summarizing is frequently necessary for analysis [54], and feature extraction is used to accomplish this for music (as well as many other types of data), particularly when machine learning is to be used in conjunction with audio content analysis. In order to facilitate learning within a realistic timeframe, the goal is to minimize the overwhelming amount of data to a manageable set of values. The Mel-Frequency Cepstral Coefficient (MFCC), a gauge of a musical composition's timbre, is one frequently extracted characteristic. To depict the piece's chords, harmonies, melody, main pitch, beats per minute, or rhythm, additional elements may be used.

2.2.3 Statistics and Machine Learning

- Computational techniques for modeling, classification and clustering, retrieval, similarity and pattern matching, extraction of musical features for both monophonic and polyphonic music are imposed.
- Formal methods and databases: these are automated methods for identifying and recognizing music, including score following, automatic accompaniment, music routing and filtering, query languages, standards and other metadata, protocols for managing and retrieving music information, multi-agent systems, and distributed search.

- Software for retrieving music information: web-based search and semantic retrieval, query by humming, acoustic fingerprinting, intelligent agents, collaborative software, and the Semantic Web and musical digital objects.
- Automatic summarization, citation, excerpting, down sampling, transformation, formal models of music, digital scores and representations, music indexing and metadata are some examples of music analysis and knowledge representation.

2.3 Issues for MIR [55]

Based on the content, as MIR differs significantly from other information retrieval tasks, processing music inquiries requires a distinctive strategy. A number of things contribute to the task's difficulty. [47,56,57,58]

- Music is polyphonic, meaning that multiple notes can be heard at once.
- A query might be shown with a key that differs from the one that is stored.
- Differences in enhancement, relaxed notes with varying pitches and repeated notes are likely to occur between the inquiry and replies. (The same composition can vary similarly in different arrangements and performances). Despite these variations in note quality, listeners typically perceive musical compositions as being very similar.
- Due to limitations in singing precision, errors are likely to occur if the analysis is conveyed through song.
- Although a request is probably going to be based on a musical composition's melody, it's unclear what that melody is. It's as ambiguous a concept as the meaning of a piece of text.
- The commonest representations of music are as audio streams, not as sequences of notes.

2.4 How Musical Are Humans? [1]

One reason why MIR may concentrate on professional approaches to music is that historically the focus of attention in both musicology and music psychology used to be on the production of music by composers and performers. Only recently the study of listeners' 'passive' competence has begun to receive similar attention. Petres [23] gives an interesting overview of passive musical competence and the problems of testing it without presupposing professional skills. In her view, 'the ordinary adult listener is a musical expert, although s/he may be unaware of it.' Most people acquire this expertise by being exposed to music, but a surprising percentage of the population (around 50% in the UK and USA) have received some musical instruction as well. Genuinely, a musicality (tone-deafness, the inability to distinguish between pitches) is genetically determined and occurs in about 4% of the population. It is not the result of lack of motivation or training. Non-trained listeners generally possess the following musical abilities, among other things:

- They are able to distinguish subtle stylistic differences;
- They are equally good at learning songs as professionals;
- They can identify out-of-key notes;
- They recognize patterns of implication and realization[19] ;
- They are able to move with music.

2.5 Impact of Music in Human Life

People of all ages and cultures use music in their daily lives; it is one of the most universal forms of expression and communication available to humanity. Thus, discussing musics in the plural as opposed to the singlular seems more suitable [113]. Moreover, studies conducted by ethnomusicologists and anthropologists indicate that music has long been associated with humanity. [113]

For the vast majority of people, engaging in activities like as singing, playing (informally or formally), improvising, exploring and composing is occurring at a common place. While listening to music is a fun activity in and of itself, its impact extends beyond light entertainment.

Individuals who participate in these activities can experience a host of good benefits in addition to being able to communicate their innermost thoughts and emotions. More and more empirical and experimental research is being done on the wider benefits of music-making, and studies conducted in the music-related sciences indicate that successful music-making can have a positive impact on a variety of aspects of human life, including the physical, social, cognitive and emotional domains as well as the educational and physical ones.[113] It is possible to learn with and through music in both formal settings (like classrooms with set curricula) and informal ones (like the family home with friends), where participation in musical activities often occurs spontaneously and without conscious aim.

2.5.1 Mood and Music

Listening to music is among the most fascinating human activities. The majority of everyday activities have a discernible benefit that can be credibly linked to the pragmatic goals of reproduction and survival. Furthermore, not many seemingly strange behaviors can rival music for snatching up so much money, time and energy. One of the most common hobbies is listening to music. People's daily lives are accompanied by music almost everywhere.

Segmental features and suprasegmental characteristics are the two categories of structural features. The distinct sounds or tones that comprise a piece of music are known as segmental features; these include acoustic components like pitch, loudness and duration. The fundamental frameworks of a composition, including melody, pace, and rhythm are known as suprasegmental characteristics.[35]

Many distinct musical elements have a strong correlation with certain feelings.[36] Tempo is usually seen to be the most significant component in influencing how emotions are expressed in music, but other elements including mode, loudness and melody also have an impact on the piece's emotional valence.

Structural Feature	Definition	Associated Emotions
Tempo	The speed or pace of a musical piece	Fast tempo: happiness, excitement, anger. Slow tempo: sadness, serenity.
Mode	The type of scale	Major tonality: happiness, joy, Minor tonality: sadness.
Loudness	The physical strength and amplitude of a sound	Intensity, power, or anger
Melody	The linear succession of musical tones that the listener perceives as a single entity	Complementing harmonies: happiness, relaxation, serenity. Clashing harmonies: excitement, anger, unpleasantness.
Rhythm	The regularly recurring pattern or beat of a song	Smooth/consistent rhythm: happiness, peace. Rough/irregular rhythm: amusement, uneasiness. Varied rhythm: joy.

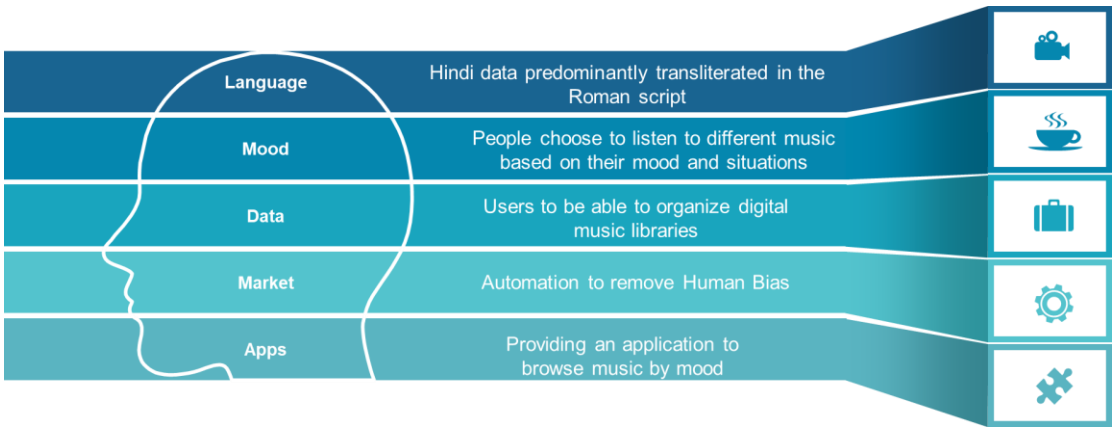
Table 2.2 Structural Feature and Associated Emotion

2.6 Indian Music Film Industry

India is a country renowned for a variety of things, including its extremes in wealth and poverty, its tolerance of other religions and even its diverse cuisine and clothing. However, all of these things merely scratch the surface of this vibrant nation's culture. This still-thriving community has a long and intricate history full with customs and distinctive spirituality. Music is among the most intriguing facets of Indian culture. When one thinks of Indian music, two distinct genres typically spring to mind: either

the pop music that permeates Bollywood, which produces the majority of the country's current musical output, or the lengthy, classical ragas.

The science of arranging tones sequentially to create compositions with coherence and unity is known as music. [1] The ancient Raga system is used to categorize Indian classical music. Every Raga is arranged according to a set of heptatonic scales or systematic orders of modes called Thaats.[2] Raga is a fixed element combination that varies according to frequency. The Sanskrit word "*Raga*" comes from the word "*Ranj*" which means "providing the colorful delights".[3, 4] Every raga has its own set of rules and laws, making them quite structured. In the past, people thought that life lacked color without music.



India's music has a vast history, a deep philosophical foundation and a difficult path to where it is now. Raga and tala are the two main components of Indian classical music. In Indian language, raga means music and tala means rhythm. The ragas are not like Western music in that the performer can add a wide variety of ornaments to the prescribed tunes. Indian classical music is not complete without ornaments, which are notes added to the melody to give the composition a spontaneous or more intriguing tone.

Another key component of music is tala, which is typically played on a small tabla drum and maintains the tempo and beat. This is different from the Western method in that the rhythm of Indian music determines the sounds to generate on the tabla by using particular patterns and onomatopoeic phrases. Generally speaking, Hindu philosophy offers a very intriguing idea that is not often comprehended in the context of music through the medium of classical Indian music.

The music from Hindi films also connects sound and vision. Indian philosophy is centered on the concept of sound, which is still important in their contemporary filmmaking. As technology advances and makes it possible for films to be shared, the use of images for instruction or sales is growing in popularity and convenience.

These screenplays are tailored to suit a particular environment and tone that is conveyed through the medium. Audiences will remember the song by the movie scene and vice versa when these songs are linked with a visual representation, which makes it easier to remember.

2.7 Western Music v/s Indian Music

Western Music Genre	**Indian Music Genre**
Polyphony	Homophonic
Harmony	Melody
12 Key Notes	7 Sur **सुर**
Uses Texture Composition	Uses Taal - a cycle of beats centered around 'Sam' that repeats itself

Table 1.3 Western Music v/s Indian Music

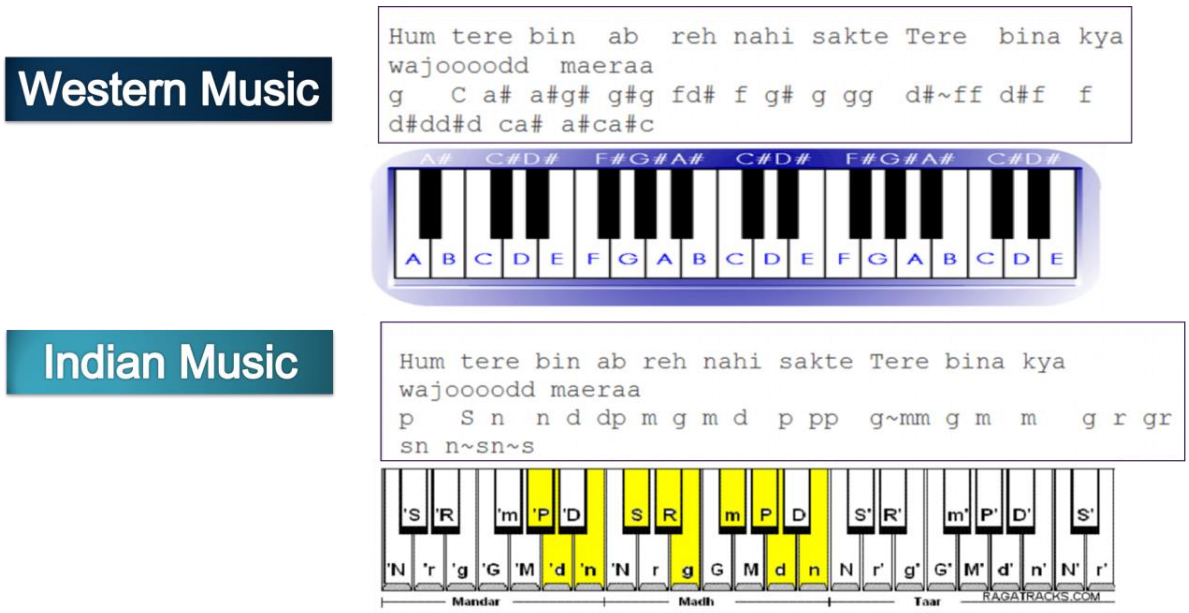


Figure 2.2 Analysis of Western Music and Indian Music Notes of Song "Hum Tere Bin"

2.8 What are Lyrics & Genre?

2.8.1 Lyrics

Sung words, often divided into verses and choruses, are called lyrics. Lyricists are those who write lyrics. a lengthy musical work, like an opera, in words. LRC is a file format for computers that allows lyrics to be synchronized with audio files (MP3, Vorbis, or MIDI) [2,11].

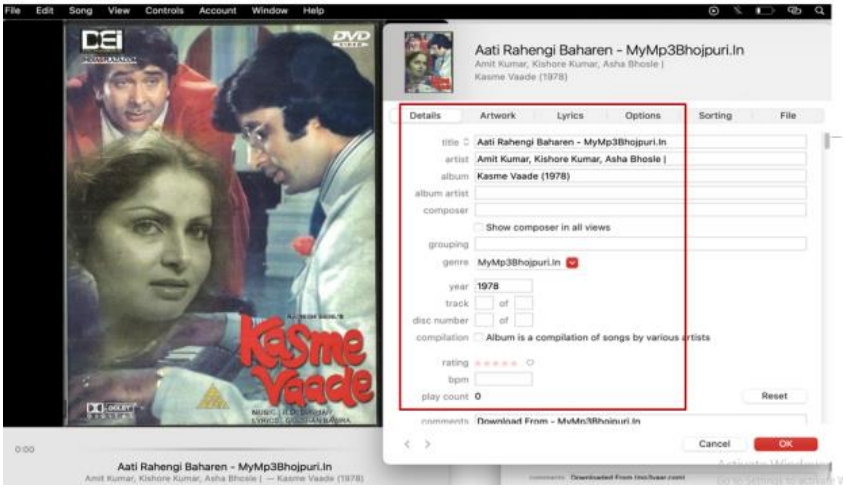


Figure 2.3 Metadata Information of Song “Aatee Rahengee Bahare”

When an audio file is played with certain music players on a computer or on modern digital audio players, the song lyrics is displayed. But in most of the case information is modified as lead by whichever website it is published. For example, as shown in above figure that shows MacOS Music Player Get Info tab where Genre of song is mentioned as website’s name, which shows data mismatch. Even Lyrics tag is available, but in most of the cases, field is Empty. However, in some cases, the lyrics file generally has the same name as the audio file, with a different filename extension. [4] For example, song.mp3 and song.lrc.

[00:12.00]Line 1 lyrics
[00:17.20]Line 2 lyrics
[00:21.10]Line 3 lyrics
...
[mm:ss.xx]last lyrics line

2.8.2 Genre

The term "genre" in music refers to the style of music. A music genre is a term used in conventions to classify musical works that are associated with a common custom or tradition. It should be distinct from musical style and form, even though these phrases are occasionally used synonymously in practice. [6]

Identifying genre will also help musicians and their representatives choose the right labels to approach with their music and can even help them decide if they need a label at all. Until now, total 125 Genres for tracks are identified but for Hindi Songs, this research will focus on 10 popular genres of tracks that are prevalent in Hindi Film Industry.

2.9 POPULAR MUSIC GENRES

- **Filmy**

Filmy ("of films") soundtracks are pieces of music composed and performed specifically for Indian cinema and the country's major motion picture industry. The majority of composers for motion pictures are music directors; playback singers provide the songs, and the genre accounts for 72% of India's music sales market. [28]

- **Romantic**

Romantic composers sought to create music that was individualistic, emotional, dramatic and often programmatic; reflecting broader trends within the movements of Romantic literature, poetry, art, and philosophy. Romantic music was often ostensibly inspired by (or else sought to evoke) non-musical stimuli, such as nature, literature, poetry, super-natural elements or the fine arts. [8]

- **Sad**

Blues is a kind of music that is renowned for being sorrowful. The blues song writers' difficult life is frequently depicted in the lyrics. [8]

- **Dance**

The type of music you typically hear at bars and clubs is known as dance music. It is excellent for dancing to because it has a strong beat and is repetitive frequently. Dance music comes in a wide variety of styles, just like rock. They all will get your feet moving, regardless of how quickly or slowly they move. [8]

- **Remix**

In modern dance music, remixes have become the norm, enabling a single song to be enjoyed in a variety of musical contexts or dancing environments. Such remixes frequently include new vocalists or musicians to the original mix as "featured" artists. Rap and hip-hop music frequently employ the remix.

- **Happy**

A mid-tempo soul and neo soul tune, Happy. Songs that make people happy.

- **Philosophical**

Songs with philosophical topics consist of, but are not restricted to, stories about the function and role of society, the meaning of life, ethics, morals, and the place of art in human lives, the value of experience and the growth of knowledge.

- **Rock**

In the rock genre, traditional Indian instruments like the tabla, sitar and dole are blended with electric guitar.

- **Pop**

Indian pop music that is danceable is referred to be "Indi pop", which is most often associated with Bollywood films. It was referred to as "sugar-coated dance melodies" and had Hindi lyrics and large orchestral film music. Indians appear to be more devoted to songs than performers.

- **Party**

Party is block-rocking, bass-driven hip-hop that only thinks about maintaining the groove. All of the lyrics are unimportant, lacking the radical undertones of hardcore rap and only a small portion of the ingenuity of old school rap. Instead, the emphasis is on the song as a whole, with the bass and percussion taking center stage.

2.10 Indian Filmy Songs

Hindi is the most broadly communicated language in India, practically 41% of the Indian people being its local speakers. It is additionally one of the Indian dialects with a decent collection of devices and assets for Natural Language Processing research. Music from Bollywood comprises practically 71% of music deals in India. This is a tremendous market and it would be basic to mark away at better association of this information.

Individuals decide to tune into various music dependent on their mind-set and circumstances. It is subsequently, of help to clients to have the option to coordinate advanced music libraries as per their opinion or utilize this data in programmed playlist age undertakings as investigated in [61]. Watchword age from verses would help recognize the opinion of the melody and furthermore in different errands, for example, subject discovery [62].

Bollywood tunes have encountered an impressive change as far as verses over the previous many years. Sometime in the past, Bollywood melodies contained generally Hindi words, yet these days lyricists use loads of English words to communicate. Till date, we don't know about any deliberate quantitative examination of how it has changed and how the advancement occurred throughout the long term.

Computational examination of Bollywood melodies has not attracted a lot of consideration by analysts. In any case, a few investigations on Music Information Retrieval (MIR), utilization of melodic scales in Bollywood music and symbolism in contemporary verse, computational examination of conventional tune assortments, tune highlights in well-known tune and corpus phonetic investigation of 50 years of Bob Dylan verses have been completed in the new many years. Much effort has been done on the music state of mind grouping dependent on sound, verses, and social labels and all together.

Programmed techniques are needed to be characterized in music as indicated by temperaments. This has been accomplished by building a framework for characterizing mind-sets of Hindi tunes which utilizes sound related highlights, for example, musicality, wood and force [64]. Music Information Retrieval Evaluation eXchange (MIREX) has been utilized during the trial. What's more, the choice tree classifier has been used as a typical method of grouping exactness of 51.56% has been accomplished [64].

Moreover, the advancement of Bollywood music concerning the utilization of melodic scales has been the theme under examination. Through an investigation, different social impacts on Bollywood music and its arrangement throughout the years were distinguished and measured [63]. From the examination, some striking realities were uncovered about the scale utilization designs that were helpful in figuring intriguing guesses which can be checked genuinely.

2.11 Evolutionary Change in Lyric Content based Analysis

As grouping a song is exclusively dependent on a challenging task, Bernard et al. [3] suggested a method to identify the avid extreme of song through verses using Natural Language Processing. A dataset comprising 420 songs was considered, with an equal number of songs representing positive and negative emotions. The Last.fm API was used to retrieve song labels. A few refrains from melody verses may establish to a specific feeling and tunes can also talk about negative feelings about good things and vice versa. As a result, estimating and analyzing music can be a complex task. Songs can start out with positive feelings but end up with negative feelings or the other way around. There were several calculations suggested to deal with each of those problems.

The simplest computation that relies on word counts is the word list. The songs are arranged according to the words that appear in each verse. Here, the computation runs over the melodic expressions. Word reference for melody characterization involves calculating the availability of each verse word in terms of both positive and negative word references, as well as single or none.

To determine how similar an input song is to the songs in the training set, the cosine similarity method which is based on phrase frequency-inverse document frequency is employed. Noisy data could not be handled by the suggested approach. The results indicate that while song lyrics by themselves may not yield very encouraging results, combining lyrical elements with other factors, including audio, increases the categorization accuracy. The characteristics and approaches utilized by different authors to categorize songs are shown in Table 2.4.

Author Name	Features/Method	Dataset	No. of Songs	Emotion
Ashley M et al. [66]	TF, IDF	Created manually	420	Positive, Negative
Yanqing Xia et al. [65]	Sentiment related words	5 SONGS	2,653 Chinese Pop songs	Positive, Negative
Corona et al. [67]	TF, IDF, BM25	Million Songs	32,302 English songs	--
Shanmugapriya et al. [68]	Hidden Markov Model	Collection of songs from Website	--	Happy,Angry,Sad
Emil Ian et al.[69]	TF, IDF, Key Graph	Collection of songs from Internet	200 Songs	--
Doran Walsten et al. [71]	K-Means Clustering	Collection of songs from Website	--	Classic rock, county, grunge, modern rock, pop, r&b, rap
Govind Sharma et al.[70]	Bag of Words/ Latent Dirichlet Allocation	Collection of songs from Internet	--	Happy, sad, angry, Tired, Love, Funny
Charulatha et al. [72]	Rakshini Algorithm	--	--	Happy, sad, calm

Table 2.4 Features & Methods for Lyrical Analysis

2.12 Literature Review

In Indian Hindi classical music, a "Raga" is a set of musical notes that create a specific mood or feeling when played. Each Raga has its own unique sound and can express different emotions like happiness sadness or calmness. "Rasa" refers to the emotional flavor that a Raga brings out. There are different types of Rasas such as joy love or anger which help listeners feel a certain way when they hear the music.

As mentioned in[31] Bi-Sampling Approach, this method involves taking two different samples or groups of music to compare and analyze them. It helps in understanding how different Raga can evoke various feelings and moods. By listening to different pieces of music and observing how they make people feel researchers can better classify these moods. The dual sampling technique enhances the accuracy of mood identification, making it a valuable tool for music analysis and recommendation systems in various applications**.**

As lyrics play major role for classifying Hindi songs, By analyzing the distribution of words across various documents, LDA effectively identifies clusters of topics, providing insights into lyrical trends and patterns. This method not only enhances the understanding of lyrical content but also serves as a valuable resource for music analysis and recommendation systems. In[73] Rafayet HossainMd, Rahmatul Kabir Rasel Sarker, Mehejabin, Bishwajeet Pandey have recommended English Songs Title Based on Latent Dirichlet Allocation applied on Lyrics.

Coupled with a verse-based emotion analysis of Bengali as well as Telugu music utilizing machine learning and neural network classifiers, these approach[29,30] not only enhance the classification of emotional states but also provides deeper insights into the lyrical content and its impact on listeners. By integrating these methodologies, researchers can uncover intricate emotional layers within the music, paving the way for innovative applications in music analysis and recommendation systems.

Various literature survey papers show Genre classification in music relies significantly on the use of Mel-Frequency Cepstral Coefficients (MFCC) features, which offer a robust representation of audio signals. One notable advantage of MFCC features is their invariance to changes in key and tempo, meaning that these characteristics remain consistent even when the pitch or speed of a piece of music is altered. This property enhances the reliability of genre classification systems, allowing them to

accurately categorize music across different styles, facilitating a better understanding of musical similarities and differences. [78,75]

Further literature study shows Mult objective Time Series Matching for Audio Classification and Retrieval represents a significant advancement in the field of audio analysis. This approach focuses on evaluating genre classification performance by analyzing lyrical features in relation to various other attributes, including audio, symbolic and cultural elements. [75,77,78] By integrating these diverse techniques, researchers can gain a more nuanced understanding of how different features contribute to the identification and categorization of musical genres. This multifaceted methodology not only enhances the accuracy of audio classification but also facilitates a richer retrieval experience, allowing users to explore music based on complex criteria.

Further combining approaches[34,39] with lyrics as well as genre classification study shows that evaluating genre classification performance involves a multifaceted approach that considers various features, including lyrical, audio, symbolic and cultural elements. Lyrical features, such as themes, vocabulary and lyrical complexity, can provide significant insights into the genre of a song. However, these aspects must be analyzed alongside audio features like tempo, key signatures, and instrumentation to create a comprehensive understanding of musical categorization. Furthermore, symbolic elements such as imagery and references within lyrics along with cultural factors that define genre boundaries in different communities add additional layers to this evaluation process. By integrating these diverse features, researchers can enhance genre classification and lyrics classification of music that reflects both individual artistry and collective cultural identity.

Author & Publication	Paper Title	Literature
C, Mohan & Arkachaari, Vinayak & N, Harsha & N, Sushmitha & K, Gayathri & S, Ullas & Rao, Pathi & G, Sudha & Darapaneni, Narayana. arXiv, 2022.	Bi-Sampling Approach to Classify Music Mood leveraging Raga Rasa Association in Indian Classical Music [31]	• Hindi Songs Mood classified with Indian Rasa • Trained Songs with Human predicted Rasa for Training Dataset • Ascertain the Rasa (mood) association with ragas in

		Indian classical music (Hindustani & Carnatic) • Collect the songs (mix of classical renditions and Indian Movie songs) for each of the ragas and Associate rasa (mood) • kNN,SVM Algorithm
Rafayet HossainMd, Rahmatul Kabir Rasel Sarker, Mehejabin, Bishwajeet Pandey,2019 IEEE	Recommendation Approach of English Songs Title Based on Latent Dirichlet Allocation applied on Lyrics [73]	• Topic Modeling on Lyrics using LDA Algorithm • Based on top most words appearing in Lyrics Title is recommended
Rifqi Abdul Aziz,Moch. Arif Bijaksana,Adiwijaya School of Computing Telkom University Bandung, Indonesia 7th International Conference on Information and Communication Technology (ICoICT), IEEE Computer Society 2019	Two Stages Song Subject Classification on Indonesian Song Based on Lyrics, Genre & Artist[49]	• Naive Bayes & SVM algorithm used for classification purpose. • Comparative result analysis between Single Flat Stage v/s Two Stage • As more features/stages classification results are improved.
Kazuyuki Matsumoto, Manabu Sasayama 2018 IEEE 4th International Conference on Computer and Communications	Lyric Emotion Estimation Using Word Embedding Learned from Lyric Corpus[32]	•Word Factorization Method •k-Nearest Neighbor Algorithm is used lyrics phase similarity •Calculation of each word vector phrase •Similarity between words and emotion in Tweets, News and Novel compared with Japanese Song

		•Emotion is assigned by Edit Distance Method
R Guru Ravi Shanker, B Manikanta Gupta, BV Koushik, Vinoo Alluri International Institute of Information Technology, Hyderabad 2023	Tollywood Emotions: Annotation of Valence-Arousal in Telugu Song Lyrics[29]	• Telugu Language • 4 class - Quadrant Classification Method • Main Class - 4 Sub Class • Valence classification (VC), Arousal clas_x0002_sification (AC) and Quadrant classification (QC). • SVM-based (context-free) classifications, • Popularly employed TF-IDF
Maraz Mia,Pulock Das and Ahsan Habib Institute of Information and Communication Technology, Shahjalal University of Science and Technology, Sylhet, Bangladesh International Journal of Computing and Digital Systems ISSN (2210-142X)	Verse-Based Emotion Analysis of Bengali Music from Lyrics Using Machine Learning and Neural Network Classifiers[30]	• Bengali Language • Main Class - 3 Sub Classes and 2 Classes • Short-Term Memory (BLSTM), BERT and Multi-Layer Perceptron (MLP).
Rafayet Hossain, Ahmed Al Marouf International Conference on Bangla Speech and Language Processing (ICBSLP), 21-22 September, 2018 IEEE	BanglaMusicStylo: A Stylometric Dataset of Bangla Music Lyrics[11]	• BanglaMusicStylo dataset, the lyrics of 2824 Bangla songs are stored • Keywords are used for Meta Searching, Genre classification

Afif Hijra Ferdinan,Andrew Brian Osmond, Casi Setianingsih IEEE-2018 International Conference on Control, Electronics, Renewable Energy &Communications (ICCEREC)	Emotion Classification in Song Lyrics Using K-Nearest Neighbor Method [33]	• In this paper POS Tagging approach is used to classify Lyrics of songs • Preprocessing and Stemming of Lyrics • TF-IDF Algorithm for word similarity and classification • 4 Emotion Label are classified : Love , Anger, Happy, Sad • Total of 48 Songs are classified • Indonesian song's lyrics database
LKanika Gupta, Monojit Choudhury, Kalika BaliNI Systems (India) Pvt. Ltd. Bangalore, Microsoft Research Labs India [76]	Mining Hindi-English Transliteration Pairs from Online Hindi	• Online Hindi Lyrics Mining • Crawling & String-Matching Algorithm
Yunjing An1, Shutao Sun, Shujuan Wang 2017 IEEE ICIS 2017 IEEE Computer Society	Naive Bayes Classifiers for Music Emotion Classification Based on Lyrics [34]	• Similar Quadrant wise four emotions are trained • 3 emotion labels are assigned Contentment (Sweet), Depression (Sad), Exuberance (Happy) • 3552 songs are analyzed • Naïve based classifier • Dataset by crawling the web of Baidu music
Matt McVicar† Daniel PW Ellis Masataka Goto†- IEEE Signal Processing	Leveraging Repetition for Improved Automatic Lyric Transcription in Popular	• Automatic lyric alignment/recognition • ROVER Word

2014[74]	Music	Translation Algorithm
Matthias Mauch, Hiromasa Fujihara, and Masataka Goto, IEEE Trans. On Audio and Speech, Language Processing, 2012[75]	Chord Information Into HMM-Based Lyrics-to-Audio Alignment	• Chord Role in Lyrics & importance of blank line • Signal Processing • Training Data Set
Tom LH. Li and Antoni B. Chan.International Conference on Multimedia Modeling, Taipei, 2011[78]	Genre classification and the invariance of MFCC features to Key and Tempo	• Prediction with various key and tempo transformations of the songs • MFCC & Key Notes • Confusion Matrix of Genres
Philippe Esling and Carlos Agon IEEE Language Processing Oct,2013[77]	Mult objective Time Series Matching for Audio Classification and Retrieval	• Mult objective Time Series Matching for Audio Classification and Retrieval
Cory McKay, John Ashley Burgoyne, Jason Hockman, Jordan B. L. Smith, Gabriel Vigliensoni and Ichiro FujinagaISMIR 2010[39]	Evaluating Genre Classification Performance Lyrical Features Relative to Audio, Symbolic and Cultural Features	• Mining of Lyrical and Audio content • Various types of Genres.

Table 2.5 Contents of Literature Survey Papers

2.13 Performance & Correctness Measure [38]

Numerous metrics have been presented to assess the effectiveness of information retrieval systems. The actions call for a document collection and an inquiry. The standard measurements outlined here are predicated on the ground truth concept of relevancy, which holds that each document is either known to be relevant or non-related to a given query. In actuality, questions could be poorly phrased and have varying degrees of relevance.

- **Precision** is the probability that a (randomly selected) retrieved document is relevant.[80]

- **Recall** is the probability that a (randomly selected) relevant document is retrieved in a search.[80]

- **F-measure** is measure that combines precision and recall is the harmonic mean of precision and recall.[81]

$$F= 2* ((precision* recall)/ (precision+ recall))$$

- **Confusion Matrix**

An indicator of performance for machine learning classification technique is a confusion matrix. It's a table that shows how well a classification model performs when applied to a set of test data whose real values are known. [83] Here's a breakdown of a typical confusion matrix:

- True Positives (TP): These are the cases where the model correctly predicts the positive class.
- True Negatives (TN): These are the cases where the model correctly predicts the negative class.
- False Positives (FP): These are the cases where the model incorrectly predicts the positive class when it's actually negative. Also known as Type I error.
- False Negatives (FN): These are the cases where the model incorrectly predicts the negative class when it's actually positive. Also known as Type II error.

- **Hamming Error**

The Hamming error provides a measure of how well a classifier performs in terms of predicting multiple labels simultaneously. A lower Hamming error indicates better performance, as it means fewer labels are predicted incorrectly. In the context of classification tasks, the Hamming error is calculated as the proportion of labels that are incorrectly predicted, relative to the total number of labels. It's essentially the proportion of incorrectly predicted labels to the total number of labels. [82]

CHAPTER-3

Implementation Tools

Implementation Libraries

Saarth Dataset

Music Classifier System

Stage 1 Genre Classification

Stage 2 Lyrics Classification

Stage 3 Lyricist Classification

CHAPTER 3

Dataset Preparation & Implementation Techniques

This chapter includes the framework which is adapted to prepare database and expanded based on the specific nature of the research for implementing proposed work plan. This chapter also highlights techniques for execution that are necessary for implementing three stage classification technique using machine learning algorithm of Java and Python programming languages.

3.1 Implementation Tools

1) Java

Java provides rich set of libraries to process on Audio songs to fulfil all mathematical equations and derivative of Digital Signal Processing. To do any kind of processing on an audio file you need the "raw" data, meaning an audio file which has uncompressed audio samples. For example, the WAV format, which normally has samples in 16-bit signed 2's complement, and for that MIR processing Java is the best one.[84]

2) Net Beans IDE

Java is used to create the software development platform in Net Beans IDE. Applications are created using a collection of modular software components known as modules used in the Net Beans Platform. Applications developed on the Net Beans Platform, like the Net Beans integrated development environment (IDE), may be expanded by outside developers as well.

Although the Net Beans IDE is primarily focused on Java development, it also supports PHP, C/C++, HTML5 and other languages.[85]

3) Python Language

Because of its ease of use, readability, and large ecosystem of libraries and frameworks designed specifically for machine learning, Python is among one of most widely accepted programming languages for this purpose. Python easily interacts with

various programming languages and machine learning tools, including Jupyter Notebooks for interactive development and documentation, NumPy and pandas for data processing, and matplotlib and seaborn for data visualization. Efficiency is increased and the machine learning workflow is streamlined by this linking.

4) Anaconda IDE

Anaconda IDE provides a comprehensive environment for Python development, particularly suited for data analysis, scientific computing and the machine learning tasks. It comes bundled with a wide usage of pre-installed Python libraries and tools commonly used in these domains. Anaconda IDE is a feature-rich and user-friendly IDE tailored for Python developers and data scientists, offering a powerful set of tools and functionalities to streamline the development and analysis workflow.[89]

5) Rapid Miner

A wide range of tools and capabilities are available for data preparation, modeling, assessment, and deployment using RapidMiner, an all-inclusive data science platform. Without the knowledge of writing code, users may create and carry out data analysis procedures using its visual workflow interface. RapidMiner is a powerful and user-friendly platform for data science and analytics, offering a rich set of features and capabilities for organizations looking to derive insights and value from their data.[90]

3.2 Implementation Libraries

3.2.1. JSoup Library

Jsoup is a java html parser and a java library which is used to parse HTML document. Jsoup provides API to extract and manipulate data from URL or HTML file. It uses DOM, CSS and techniques for extracting and modifying files that resemble Jquery.
The permissive MIT license is used to distribute Jsoup, an open-source project. Soup parses HTML to the same Document Object Model (DOM) as contemporary browsers and implements the WHATWG HTML5 standard.[88]

- Use DOM traversal or CSS selectors to find and extract data;
- Scrape and parse HTML from a file, URL, or string;

- Modify HTML elements, attributes, and text;
- Check user-submitted content against a safe white-list to avoid XSS attacks;
- Produce clean HTML

Jsoup is made to handle any kind of HTML that might be encountered in the real world; it can handle data cleaning, validate HTML and creates invalid tag-soup HTML also can provide, a logical parse tree. [13]

Jsoup is available as a downloadable .jar java library. The current release version is 1.13.1

- jsoup-1.13.1.jar core library
- jsoup-1.13.1-sources.jar optional sources jar
- jsoup-1.13.1-javadoc.jar optional javadoc jar

As implementation perspective if Maven is used then Maven is to manage the dependencies in your Java project, you do not need to download; just place the following into your POM's <dependencies> section, soup is entirely self-contained and has no dependencies. jsoup runs on Java 7 and up, Scala, Kotlin, Android, OSGi, Lambda, and Google App Engine.

```
<Dependency>
  <! -- Jsoup HTML parser library @ https://jsoup.org/ -->
  <groupId>org.jsoup</groupId>
  <artifactId>jsoup</artifactId>
  <version>1.13.1</version>
</dependency>
```

3.2.2 Apache Tika Server 1.7.jar

Almost one thousand distinct file formats (including PPT, XLS, and PDF) include text and information that may be found and extracted using the Apache Tika™ toolkit. Tika is helpful for a variety of tasks, including content analysis, translation, search engine indexing, and much more, because it can parse all of these file types through a single interface.[86,87]

3.2.3 Python Libraries

Python offers several libraries and tools for audio feature extraction, used as choice for processing and analyzing audio data in various applications, including music information retrieval with speech recognition, and sound classification.

A) Librosa

Librosa is a Python library for music and audio analysis. It provides functions to load audio files, extract various audio features (such as Mel-frequency cepstral coefficients (MFCCs), chroma features, spectral features), compute spectrograms, and more. Librosa is widely used in music information retrieval tasks.[92]

B) Numpy

A core Python package for numerical computation is called NumPy. In order to work effectively on multi-dimensional arrays (matrices), it offers support for a many number of mathematical functions. Numerous additional Python libraries and tools in the scientific computing ecosystem rely on NumPy as their base. It easily combines with a variety of libraries, including pandas library for data analysis and matplotlib for data visualization, scikit-learn for machine learning, and SciPy for scientific computing. [93]

C) Scipy

The SciPy.Signal module in SciPy provides functions for signal processing tasks including filtering, convolution, spectral analysis, and feature extraction. While it's not specifically designed for audio analysis, it can be used for basic feature extraction and signal processing tasks.[94]

D) Sklearn

Scikit-learn is a Python machine learning tool based on SciPy, available under the 3-Clause BSD license. Originally released as scikits.learn and also known as sklearn, the Python machine learning package scikit-learn is open-source and free to use.[91] It is compatible with NumPy and SciPy scientific and numerical libraries for Python and supports several classification, regression, and clustering algorithms such as support-vector machines, random forests, gradient boosting, k-means, and DBSCAN.

3.3 Sanchay Dataset

For Lyrics Extraction there, is a must requirement for collecting metadata of all songs, and as such it is already available in metadata to prepare dataset for that metadata information which is easy to build.

Metadata extraction procedure will be useful to extract various Meta information of song paired with music, which will be useful to prepare repository with essential song metadata. The prepared dataset given Hindi transliterate named "*Sanchay*" means a repository will be helpful in area of music segmentation, digital signal processing, songs feature extraction, emotion classification, lyrics mining, playlist generation, artist-genre paired clustering with commercial applications too.

There are large datasets of royalty-free music such as Jamendo (often used [95,96]) or others found in the DB Tunes link directory focus on the ones that cover commercial popular music and we will see that few proposed metadata on cultural aspects, lyrics and audio altogether. However, these all dataset contain popular English based songs, in addition to that, UCI has song lyrics with the fewest possible characters from several Hindi-Bollywood datasets that are accessible on Kaggle. More than 2500 songs are needed for research and commercial applications in order to train artificial neural networks. For example, MusicBrainz has a lot of cultural information, but not lyrics. Several academics have utilized the tags in the Last.fm dataset to compute moods and emotions [97, 98].

Lyrics are the most important component of songs, more so than beat, melody, fusion, performer, or genre, as they directly influence the listener's selection and attitude. To the best of our knowledge in this study, we presented the very first *Sanchay* – Hindi word of synonym as collection of data. Dataset having 5076 Songs lyrics of 257+ lyricists.

As mentioned in "Related Work", any mp3 song contains LRC file, that is embedded with mp3 song with Lyrics of that song in particular format. Now as such every Hindi song does not have LRC file, but in part of mp3 song it contains Metadata where it's having Lyrics metadata tag, and extraction of that part will be carried out by using metadata extraction and crawling over lyrics' site using various java in-built library for metadata extraction as well as crawling.

Field	Length	Description
Header	4	"TAG+"
Title	60	60 characters of the title
Artist	60	60 characters of the artist's name
Album	60	60 characters of the album name
Speed	1	0=unset, 1=slow, 2= medium, 3=fast, 4=hardcore
Genre	30	A free-text field for the genre
Lyrics	-	A free text field for lyrics
start-time	6	the start of the music as mmm:ss
end-time	6	the end of the music as mmm:ss

Table 3.1 Metadata of Mp3 Song (Properties) [20]

Currently, when a user searches for a song using its lyrics, it is possible that the search query word could include terms that are not only in English word format but also, primarily, in Hindi word cross language. This creates a cross language migration issue, which is why NLP is utilized. Primarily Extracting Lyrics Natural Language Processing (NLP) is a branch of computer science, artificial intelligence, and linguistics which studies how computers and human (natural) languages interact. NLP and the field of human with computer interaction is so linked. Algorithms for unsupervised and semi-supervised learning have received more attention in recent times. These algorithms may be used to learn from both non-annotated and annotated data, as well as data which have not been manually annotated with the required answers. [66]

NLP assessment seeks to quantify one or more attributes of an algorithm or a system, in order to determine whether (or up to what extent) the system answers the goals of its developers, or meets the needs of its users. Stepwise procedure is described further as follows

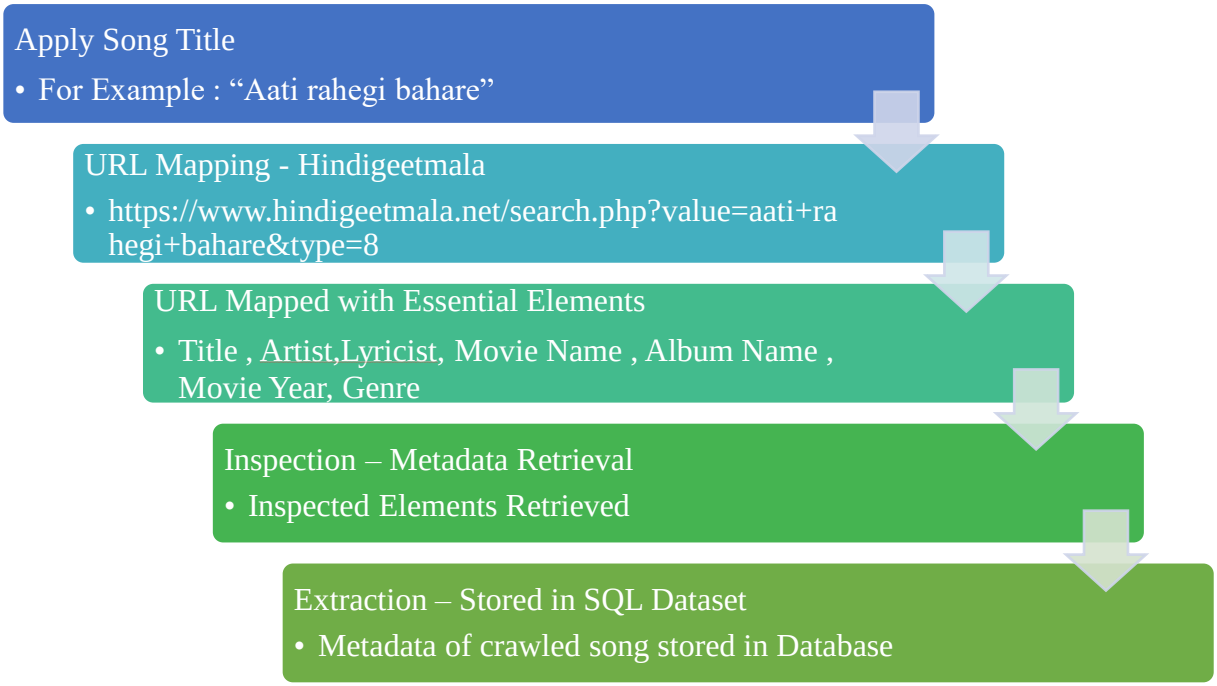


Figure 3.1 Proposed Methodology of Dataset Retrieval

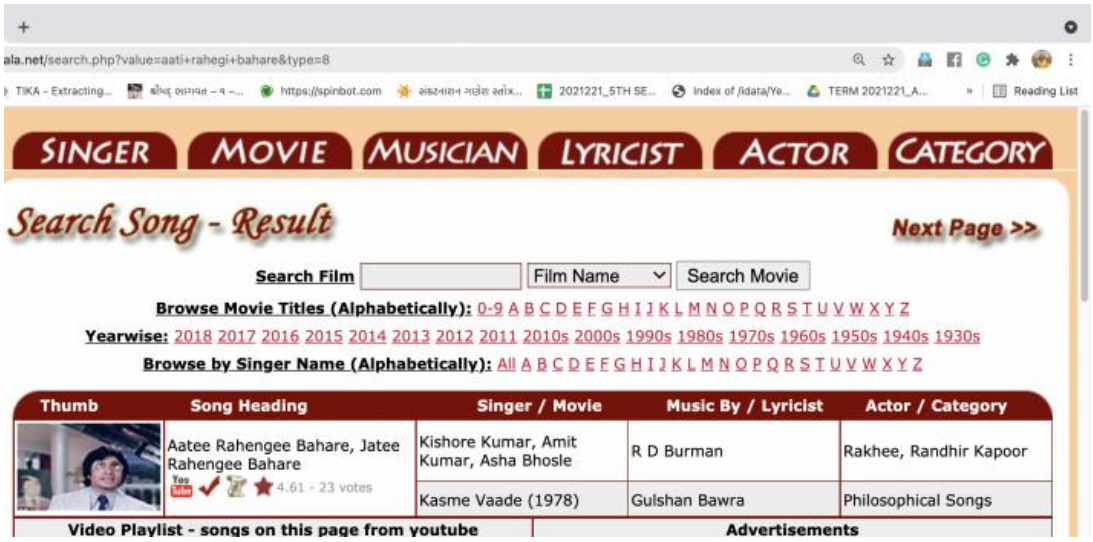


Figure 3.2 URL Mapping - Hindigeetmala Song “Aatee Rahengee Bahare”

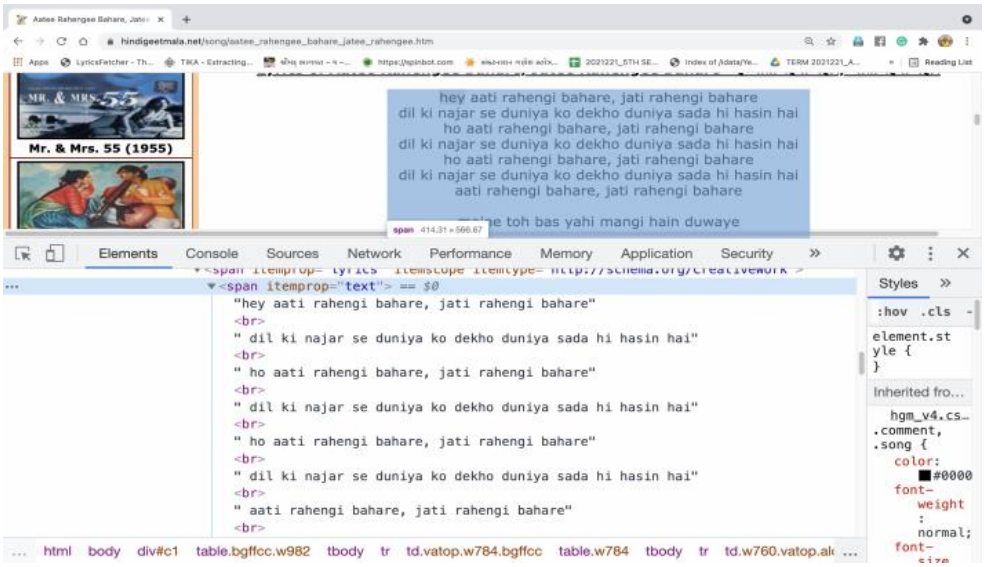


Figure 3.3 Lyrics with Tags available on - Hindigeetmala Song " Aatee Rahengee Bahare"

3.4 Dataset

For Lyrics Extraction there is must requirement to collect metadata of all songs, and as such, it is already available in metadata thus to prepare dataset for that metadata information which is easy to build. Now based on that metadata, there is primary need to crawl for lyrics of song, and thus based on that Dataset of Lyrics is prepared which is built by proposed methodology in which as initial stage metadata Tag of Song Title is passed and as stored in MySQL dataset successfully retrieved. Based on Metadata Song titles are crawled to extract lyrics over internet. There are many Hindi lyrics website available like, lyricsmint.com, geetmanjusha.com, smiriti.com, songlyrics.com, lyricsmasti.com. As per structural analysed web page information of these sites and found that *www.songlyrics.com* contains proper LRC file. Thus, to extract Lyrics, one of above-mentioned websites is used for crawling purpose. Also to highlight that for some latest movie's song retrieval *www.hindigeetmala.com* is used. As mentioned in proposed system for lyrics extraction procedure, work flow of algorithm proceeds as follow.

Algorithm Sanchay ()
Step – 1: Extract Metadata from mp3 song by mp3parse ().
Step – 2: Store Metadata in database.
Step – 3: Apply Lyrics Gatherer (song title, song artist) by using metadata retrieved information.
Step – 4: IF Lyrics Found
Store it in database
ELSE
Return Unsuccessful retrieval.

Above mentioned code is implemented in Java using Net beans IDE. Retrieved result is stored in MySQL database. Libraries that have been used Jsoup for crawling of lyrics online. Also, other libraries like Apache Tika have been used to extract metadata of mp3 song. Using these libraries above mentioned algorithm is implemented and lyrics are retrieved.

Column	**Type**	**Null**
song_id	int (11)	No
song title	varchar (100)	No
song_lyricst	varchar(100)	No
song_lyrics	Text	No
song_genre	varchar (100)	No
song_artist	varchar(100)	No
movie_name	varchar(100)	No
Year	varchar(5)	No

Table 3.2 MySQL Data Table Structure

18	Aatee Rahengee Bahare, Jatee Rahengee Bahare - हे आती रहे ` गी वहारे ` , जाती रहे ` गी वहारे `	Gulshan Bawra	hey aati rahengi bahare, jati rahengi bahare dil ki najar se duniya ko dekho duniya sada hi hasin hai ho aati rahengi bahare, jati rahengi bahare dil ki najar se duniya ko dekho duniya sada hi hasin hai ho aati rahengi bahare, jati rahengi bahare dil ki najar se duniya ko dekho duniya sada hi hasin hai aati rahengi bahare, jati rahengi bahare maine toh bas yahi mangi hain duwaye phulo ki tarah ham sada muskuraye maine toh bas yahi mangi hain duwaye phulo ki tarah ham sada muskuraye gate rahe ham khushiyo ke git yuhi jaye bit jindagi aati rahengi bahare, ho jati rahengi bahare dil ki najar se duniya ko dekho duniya sada hi hasin hai aati rahengi bahare, hey jati rahengi bahare tumse hain jab jivan me sahare jahan jaye najre wahi hain najare tumse hain jab jivan me sahare jahan jaye najre wahi hain najare leke aayegi har nayi bahar rang bhara pyar aur khushi aati rahengi bahare, hey jati rahengi bahare ha dil ki najar se duniya ko dekho duniya sada hi hasin hai aati rahengi bahare, jati rahengi bahare hey ham jo mile ho toh dil ko yakin hai dharati pe swarg jo hai toh yahi hai ham jo mile ho toh dil ko yakin hai dharati pe swarg jo hai toh yahi hai bhule se bhi gum aaye na waha, pyar hai jaha bandagi aati rahengi bahare, jati rahengi bahare dil ki najar se duniya ko dekho duniya sada hi hasin hai aati rahengi bahare, jati rahengi bahare	Philosophical Songs	Kishore Kumar, Amit Kumar, Asha Bhosle	Kasme Vaade	1978

Table 3.3 Sample Data Value of Song "*Aatee Rahengee Bahaare*"

3.5 Dataset Statistics

Dataset Sanchay prepared from hindigeetmala.com website retrieval using proposed algorithm that shows result statistics of data as mentioned in table.

Computation time of retrieval of 100 songs takes 4 minutes 17 sec and for 5076 Songs it takes Time Complexity measure as 2 hours 17 minutes (based on Internet availability and speed) and Size of dataset is: 73,93,647 bytes. (Sample Size 7.4 MB on disk).

No. of Songs	5076
No. of Lyricist	715
No. of Movies	2728
No. of Artist	1987
No. of Genres	57
Year Range	1940 - 2018

Table 3.4 Dataset Overall Statistics

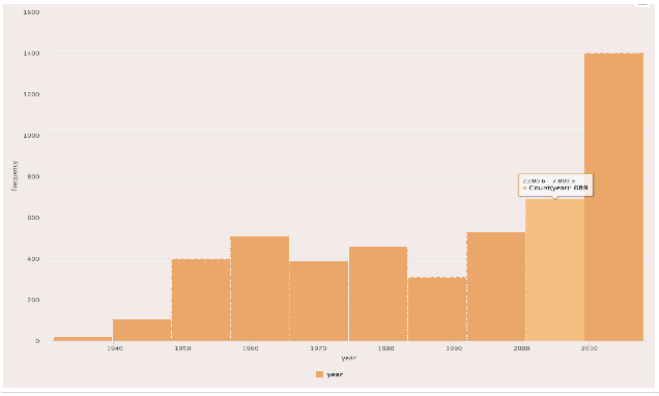


Figure 3.4 Year wise collected Song Statistics

```
Document doc = Jsoup.connect ("https://hindigeetmala.net/").get ();
log(doc.title());
Elements newsHeadlines = doc.select("#mp-itn b a");
for (Element headline : newsHeadlines) {
  log("%s\n\t%s",
    headline.attr("title"), headline.absUrl("href"));
}
```

Now based on that metadata, there is primary need to crawl for lyrics of song and thus based on that Dataset of Lyrics is prepared which is built by proposed methodology in which as initial stage metadata Tag of Song Title is passed and as stored in MySQL dataset successfully retrieved metadata and based on those crawled lyrics over internet. There are many Hindi lyrics website available like, lyricsmint.com, Geetmanjusha.com, smiriti.com, songlyrics.com, lyricsmasti.com. As per structure of website analyzed, the web page contains metadata information of these sites and data extracted from www.songlyrics.com which contains proper LRC file, meaning lyrics of song, and thus use this website for crawling purpose. Dataset Sanchay prepared from hindigeetmala.com website retrieval using proposed algorithm shows results statistics of data as mentioned in table.

Computation time of retrieval of 100 songs contains almost Time: 4 minutes 17 sec and of 5076 Songs it takes Time Complexity measure as 2 hours 17 minutes (based on Internet availability and speed) and Size of dataset is: 73,93,647 bytes. (Sample Size 7.4 MB on disk)

Sanchay dataset is useful as training model for researchers who want to classify songs based on genre and Lyricist wise as it covers almost all years of Bollywood Hindi songs. Total of 57 genres are enlisted in retrieval process but in India 125 types of genres are available, some of the top 10 genres are enlisted as below. Result shows Bollywood filmy songs mostly contain Romantic and Dance songs, while others are followed as Sad, Philosophical, Dard bhare, Masti bhare etc.

3.6 Music Classifier System

The technique presented over here is using implementation of kNN, SVM, Naïve Bayes and Random Forest training algorithm model and testing as well. Here in WAV format audio files are given and which are then passed through sequential set of steps on which various operation such as Labeling, Feature extraction then preprocessing training and Multi model training is implemented through which final prediction of genre of song is carried out as label genre of song.

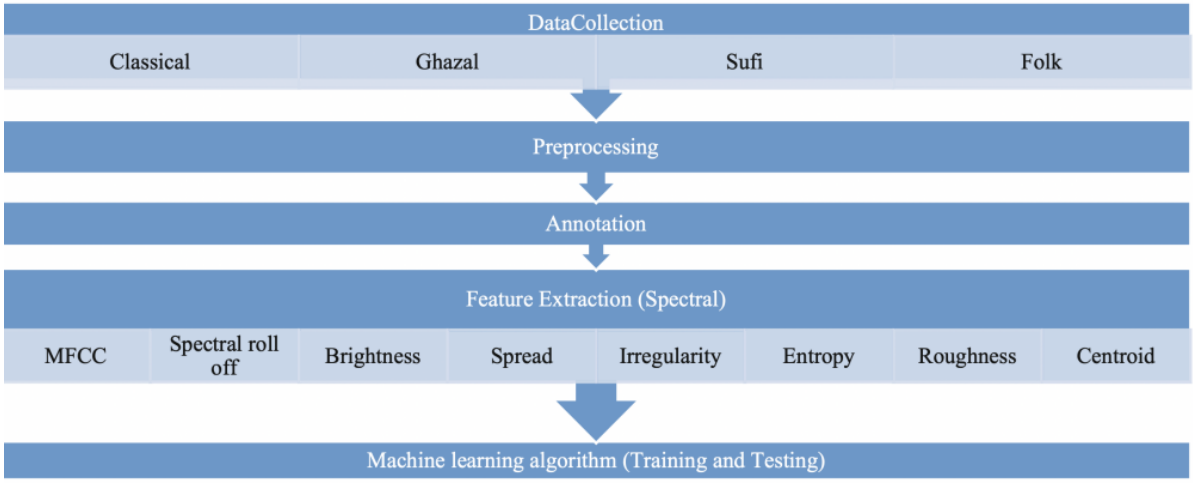


Figure 3.5 Music Classifier Machine Learning Algorithm Steps

Step 1: The songs of 10 genres Classical, Tareef, Philosophical and Romantic etc. are considered.

Step 2: The database collected and sampled at a rate of 44100 Hz and 16-bit precision.

Step 3: 917 songs consisting of approx. 90 songs of each genre are considered in this research work. (100 Genres)

The music clips considered are downloaded from freely available online songs.

The training dataset consists of 917 songs and test data contains another 900 songs.

Spectral feature is used to differentiate the defined classes.

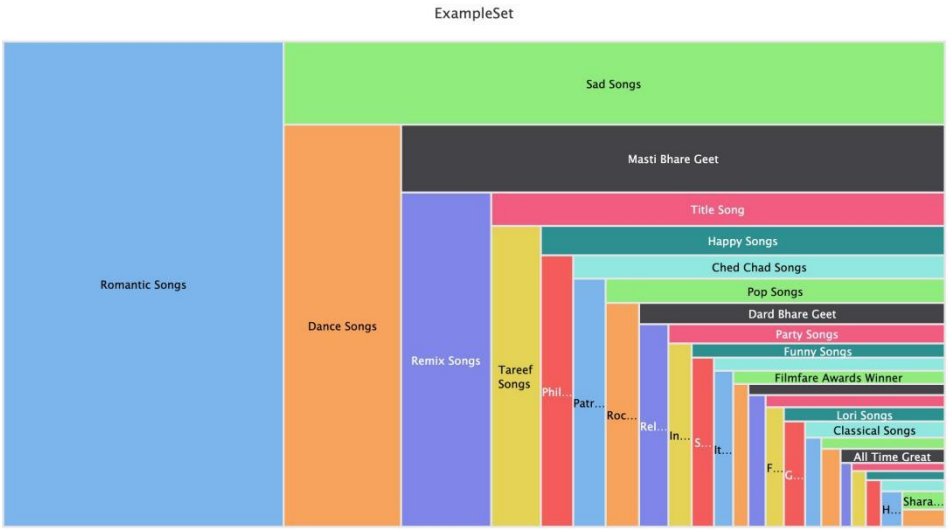


Figure 3.6 Collection of Majority of Hindi Film Industry Songs
(as per Genre wise)

3.6.1 Stage 1: Genre Classification

Literature Review as mentioned in earlier chapter for Track Extraction is done on Western Music Dataset, while as in Hindi, whole structural composition changes due to music notes structure, instrumental variations, segmentation features and composition of song made, thus, the first and foremost question arises to study difference of western and Indian music, hence the feature to extract track plays an important role.

Western Music	**Indian Music**
Polyphony	Homophonic
Harmony	Melody
Composed	Improvised
Instruments are “accompaniments”	Vocals are “accompaniments”
Uses Texture Composition	Uses Taal - a cycle of beats centered around ‘Sam’ that repeats itself

Table 3.5 Western Music v/s Indian Music [99]

Binary classification and multiclass classification are two distinct challenges that fall under the umbrella of genre categorization. Multiclass classification entails assigning an object to several classes, while binary classification, which is a well understood operation, includes only two classes. Multiclass classification frequently necessitates the simultaneous use of various binary classifiers because many classification techniques that have been created particularly for binary classification. [79] There are several varieties of audio characteristics which are helpful for categorization in addition to the feature extraction procedure. The following features are the best ones that primarily concentrate on the structural depiction of genres. [100]

3.6.1.1 Timbre features

Pleasant tunes often have a lively and bright tone, whilst sorrowful ones tend to be reflective and dark. It is possible to determine the emotion's polarity using timbre characteristics. The following is a list of the timbre characteristics used: Spectral smoothness, MFCC, peak based, LPC, spectral smoothness, centroid, Rolloff point, flux, zero crossing, strongest frequency via zero crossing, strongest frequency via spectral centroid, strongest frequency via FFT maximum. The applied technique determines the average and standard deviation for every frame.

3.6.1.2 Intensity features

It is possible to determine if an emotion is strong or not by looking at intensity aspects. For instance, we may determine whether a music is energetic or enthusiastic by utilizing intensity characteristics if the song expresses a favorable mood. The intensity features presented in this article are Fraction of Low Energy Windows and RMS. The calculation of two important measures of mean and standard calculations gave us four intensity characteristics.

3.6.1.3 Rhythm features

We may also infer to some information about the good or negative sentiment of the music through rhythmic elements. Rapid songs are usually more upbeat than sluggish ones. Strongest Beat, Beat Sum and Strength of among all Strongest Beat are the rhythm characteristics that has been retrieved.

Six rhythm characteristics were obtained by computing the mean and standard as well. To facilitate the mining of the relationships between these variables and the related mood of the audio recordings, the features for each piece of music are extracted and collected into a uniform file format.

These characteristics are shared by all classes, which means they don't provide much information to the categorization process. While low information characteristics don't harm a system alone, they might reduce system performance when combined.

By reducing noisy data, low information features help the proposed model become clearer. Using the higher information features improves performance and reduces model size, which leads to quicker training and classification and reduced memory usage.

For the previously stated rationale, only timbre elements were used for track extraction and genres were categorized accordingly. These days, any qualities other than MFCC, OSCC, and LPC are only helpful for psychological purposes, such as mood-based music classification. So, it becomes confusing to decide which feature to use.

3.6.1.4 LPC v/s MFCC

LPC is typically utilized for resynthesis and speech analysis. Vocoders that employ musical instruments as an excitation signal to a time-varying filter calculated from a singer's voice can be created via LPC synthesis. LPC must be tolerant of transmission faults as it is commonly used to communicate spectral envelope data. Since the filter coefficients are extremely error-prone, it is not recommended to transmit them directly (see linear prediction for a definition of coefficients). Put differently, even a tiny inaccuracy has the potential to completely distort the spectrum, or even worse, to cause instability in the prediction filter. [102]

However, Mel-frequency cepstral coefficients, or MFCCs, are the coefficients that come together to form an MFC. They originate from a particular kind of cepstral representation (a nonlinear "spectrum-of-a-spectrum") of the audio sample. The Mel-frequency cepstrum (MFC) differs from the conventional cepstrum in that its frequency bands presented are evenly spaced on the Mel scale, almost emulating the response of the human auditory system. The regular cepstrum's frequency bands which are spread linearly. When it comes to audio compression, for instance, this frequency warping can enable improved sound representation. [101]

3.6.1.5 MFCC v/s OSCC

OSCC	MFCC
Octave Scale Cepstral Co-efficient	Mel Frequency Cepstral Co-efficient
Having Octaves of Liner Frequency	Sine & Cosine of Discrete Frequency
Filter out instrumental part	No Filtering process
Works Best n Vocal Instruments	Compatible with musical instruments
Computation Complexity is Higher	Computation Task Is Easier

Table 3.6 MFCC v/s OSCC [45,102]

According to the abovementioned analysis, the proposed methodology selected to identify genres using MFCC rather than OSCC due to cause of the latter's shortcomings.

Numerous studies have been conducted over the past three decades to identify the most usable techniques for extracting numerical characteristics from audio recordings and the best-performing classification algorithms for audio classification, as machine learning and deep learning have progressed. For this specific field, "music information retrieval systems" is a good word. In a comparative investigation, Tao Li et al. recognized Mel Frequency Cepstral Coefficients (MFCC) as one of the content-based acoustic characteristics extraction techniques from audio. [15] The researchers also proposed a unique methodology called Daubechies Wavelet Coefficient Histograms (DWCH).

The requirement for a novel method proposal is established by the authors' observation that conventional feature extraction techniques largely capture partial information of musical signals. The researchers compared other [78], gave a thorough analysis of the most current audio approaches (as of 2016), classifying them first into the Physical and Perceptual categories, then into different seven types based on domain.

Mel Frequency Cepstral Coefficients (MFCC) were classified as Wavelet Based Perceptual Features in the study's analysis of audio feature extraction methods. The methods for extracting musical features have developed further. On a set of unrelated music regression and classification tasks, K Choi conducted a comparison analysis between the traditional MFCC feature extraction technique and a pre-trained convnet

feature. He discovered that the convnet feature outperformed the baseline MFCC feature in all the tasks taken into consideration [21], providing a foundation and justification for the success of the application of the transfer learning concept on music classification tasks.

A) MFCC

The real cepstrum of a windowed short-time signal obtained from the fast Fourier transform of the speech signal is known as the MFCC representation. This simulation differs from the real cepstrum in that it makes use of a nonlinear frequency scale which mimics the activity of the available auditory system. The MFCC might resemble this: [102] MFCCs are frequently employed as features in speech recognition systems, including those that can automatically identify phone digits that are spoken into a handset. Applications for retrieving music information, such genre classification and aural similarity measurements, are increasingly using MFCCs.

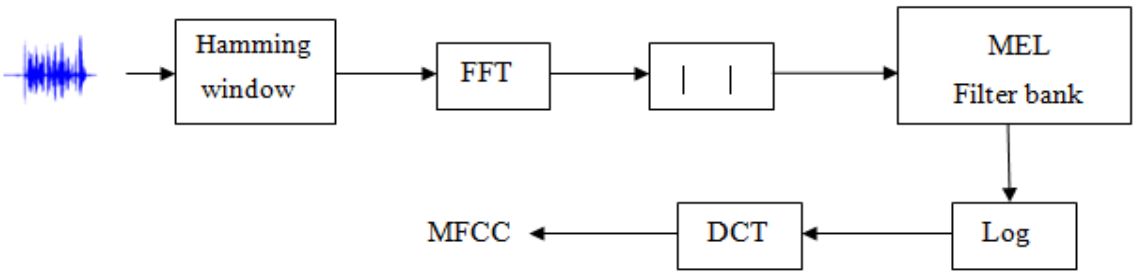


Figure 3.7 MFCC Calculation [6]

MFCCs is a normalized energy parameter of audio**.** A popular formula to convert f hertz into m Mel is:[102]

$$m = 2595 \log_{10}\left(1 + \frac{f}{700}\right)$$

MFCCs are often obtained in this way:[102][103]

1. Take a signal and its Fourier transform (a windowed snippet).

2. Use triangle overlapping windows to map the powers of the spectrum that were acquired above onto the Mel scale.

3. Calculate the power logs for each of the Mel frequencies.

4. Consider the list of Mel log powers and its discrete cosine transform as a signal.

5. The amplitudes of the resultant spectrum are the MFCCs.

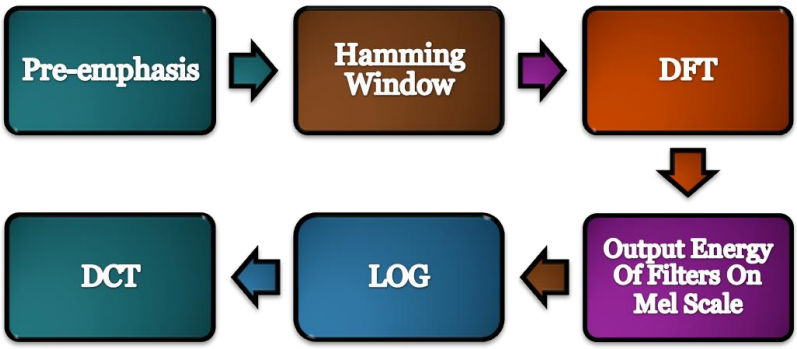


Figure 3.8 Process of Calculation Of MFCC

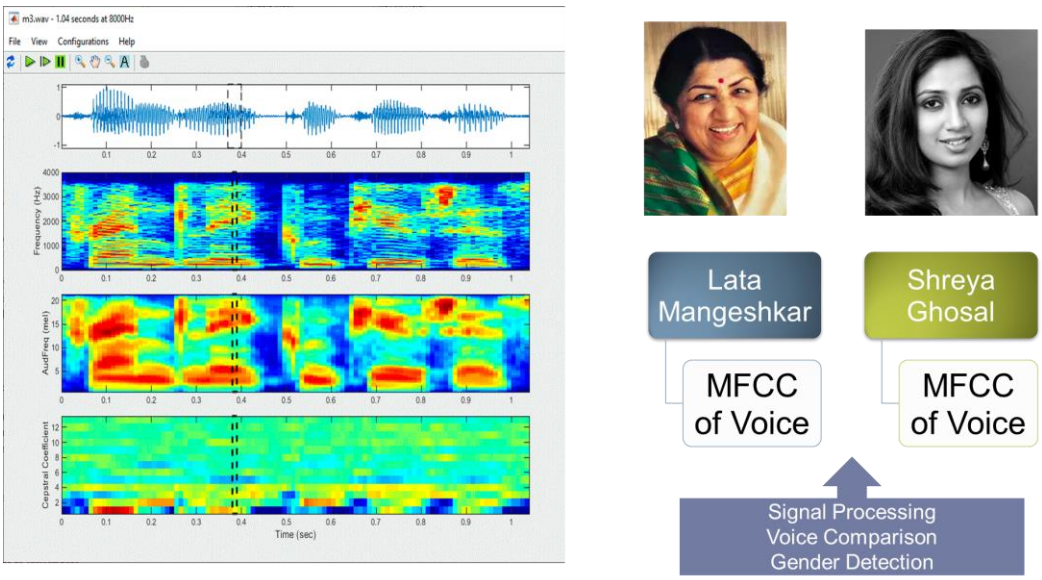


Figure 3.9 MFCC Voice Comparison Signal Chart of Different Singers

For implementation point of view Songs in .WAV Format [78] with 16 Window Sample Rate and 13 DCT co-efficient are given are input set for feature extraction procedure. The reason to keep audio signal in WAV file is a lossless audio format that does not compress the original analog audio recording from which it is derived. WAV files objectively have better quality and provide more true and accurate audio clips. The Converter class implements the conversion of an MPEG audio file to a

.WAV file. To convert an MPEG audio stream, just create an instance of this class and call the convert () method, passing in the names of the input and output files.

- Size Of Song varies from 1.2 MB to 9MB
- Default Audio Sample Rate is
    - 48kbps to 128 kbps – MP3
    - 32kHz to 42 kHz –WAV
    - Thus, Taken Max 44.1 kHz = 16-bit Samples
- 16-bit Samples Capability
    - Size Up to 10.584MB, No Loss of Information, CD - Quality

Thus 16-bit sample rate is kept while calculating MFCC audio feature. MFCC having 40 Filters. If DCT=13 then, 13 Linear Filters & (40-13) =27 Logarithmic Filters. By Standard 13 Linear is good, but it is variable. DCT 13 means 13rd level of linear smoothening.[78]

B) ZCR

An audio frame's Zero-Crossing Rate (ZCR) measures how frequently the signal's sign shifts across the frame. In other words, it is determined by dividing the number of times the signal's value shifts from positive to negative to vice versa by the frame's duration. The ZCR is defined according to the following equation [104]:

$$z_i = \sum_{m=1}^{n-1} sign[x_i(m-1)x_i(m)]$$

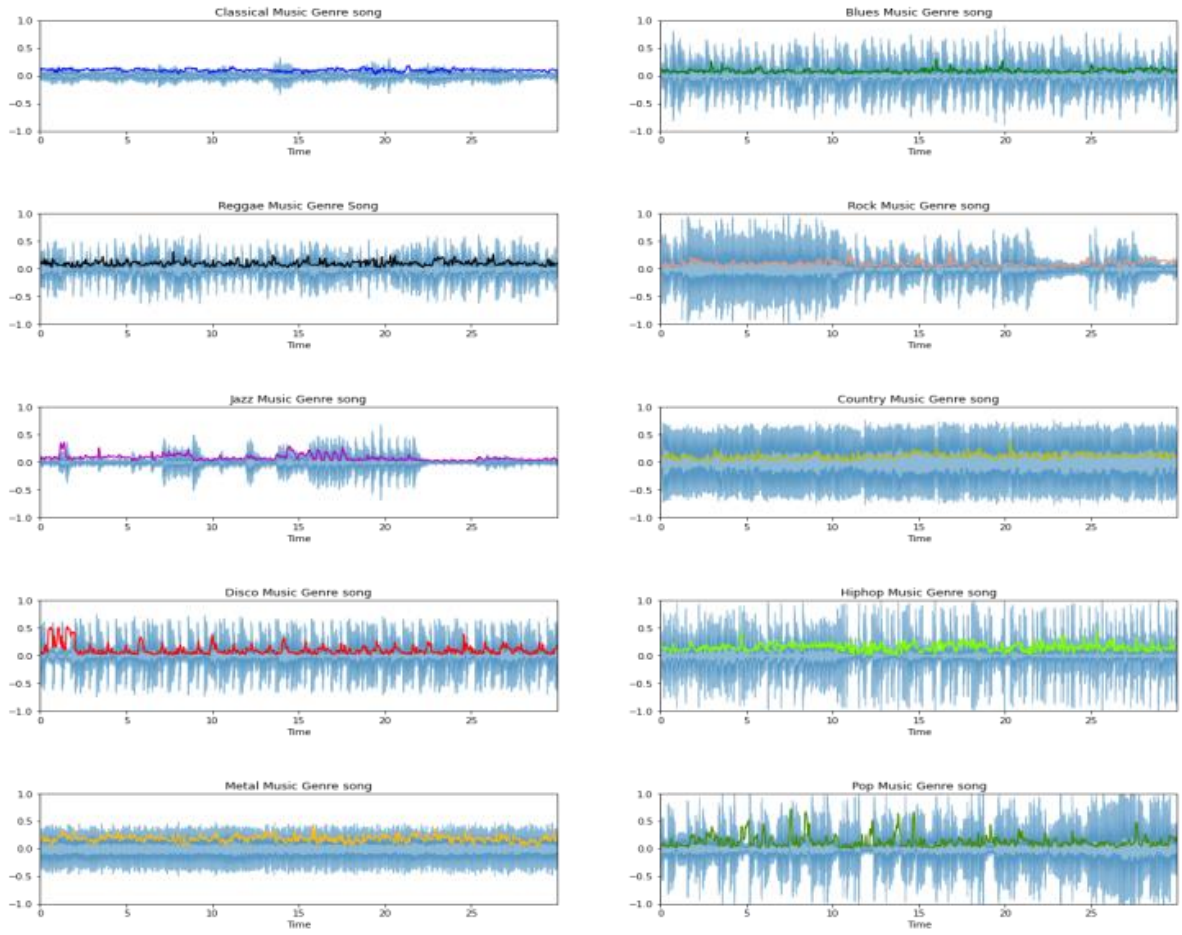


Figure 3.10 Sample ZCR values of different Genres

C) Centroid

A measurement used in digital signal processing to describe a spectrum is called the spectral centroid. It reveals the location of the spectral center of mass. It is strongly related to the perception of brightness in a sound perceptually. It is often referred to as the spectral mass center. It is determined using a Fourier transform to determine the weighted mean of the frequencies present in the signal, with their magnitudes serving as the weights. [104]

$$Centroid = \frac{\sum_{n=0}^{N-1} f(n)x(n)}{\sum_{n=0}^{N-1} x(n)}$$

Where x(n) represents the weighted frequency value, or magnitude, of bin number n, and f(n) represents the center frequency of that bin.

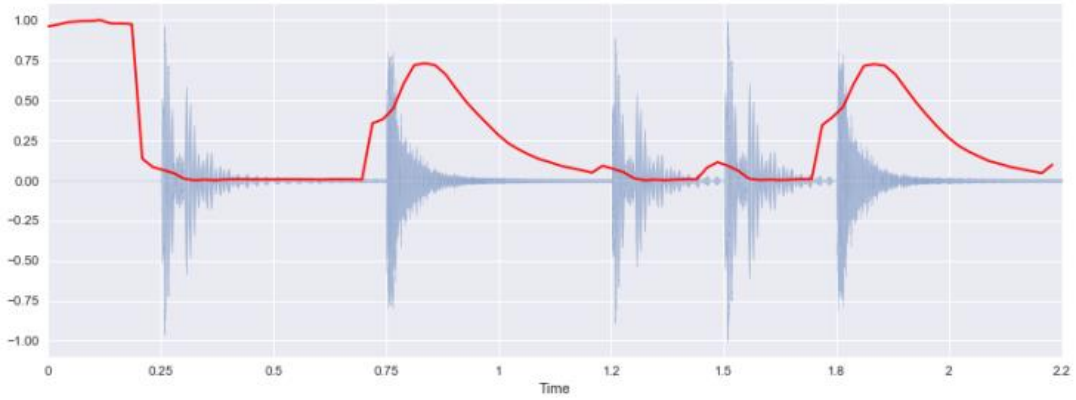


Figure 3.11 Centroid Value Signal Map

D) Roll Off

Specifically in electrical network analysis, and specifically in conjunction with filter circuits in the transition between a passband and a stopband, roll-off is the steepness of a transfer function with frequency. Although it is most frequently used to describe the network's insertion loss, it can theoretically be used to describe any significant function of frequency and any technology, not just electronics. Typically, roll-off is calculated as a function of logarithmic frequency.[104]

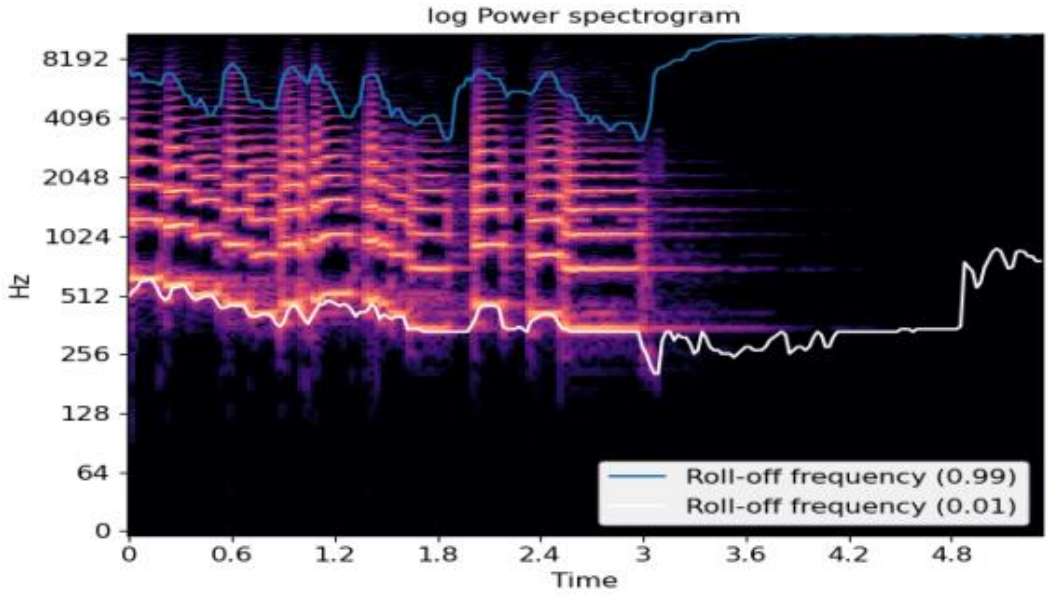


Figure 3.12 Log Power Spectrogram

3.6.1.6 Methodology of Proposed Research

The technique presented over here is using kNN training model and testing as well. Here in WAV format audio files are given which are then pass through a set of steps on which various operation such as Labeling, Feature extraction then preprocessing training and Multi model training is implemented through which final prediction of genre of song is carried out as label genre of song.

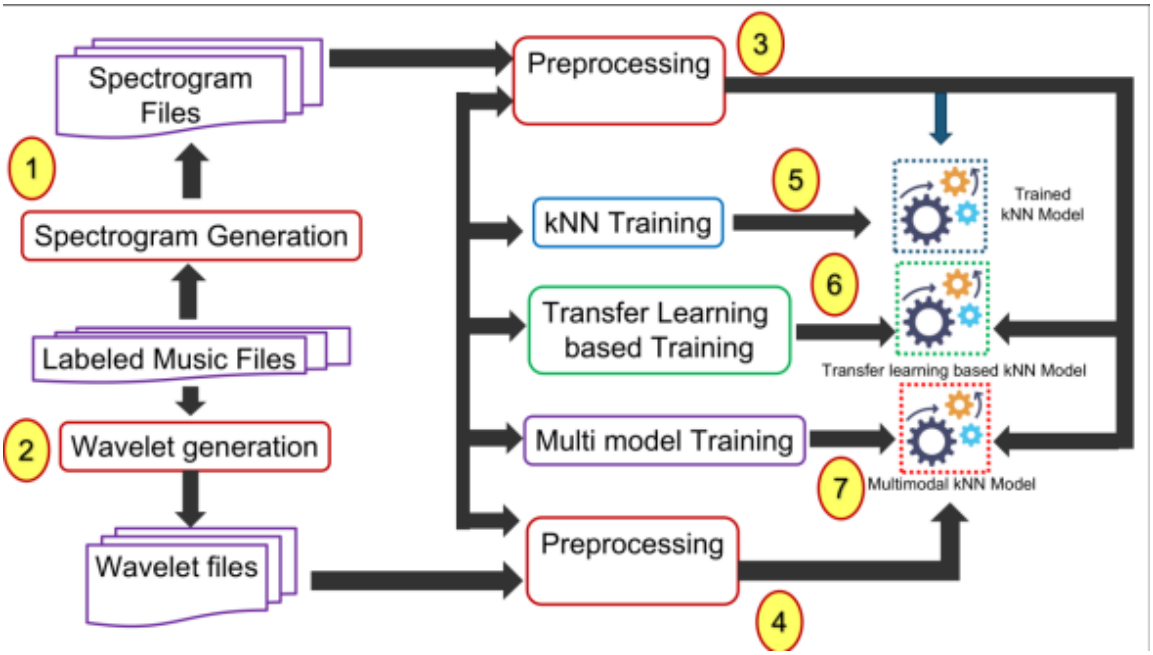


Figure 3.13 Proposed Architecture Training Testing Model

Step 1: The songs of 10 genres Romantic, Pop, Philosophical and Filmi Sugam etc. are considered.

Step 2: The database collected and sampled at a rate of 44100 Hz and 16-bit precision.

Step 3: 926 songs consisting of approx. 90 songs of each genre are considered in this research work. (10 Genres)

The music clips considered are downloaded from freely available online songs.

The training dataset consists of 917 songs and test data contains another 900 songs.

Spectral feature is used to differentiate the defined classes, further to address classification of songs based on spectral features random forest algorithm is implemented, comparative analysis between other machine learning algorithm is carried out during training and testing phase with different ratios as well.

3.6.2. Stage 2: Lyrics Classification

A song's lyrics are a crucial part of it. A song's lyrics are a set of words used by the songwriter to convey their emotions and ideas so that listeners may decipher the message the song is trying to convey. [3]

Given this, it is possible to employ song lyrics as a study object when classifying songs according to their subjects. [4] Several additional song components, such as user interpretation, which has been employed in prior research, can be used as parameters of song classification in the study of song classification based on major subject. [49] The researcher was able to demonstrate that adding user interpretation to poetic lyric difficulties may be solved, based on the findings of a prior study. [49] By comparing solely lyrics, the algorithm achieves superior performance using user interpretation data and lyrics.

As research study, we continue to employ lyrics as one of our research objects in this investigation. Since there isn't presently a website such to songmeanings.com for Indonesian music, we are unable to employ user interpretations; nevertheless, we do provide genre and artist information to get over the poetic difficulties associated with categorizing songs according to themes.

Furthermore, utilizing song lyrics, genre, and artist information—a novel idea not explored in earlier topic classification research—this study developed a system that can classify songs based on their subjects in two steps. Some song themes can be further subdivided into more specialized subjects, which is why this two-stage approach is necessary.

"*What kind of love song*?" can be used to categorize songs about falling in love or songs about shattered hearts. As an additional illustration, songs about religion may be categorized according to their respective faiths (*Islam, Christian, Hindu, Buddhist*). Listeners will be able to locate songs with more specialized themes with the aid of this two-stage categorization. Generally speaking, there are many steps to this research: data preparation, feature extraction, classifier construction, classification, and system assessment.

Case folding, tokenization, stop word elimination, and stemming are the four steps involved in data processing. Nazief-Andriani method is the stemming algorithm utilized in this preprocessing data since this the label encoding technique is used to preprocess artists and genres, converting the category data about them into numerical

data. Using the bag of words model, feature extraction is carried out following data pretreatment.

3.6.2.1. Expressiveness in Lyrics

Across the subcontinent, expressive are typically used in elegant articulation and academic categories. About 1,000 movies are released in Hindi each year, making it one of the largest entertainment industries in the world. [105]

3.6.2.1.1. Reduplication in Hindi

Regardless of linguistic family, reduplication is a common and advantageous feature of all South Asian languages [106]. The common instances of reduplication in the language should be examined first in order to demonstrate how expressive in Hindi are new. The accompanying reduplication designs are shown in Hindi in [107]:

1) Full lexical reduplication: is the presentation of durative or iterative circumstances through the reduplication of lexemes.
baiʈhe-baiʈhe 'while sitting'
da:ne 'grain' da:ne-da:ne 'each grain'

2) Partial lexical reduplication (echo-formation): When the reduplicate base's first consonant is changed to "v," the meaning becomes "etc."
kha:na: 'food' kha:na:-va:na: 'food, etc.'
roti 'flatbread' roti-voti 'flatbread, etc.'

3) Redundant Compounds: When two words with similar semantic meanings are combined, they also have certain phonological characteristics, which adds the property of "etc."
dhən 'wealth' dhən-dəulət 'wealth-wealth'
dhərm 'religion' dhərm-ima:n 'religion-religion'

4) Expressive reduplication: "segmental indivisible morphemes with minimal meaning that are composed of repeated syllables" (Mohan 2015, 119).

ʈəp-ʈəp 'sound of water dripping'

cit-cit 'sticky'

Only the most basic expressive form—in which the reduplicant copies a monosyllabic base—is demonstrated in Mohan's examples. On the other hand, the structure of expressive in Hindi is highly flexible, which sets it apart from the prosaic grammar's regulated forms of reduplication (1-3). For example:

1) $C_1VC\text{-}C_2VC$

- *rim-jhim* 'sound of rain falling'
- *jhil-mil* 'shining'

2) $CV_1C\text{-}CV_2C$

- *dhək-dhuk* 'heartbeat'
- *ʈa:pur-ʈupur* 'rain falling'

3) $C_1V_1C_2V_1C_2V_1\text{-}C_1V_1C_2\ (V_1)$

- *jhənənə-jhən* 'sound of foot anklets'
- *ghənənə-ghənə* 'sound of thunder'

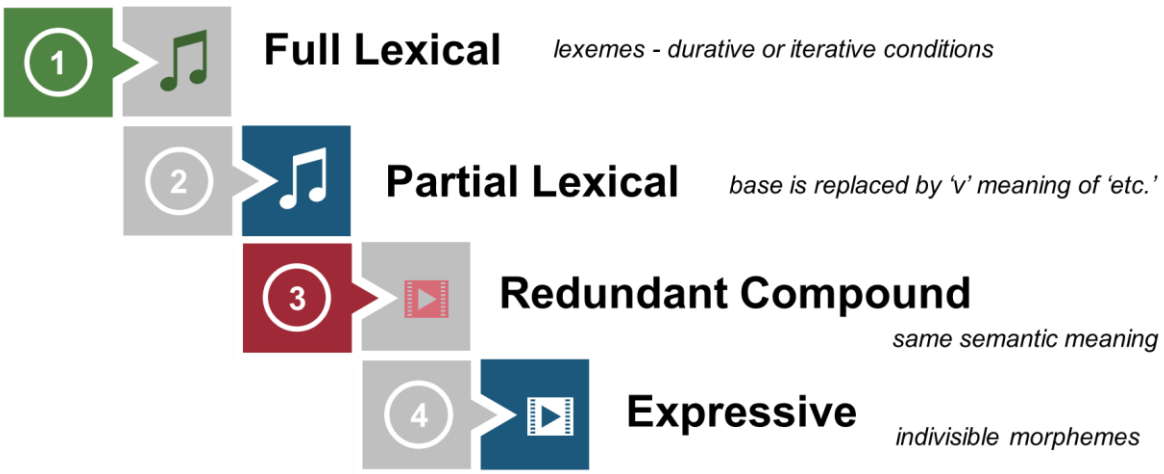


Figure 3.14 Reduplication in Hindi Songs [106,107]

In this way, expressiveness demonstrates a vast array of possible morphological and phonological instances that fundamentally depart from the predefined examples of excess, incomplete, and full reduplication in Hindi as defined by Mohan. [31]

Given their extreme structural flexibility and ability to defy categorization in terms of morphological and phonological design, expressive in Hindi should be examined separately from monotonous reduplication and treated as a distinct word class.

Hindi cinema would not be the same without its soundtrack, which dates back to the middle of the 20th century. But as Sarasin points out, movie melodies are more than just a component of movies; they constitute the primary melodic category in South Asia. Within the films itself, songs are performed in accordance with a variety of themes, such as festivals of occasions, weddings, or social gatherings, or to incite people to fight or celebrate other events. In any case, emotional love is the most common topic for a song in a Hindi movie.

Here, expressive words have a special function since it is in expressions of love that representation, metonomy, and other lovely devices are most frequently employed, and here is where the expressive word class's syntactic flexibility really shines. Expressives, for instance, are typically heard in reference to explicit objects, such as a woman's anklets (pa:yəl), which convey—or "talk"—emotions that are inappropriate to express verbally through dancing activities.



Figure 3.15 Expressiveness in Hindi Film Song

Expressive also promote emotionally charged interactions between the beloved and their darling and the ordinary environment, such as climax moments like a rainfall or the stars sparkling in the night sky.

Finally, it is essential for certain bodily parts, such as the "heart" or the "eyes," to metaphorically represent the human person themselves, conveying a wide range of emotions. This lovely aspect is further supported by the use of “expressivization”.

On the other hand, It refer to this creative interchange between the two in each cycle of Hindi movie song verses as "expressivization". Reworking noteworthy or phonesthetic elements in ordinary action word structures into expressive and vice versa is known as expressivization.

Expressivization in Hindi occurs when the linguistic basis is repeated. Hindi movie tunes appeal to a broad, semantically diverse audience, and the frequent use of expressive words implies that even non-Hindi speakers are likely to understand the expressive connotations. This volume aims to demonstrate that expressive are widely recognized throughout the South Asian etymological zone. Hindi cinema songs have recently tended to blend more significantly across regional phonetic assortments.
Expressives are common in such multilingual songs because they are anticipated to have cross-semantic conceivability. For instance, the battle song arrangement in the recently released film Bajirao Mastani blends together a number of Marathi patterns; nevertheless, expressive are used to reduce semantic differentiation and highlight the beautiful elements. The Marathi frameworks are highlighted.

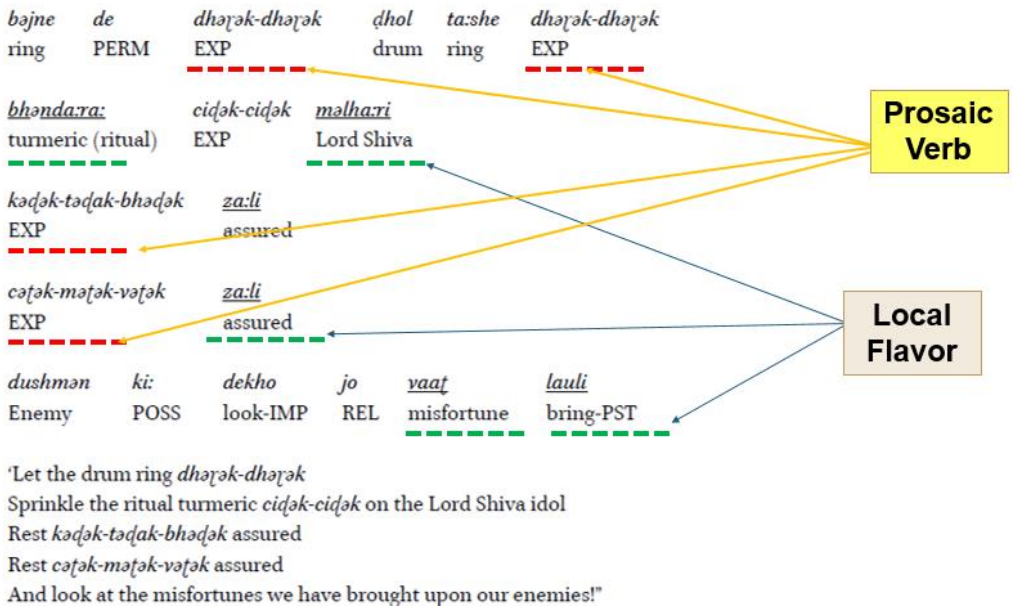


Figure 3.16 Expressivization of prosaic verb and local flavour in Malhari of movie Bajirao Mastani

3.6.2.2. Similarity Metrics for Lyrics Classification

When it comes to classifying lyrics, measuring similarity between texts is crucial. Several similarity metrics can be applied, based on the particular demands of the categorization assignment you're performing. These are a few typical similarity measures for categorizing lyrics [6]:

- **Cosine Similarity:** The cosine of the angle formed by two vectors in a multidimensional space is measured by cosine similarity. Regarding the categorization of lyrics, each song's lyrics may be represented as a vector in a high-dimensional space, with each dimension being a term (word) in the lexicon. The similarity between the lyric vectors of two songs is then quantified using cosine similarity. It is frequently used in text categorization jobs and is resilient to text length.

- **Jaccard Similarity:** Jaccard similarity measures the similarity between two sets by comparing their intersection to their union. In lyrics classification, you can treat the set of unique words in each song's lyrics as sets and compute the Jaccard similarity between them. This metric is particularly useful when one is interested in the presence or absence of words rather than their frequencies.

- **Edit Distance (Levenshtein Distance):** The term "edit distance" refers to the least number of modifications (replacements, insertions, and deletions) needed to change one text into another. When it comes to classifying lyrics, edit distance may be used to quantify the degree of textual content similarity between two songs in order to determine how similar their lyrics are. This statistic is especially helpful for songs that may have distinct phrasing but comparable ideas or topics.

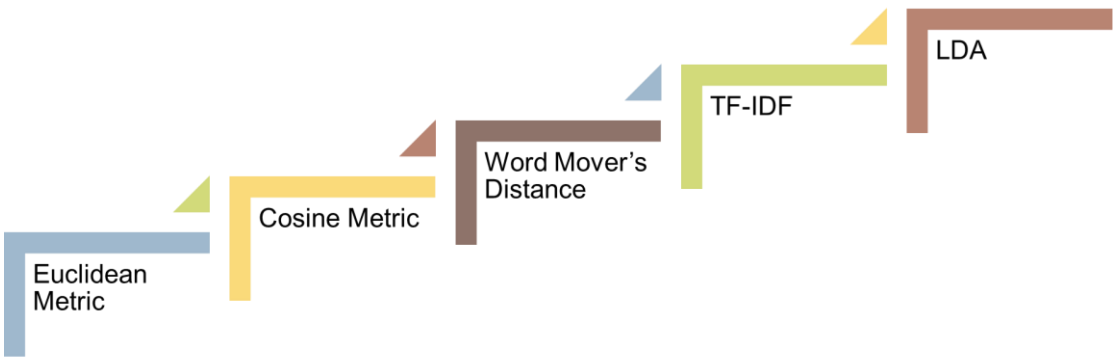


Figure 3.17 Word Similarity Metrics

- **TF-IDF (Term Frequency-Inverse Document Frequency) Similarity**: The significance of a phrase in a document with relation to a corpus of documents is represented by TF-IDF. To classify lyrics, you may generate TF-IDF vectors for each song and use cosine similarity or other distance metrics on these vectors to determine how similar the songs are to one another. TF-IDF takes into account both the frequency of terms in a song and their rarity across the entire corpus, thus capturing the importance of terms in distinguishing between songs.

- **Word Embedding Similarity (e.g., Word2Vec, GloVe):** Words are represented using word embedding techniques as dense vectors in a continuous vector space, with related words grouped together. For lyrics classification, you may encode each song's lyrics as a vector using pre-strained word embeddings such as Word2Vec or GloVe, and then apply cosine similarity or other distance metrics on these vectors to determine how similar the songs are to one another. Word embeddings are a useful tool for capturing semantic links between words and the semantic similarity between lyrics in songs.

- **LDA – Latent Dirichlet Allocation:** A generative probabilistic model called Latent Dirichlet Allocation (LDA) is employed in topic modeling, a method for identifying latent theme patterns in a set of documents. According to LDA, every document in the collection is derived from a variety of themes, and every topic has a distribution of words that define it. The model uses the observed word frequencies in the documents to infer these underlying topic distributions.[73]

  Natural language processing (NLP) activities such as text summarization, content recommendation, information retrieval, and document clustering all make extensive use of LDA. It offers a helpful framework that reveals latent topic structures to facilitate the exploration and comprehension of big text collections.

3.6.2.3. Algorithms for Lyrics Classification

A) SVM

SVM is a potent machine learning model that is applied to regression and non-linear classification problems [108]. It uses "hyperplanes," or boundary lines, to separate the samples of each class. For a given set of data, there could be several hyperplanes for every kind of data. Finding the optimal hyperplanes to optimize the distance between the class borders is the goal of the optimization.

B) Naive Bayes

NB is based on the Bayesian theorem that calculates the probability of a given sample based on the prior probabilities. Despite being simple, NB often provides better results as compared to sophisticated classifiers. Assuming that the attributes are conditionally independent, a sample is attributed to a class with the highest posterior probability. [109]

C) kNN

For supervised learning applications, the k-Nearest Neighbors (kNN) algorithm provides a straightforward and understandable regression or classification method. Being a non-parametric and lazy learning method, it doesn't learn a model while it is being trained and doesn't make any assumptions about the distribution of the underlying data. Rather, it retains a memory of the training dataset and forecasts by comparing incoming data points to those that already exist in the training set.[7] Although kNN is simple to understand and apply, its effectiveness varies based on the parameters selected, the distance measure used, and the properties of the dataset. It is frequently employed as a starting point method or in situations where interpretability is more important than forecast accuracy.

D) Random Forest

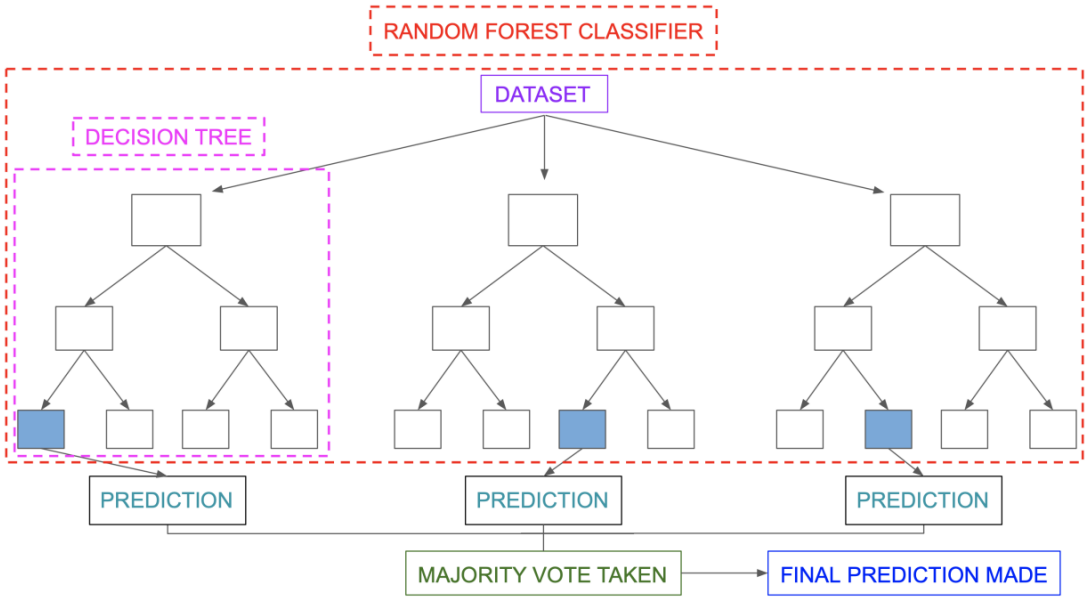


Figure 3.18 Random Forest Classifier

Decision trees based on the qualities of the data are used in the ensemble model RF, which is tree-based. From the input, a random vector is used to construct each tree. Every tree cast one vote for the ultimate prediction, which is determined by the ensemble architecture. [40] Data that are noisy or erroneous, have missing values, or are sparse may all be handled via RF.

A random forest is created by combining several tree predictors so that every tree in the forest is reliant on the values of a random vector that is randomly sampled and has the same distribution for every tree.

The generalization error approaches a limit as a forest's tree count grows. The generalization error of a forest of tree classifiers is determined by the strength of each individual tree in the forest and the correlation between them. A random subset of characteristics may be used to split each node, resulting in error rates that are more noise-resistant and compare well to Adaboost. [41] Internal estimates track correlation, inaccuracy, and strength and are used to illustrate how the number of features employed in the splitting changes. Variable significance is also measured using internal estimations. These concepts also hold true for regression.

A random forest is a classifier consisting of a collection of tree structured classifiers $\{h(x, \Theta_k), k=1, ...\}$ where the $\{\Theta_k\}$ are separate random vectors with identical distributions, and every tree votes one unit for the class that is most popular at input x.

Unpredictable Features There is a persistent difference in the generalization error of random forests described in the literature and others. Random split selection [39], for example, performs better than bagging. Additionally, Breiman's addition of random noise to the outputs [42] performs better. However, Adaboost[41] and other algorithms that reweight the training set adaptively (arcing) perform better than any of these three forests (see [40], [39], and [43]).

The injected randomness must decrease the correlation ρ while retaining strength in order to increase accuracy. In the forests under study, each tree is grown by applying combinations of randomly chosen inputs at each node. The forests that are produced provide accuracy that is comparable to Adaboost.

These techniques belong to a desired class: i) They have accuracy comparable to, or occasionally better than, Adaboost. ii) It can withstand noise and outliers really well. iii) It's more rapid than boosting or bagging. iv) It provides helpful internal estimates for the variables' relevance, correlation, strength, and error. v) It's straightforward and parallelizable. Using random selection from a large number of geometrically specified characteristics to create the split at each node, [43] constructed shallow trees for handwritten letter recognition. In classification, random features and inputs yield good results; in regression, less so. In this study, bagging and random features are the only kinds of randomness employed. It's possible that different kinds of introduced randomness provide superior outcomes.

- **Finding Optimal Number of Trees**

The average number of nodes in a tree is expected to stay constant as the number of trees in the forest increases in order to determine the ideal number of trees. Our goal is to raise random forest's categorization accuracy. As [110] noted, the accuracy of the categorization is dependent on a number of criteria, such as the strength and correlation of the trees, both of which are influenced by the likelihood of a successful split, as seen in the picture.

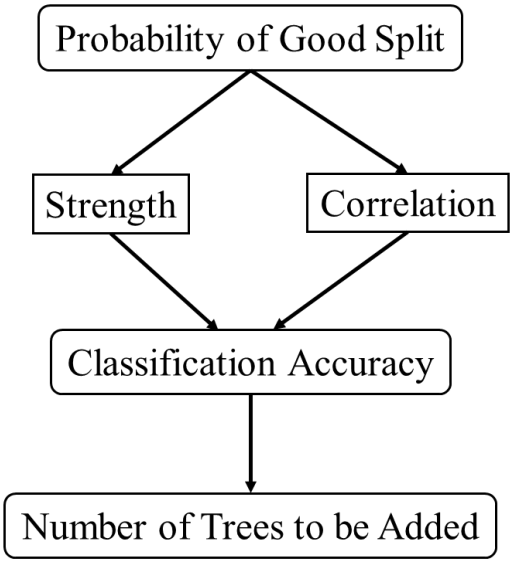


Figure 3.19 Probability of Good Split Random Forest

3.6.2.4. Good Split

If a node is split by an important feature, then there is a possibility of a good split. A good split occurs if at least one important feature is present in a set of features from which we select features randomly for node splitting. As said earlier, the performance (classification accuracy) of RF depends on the strength and correlation of trees, and strength and correlation depend upon the probability of a good split.

The minimum probability of a good split in node i of tree τ is same as the probability of picking up at least one important feature in corresponding Λ. Let us denote this probability as q.

Hence q is the minimum probability of good split in a node. Then r = (1 − q) denotes the probability that no important feature is present in Λ. Bag Γ`(#Γ` = v) contains all the unimportant features and may also contain some important features (which may be promoted to bag Γ in subsequent passes).[13] So, the maximum value of r is

$$r = 1 - q = \frac{\binom{v}{f}}{\binom{u+v}{f}}$$

For each decision tree, Scikit-learn calculates a nodes importance using Gini Importance, assuming only two child nodes (binary tree)[13]:

$$ni_j = w_j C_j - w_{left(j)} C_{left(j)} - w_{right(j)} C_{right(j)}$$

The node j's significance is denoted by ni sub(j). The weighted number of samples that reach node j is represented by w sub(j).

The impurity value of node j is denoted by C sub(j). The left(j) split node is the offspring of the left and right(j) splits on node j.

3.6.2.5. Lyrics Preprocessing

The first step that had to be taken in order for the system to computerize text data and eliminate less important or noisy data, including stop words, was preprocessing the input.

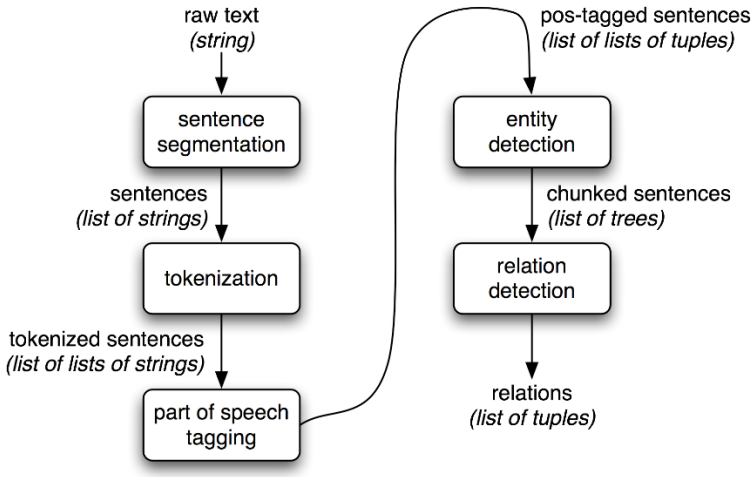


Figure 3.20 NLTK Processing

The stages of preprocessing data in this study consist of several stages as follows:

1. Case Folding

The song lyrics data were first preprocessed by converting all of the letters to lowercase. Non-letter characters were eliminated since they served as a separator.

2. Tokenization

At this point, case folding has been done on the song lyrics. The lyrics were fragmented into individual words (tokens).

3. Stop word Removal

Subsequently, the tokens generated in the earlier phases were chosen by eliminating terms deemed less significant (stop words), for instance, stop words in the Indonesian language such *"apa",*
"untuk", "yang", "ke", and *"di"*.

4. Stemming

In this last stage of pre-processing the data, the chosen words were changed into main/basic terms. Nazief Andriani produced the stemming algorithms.

The data is saved as a preprocessed dataset once all preparation steps have been completed.

5. Label Encoder Genre & Artist

Label encoding also preceded the genre and artist characteristics. Numerical data were generated from category artist and genre data during the label encoding procedure.

6. Splitting Dataset

The data used in this procedure were split into two categories: training data and testing data. The purpose of the training data was to construct a classifier. Data testing with a system testing goal. The research employed two sets of splitting data: 50%-50% and 75%-25%. This was carried out in order to ascertain how much training data affected the system's performance.

7. Feature Extraction

Preprocessed data were retrieved using a bag of words model to extract features. In the training data, this procedure tallied the attendance of every word in its class.

8. Classifier Building

A Naïve Bayes classifier was created in two phases at this point. Testing data is classified using this two-stage classifier construction process. In order to ascertain the probability value of a document against a class, supervised learning classification techniques such as Naïve Bayes are employed, according to Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze [13]. The primary feature of a strong

(naïve) assumption in this machine learning approach is independence. This approach for text categorization is based on two assumptions: Conditional Independence, which holds that each feature's likelihood is independent of its neighbor, and Bag of Words, which holds that a word's position within a document has no bearing. Two computations, namely likelihood probability and prior probability, were used to create the classifier.

Prior probability calculations for each class were calculated using equation.

$$P(A|B) = P(B)P(A \cap B) = P(B)P(A) \times P(B|A)$$

where : $P(A)$ = the prior probability of A occurring

$P(A|B)$= the conditional probability of A given that B occurs

$P(B|A)$ = the conditional probability of B given that A occurs

$P(B)$ = the probability of B occurring

In terms of probability distributions, the Bayes theorem looks like this:

$$f(\theta|data) = \frac{f(data|\theta)f(\theta)}{f(data)}$$

where the sampling density for the data is f(data|θ) and is proportional to the Likelihood function, varying only by a constant to be a correct density function. where f(θ | data) is the posterior distribution for the parameter θ.—f(data) is the marginal probability of the data, and f(θ) is the prior distribution for the parameter. This marginal probability for a continuous sample space may be calculated as follows:

$$f(data) = \int f(data|\theta)f(\theta)d\theta$$

the sampling density integral across the sample space for θ times the previous. This number, which functions as a normalizing constant to make the posterior density appropriate, is frequently referred to as the "marginal likelihood" for the data. For probability distributions, Bayes' Theorem is commonly expressed as follows: the sampling density is proportional to the likelihood function and this denominator only scales the posterior density to make it a correct density.:

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

3.6.3. Stage 3: Lyricist Classification

Lyricist as per Artist wise classification approach targets of unigram algorithm-based classification.

3.6.3.1. Uni-gram Algorithm

Text classification using unigram models is a common technique in natural language processing (NLP). In this approach, each document or text is represented as a bag of words, and the frequency of each word (unigram) in the document is used as a feature for classification. Here's a basic outline of how you could perform text classification using unigram models [112]:

A) Preprocessing:

- Tokenization: Split the text into individual words or tokens.
- Lowercasing: Convert all words to lowercase to ensure consistency.
- Removing punctuation and special characters: Strip away non-alphanumeric characters.
- Stop word removal: Remove common words that do not carry much meaning (e.g., "the", "and", "is").

B) Feature Extraction:

- Create a vocabulary: Build a list of unique words (unigrams) present in the training dataset.
- Count Vectorizer: Convert each document into a vector of word counts. Each feature represents the frequency of a unigram in the document.

C) Training:

Select an algorithm for classification, such as Logistic Regression, Support Vector Machines, or Naive Bayes. Utilizing the feature vectors and associated labels that were extracted from the training set, train the model.

D) Evaluation:

Test the trained model on a separate validation or test dataset. Evaluate the performance of the model using metrics such as accuracy, precision, recall, and F1-score.

E) Predictions:

Once the model is trained and evaluated satisfactorily, it can be used to classify new unseen documents.

3.6.3.2. TF-IDF Algorithm

A statistical technique called the TF-IDF algorithm is employed to evaluate a word's significance for a document or category inside a file set or corpus. The primary principle is that a word or phrase is deemed to have a strong class distinguishing capacity and is appropriate for classification if it appears frequently in one article and infrequently in other articles. In the present vector space model, it is the feature word weight calculation function that is most often utilized.

It is primarily made up of two components: word frequency and inverse text frequency. The term "word frequency" describes how often a certain word appears in a file. The general significance of a word can be gauged by looking at its inverse file frequency. The quotient logarithm is obtained by dividing the frequency of a word's inverse by the total number of documents, which is then divided by the number of documents that include the term.[111]

$$tf_{i,j} = \frac{n_{i,j}}{\sum_k n_{k,j}}$$

$n_{i,j}$, is the number of occurrences of the word ti in file d_j , $\sum_k n_{k,j}$, is the sum of the occurrences of all words in the file d_j.

$$idf_i = log \frac{|D|}{\left|\{j: t_i \in d_j\}\right|}$$

|D| is the total number of files in the corpus, $|\{j: t_i \in d_j\}|$ is the number of documents containing the word t_i. If the word is not in the corpus, it will cause the dividend to be zero. So in general use $1+\left|\{j: t_i \in d_j\}\right|$.

$$tfidf_{i,j} = \frac{tf_{i,j}\ x\ idf_i}{\sqrt{\sum_{t_i \epsilon d_j}[tf_{i,j} x\ idf_i]^2}}$$

$tf_i df_{i,j}$ is what makes the term ti heavy. It is evident that high-weight TF-IDF can be produced by a combination of low word frequency over the whole file set and high word frequency in a specific file. Nevertheless, it is easy to determine using formula (3) analysis that the TF-IDF does not accurately represent the distribution of each word inside each category and its distribution overall. The following presumptions apply: The final calculation indicates that the TF-IDF values are identical. The words t1 and t2 have the same frequency in the file d. The word t1 appears in most of the text in a particular kind of text set, whereas the term t2 appears in all text sets.[111]

Nonetheless, the term "t1" ought to be given greater weight because it is more concentrated in one area and has more information in that category. Regarding the term "t2," it should be assigned a lower weight because it is frequently encountered in all categories and carries less category information. That example, if a term is more sporadically distributed throughout classes, it should be given a larger weight.[111]

3.6.3.3. Posterior Probability

Using data testing, the classification procedure is executed by computing posterior probability values for each class. Because of the prior classifier development, the posterior probability calculation method employs likelihood and prior probability values that are pulled from the database. Each class's posterior probability is determined using equation .[60]

$$P(c|d) = log\ P(c) + \sum_{1 \le k < n_d} log\ P\ (t_k|c)$$

P(c|d) : Posterior probability document for a class

P(c) : Prior probability a class

$P(t_k|c)$: Conditional/Likelihood probability

n_d : The quantity of terms in the written work Equation is used to identify which class has the highest posterior probability value (Max a Posterior) after posterior probability for each class has been computed. Calculating posterior probability was used in this study to identify the class and subclass.

$$C_{map} = \underset{c \in C}{\text{argmax}} \left[\log P(c) + \left(\sum_{1 \le k < n_d} \log P(t_k|c) \right) \right]$$

For all parameters used, posterior probability was calculated using following equation.

$$C_{map} = argmax_{c \in C} \left[\log P(c) + \sum_{1 \le lyric < n_d} \log P(t_{lyric}|c) + \log P(t_{artist}|c) + \log P(t_{genre}|c) \right]$$

CHAPTER-4

Improved Random Forest Algorithm

Stage 1 Genre Classification

Stage 2 Lyrics Classification

Stage 3 Lyricist Classification

CHAPTER 4

Training Model Structure

This chapter covers training model structure of three stage classification on Sanchay Dataset. Preprocessing of the system overview is created in the analysis that contains audio features extraction, lyrics content-based filtering, lyrics translation, classification, feature extraction, classifier design, dataset splitting, label encoder and system evaluation.

4.1 Improved Random Forest Algorithm

A supervised learning approach called random forest is used for regression as well as classification. When it comes to text categorization, the Random Forest (RF) classifiers work well with high dimensional noisy data. A set of decision trees, each trained with random selections of features, make up an RF model. [13]

Improved Random forests use an ensemble approach with similarity match to classify songs in order to achieve classification results. More than one decision tree is trained using the training data. This dataset includes attributes and observations that were chosen at random and will be used for node splitting.

Decision trees play a key role in the ecosystem of a rainforest where group of decision trees are constructed for purpose of classifying dataset. A root node, leaf nodes, and decision nodes make up a decision tree. Each tree's terminal node denotes the final outcome that the specific decision tree is produced.

Stepwise procedure has been observed while implementing classification method on dataset which includes following steps.

1. Bootstrap Sampling: Random sampling from the original data with replacement creates a distinct training set for each tree. As a result, some data points might be used twice while others might not be used. Hence each word of lyrics will have only one entry as node.

2. Selecting Features at Random: Each tree only takes into account a random subset of features when splitting (usually the square root of the total features).

3. Growing Trees: Until it reaches a stopping point (such as pure groups or minimum sample size), each tree produces splits based solely on its bootstrap sample and chosen features.

4. **Final prediction:** The ultimate prediction is determined by the collective vote of

all the trees. Regression uses the average of all trees' predicted values, while classification uses the majority vote of class predictions, where at the end based on posterior probability final stage of predicted class will be assigned to song.

Random Forest pseudocode:

1. Building a new data set from the original data set with bootstrapping. Randomly select "k" features from total "m" features. Where k< <m
2. Creates new decision trees for each of the newly created data set Among the "k" features, get the node "d" using the best decision split point. Splits the nodes using information gain, the decision split which gives the greatest Information gain is used for the split.

$$IG = E(parent) - \sum w_i E(child_i)$$

Split the node into child nodes using the best split.

3. Repeat steps 1 and 2 until "i" number of nodes are created.
4. Build forest by repeating steps 1 to 3 for "n" number times to create a tree.

Once ecosystem of forest is made, subset features are chosen to perform random for each tree which is used for training step that further leads to generate results for final predicated class.

Random Forest Predication algorithm:

1. A subset of features is chosen at random for each tree and used for training.
2. Each decision tree is run through a number of test sets to produce expected results.
3. To determine the final prediction, voting classifiers are used to determine which prediction has the majority.

Improving the model's accuracy is primary goal as researcher. The process is to choose the parameters of the provided random forest classifier which are thus adjusted using hyperparameter tuning. For Random Forest, there are numerous hyperparameters. These are a few significant ones:

1. max_depth: The max_depth of a tree in Random Forest is known as the longest path from the root node to the leaf node.

2. min_sample_split: This parameter tells us the minimum number of observations in any node in order to split it in a decision tree.

3. min_samples_leaf: This hyperparameter tells the minimum number of samples that should be present in the leaf node after splitting a node.

4. n_estimators: n_estimators are the number of trees one wants to build before taking the maximum voting or averages of predictions.

5. max_features: This hyperparameter resembles the number of maximum features provided to each tree in a random forest.

As per study, the best-known optimization for Random Forest is hyperparameter tuning, Cross Validation. In the Model, there are default parameters given in Scikit-learn's random forest. With proposed random forest algorithm and with improved version hyperparameter tuning is implemented using similarity technique for parameter tuning.

For better understanding, various instances are performed for analysis from the point of view to perform classification, among which following three parameters are given a set of values to test out each of them.

n_estimators=150,1000

max_depth=6,8

max_feature=4

class_weight= balanced

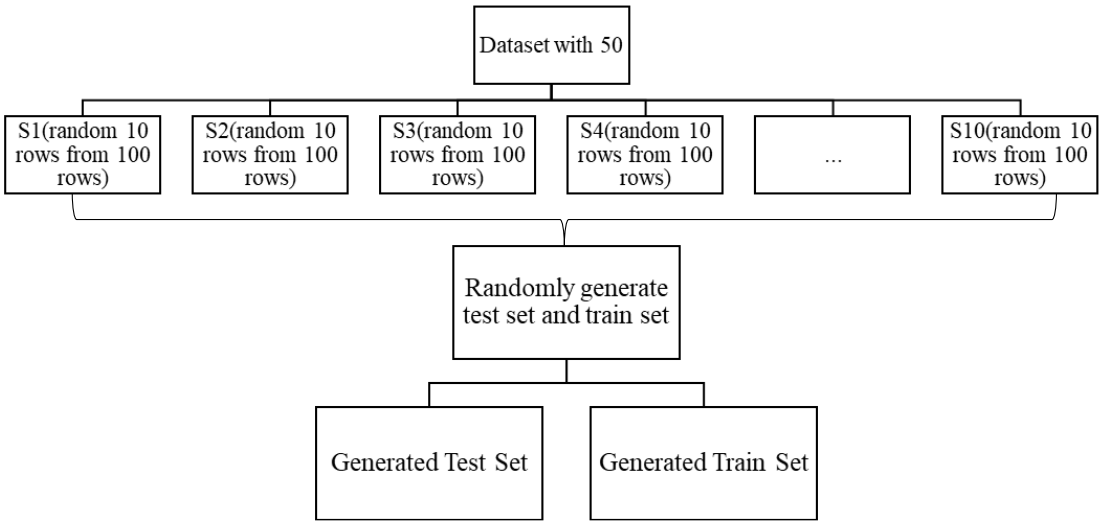


Figure 4.1 Improved Random Forest Algorithm Train-Test Model

With the training phase started using 50 songs to evaluate 1000 songs performance of random forest algorithm is carried out which showcases classification of Hindi songs in three different stages by using improved random forest algorithm. In cross-validation, the original data is split into K number of subsets with a random selection of rows. These subsets are fit into the model. The subsets are then split into a training set and a test set in each cross-validation, For Example, consider data with 50 rows, and divide it into 10 subsets as shown in Figure. Thus, each subset contains 10 random rows in it. Supposing the subsets s1, s2, s3, s4….s10. Here, the algorithm takes s1, s2, s3 and s4 as the training set and s10 as the test set. In the next cross validation, the algorithm randomly takes may be s2, s3, s4, s5…s10 as the training set and s1 as the test set and so on for each validation n number of times .

SARTH_Algorithm (S, x_i, B, D)

- // Input: A song represented by S and computed audio feature vectors x_i with
- // Improved Random Forest Multi-label classifier B, a set of ten-class C
- // verifiers with data cleaning,tokenization,stemming dataset D
- // Output: A music genre assigned to song S or the
- // unlabeled of song S

1) Input song S, with measured audio features MFCC,ZCR,Roll-off and Centroid as x_i
2) Train model train_test_split A labeled training set to assigning label genre with weight to each audio feature using maximum likelihood training is to evaluate a closed-form expression

$$P(y = C|x) \propto \sum_{x_i \epsilon N_k(x)} (y_i = C).P(x|x_i)$$

- $N_k(x)$: set of nearest neighbors of x
- $P(x|x_i)$ =likelihood of x under a local model centered at neighbor
- $y_i = C$ indicator if neighbor's label is C

$$Similarity(S, Class_i) = \log P(S|weight\ x_i)$$

3) Assigning label lyrics with parameters as n_estimators=1000, max_depth=8, max_feature=4, class_weight= balanced to Song S after stemming and translation phase.

4) Split the node into child nodes using the best split

$$IG = E(parent) - \sum w_i E(child_i)$$

5) Next, Decision is to be made according to presence word in lyrics

$$dk(x_i, x_j) = \begin{cases} 0, if\, l_l(x_i) = l_k(x_j) \\ 1, \quad otherwise \end{cases}$$

6) Assigning label_artist using term frequency-inverse document frequency to Song S

$$tfidf_{i,j} = \frac{tf_{i,j}\ x\ idf_i}{\sqrt{\sum_{t_i \epsilon d_j}[tf_{i,j} x\ idf_i]^2}}$$

7) classify S with C_{map} (Class mapping to instance S)
8) return

$$C_{map} = argmax_{c \in C}\left[\log \mathrm{P(c)} + \sum_{1 \le \mathrm{lyric} < n_d} \log \mathrm{P}(t_{lyric}|\mathrm{c}) + \log \mathrm{P}(t_{artist}|\mathrm{c}) + \log \mathrm{P}(t_{genre}|\mathrm{c})\right]$$

- P(c|d) : Posterior probability document for a class
- P(c) : Prior probability a class
- $P(t_k|c)$: Conditional/Likelihood probability

9) as the music genre
10) else
11) unlabeled the instance S
12) endif

4.2 Stage 1 Genre Classification

As per proposed algorithm, training phase of model starts by classifying input dataset songs by audio features weighted maximum likelihood for evaluating closed-form expression.

```
[23]: import scipy, matplotlib.pyplot as plt, IPython.display as ipd, sklearn
      import librosa, librosa.display
      import numpy as np
      plt.rcParams['figure.figsize'] = (14, 5)
      import mysql.connector

      mydb = mysql.connector.connect(
        host="localhost",
        user="root",
        password="",
        database="ganesha"
      )
      mycursor = mydb.cursor()


      mycursor.execute("SELECT song_id,path FROM audio_features WHERE song_id BETWEEN 2190 AND 2200")
      myresult = mycursor.fetchall()
      for p in myresult:
        audio_path = p[1]
        song_id=p[0]
        print(audio_path)
        x, sr = librosa.load(audio_path)
        print("MFCC")
        mfccs = librosa.feature.mfcc(x, 42000,n_mfcc=13)
        mfcc_val=np.mean(mfccs.shape)
        print(mfcc_val)
        print('Zero Crossing')
        zcrs = librosa.zero_crossings(x, pad=False)
        zcrs_val=np.mean(zcrs)
        print(zcrs_val)
        print("Spectral Centroid")
        cent = librosa.feature.spectral_centroid(y=x, sr=sr)
        cent_val=np.mean(cent)
        print(cent_val)
        print("Spectral Roll Off")
        S, phase = librosa.magphase(librosa.stft(x))
```

Figure 4.2 Genre Classification Audio Features Extraction

```
/Users/darshitapathak/PhD/Saarth/part16/Tere Mere Sapne Ab Ek Rang Hain(Djjohal.Com).mp3.wav
MFCC
5607.0
Zero Crossing
0.062087072005714034
Spectral Centroid
1137.3143018840055
Spectral Roll Off
2122.1806978032096
Rows updated =  1
/Users/darshitapathak/PhD/Saarth/part16/Rishta(Djjohal.Com).mp3.wav
MFCC
5342.5
Zero Crossing
0.10671098636587252
Spectral Centroid
```

Figure 4.3 Extracted Audio Features of Song "Tere Mere Sapne Ab Ek Rang Hain"

As mentioned in above screenshots from Input Dataset audio features for all songs available in MySQL Database are computed with weighted values of MFCC, Zero Crossing, Centroid and Roll-off by equations mentioned in previous chapter and are stored in MySQL Database for further step.

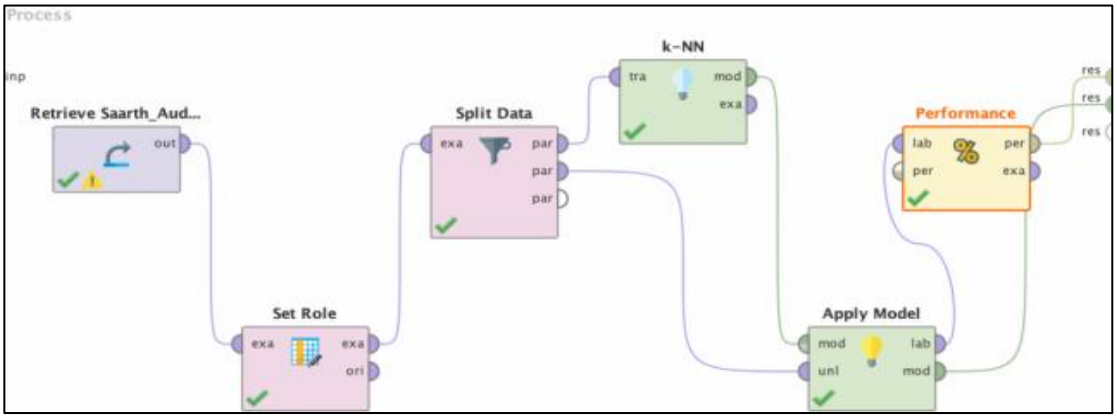
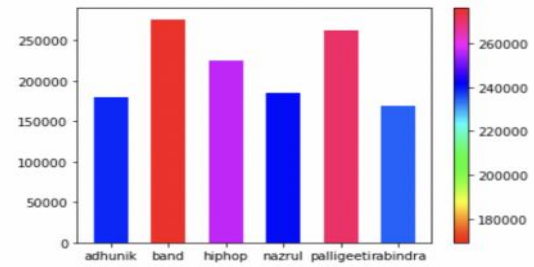


Figure 4.4 Training Process Model

```
zero_crossing
179833.67137809188
275995.7152542373
225064.01016949152
184407.72435897434
261663.73846153845
169258.16498316498
```

```
/var/folders/14/qpbkwlv55kn2g589j7g53yr40000gn/T/i
ing from Matplotlib 3.6, colorbar() will steal spa
place the colorbar.  To silence this warning, expl
  plt.colorbar(plot)
```




```
zcr
0.10345544211618259
0.10354173884514435
0.09634161292134831
0.12125388524590162
0.10808647362637365
0.11037876052631578
```

```
/var/folders/14/qpbkwlv55kn2g589j7g53yr40000gn
ing from Matplotlib 3.6, colorbar() will steal
place the colorbar.  To silence this warning,
  plt.colorbar(plot)
```

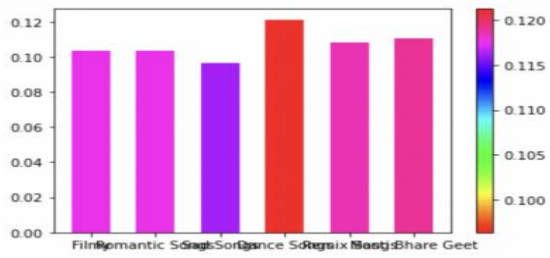
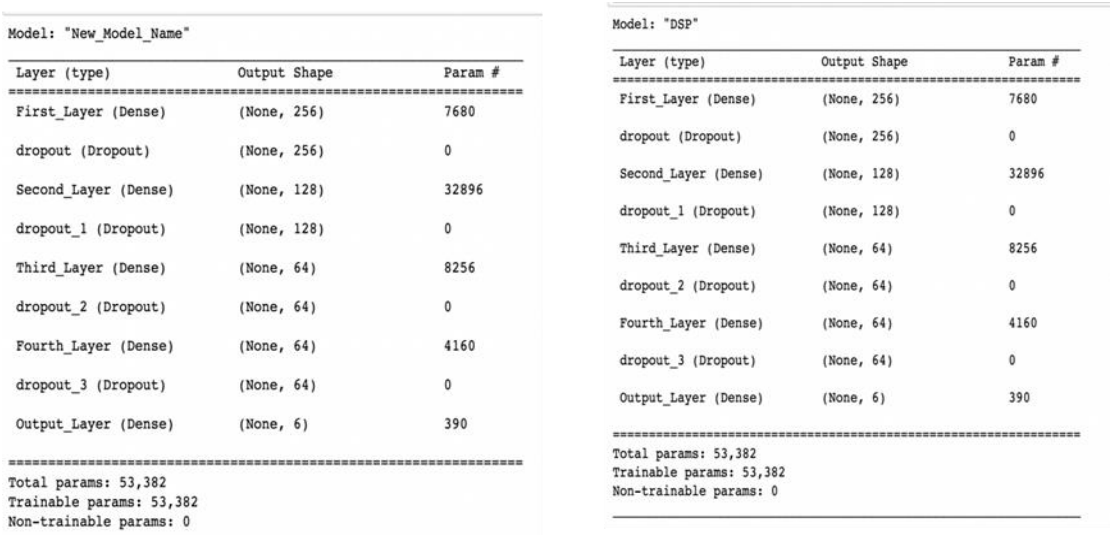


Figure 4.5 ZCR Values of Bangla Dataset & Saarth Dataset



Model: "New_Model_Name"

Layer (type)	Output Shape	Param #
First_Layer (Dense)	(None, 256)	7680
dropout (Dropout)	(None, 256)	0
Second_Layer (Dense)	(None, 128)	32896
dropout_1 (Dropout)	(None, 128)	0
Third_Layer (Dense)	(None, 64)	8256
dropout_2 (Dropout)	(None, 64)	0
Fourth_Layer (Dense)	(None, 64)	4160
dropout_3 (Dropout)	(None, 64)	0
Output_Layer (Dense)	(None, 6)	390

Total params: 53,382
Trainable params: 53,382
Non-trainable params: 0

Model: "DSP"

Layer (type)	Output Shape	Param #
First_Layer (Dense)	(None, 256)	7680
dropout (Dropout)	(None, 256)	0
Second_Layer (Dense)	(None, 128)	32896
dropout_1 (Dropout)	(None, 128)	0
Third_Layer (Dense)	(None, 64)	8256
dropout_2 (Dropout)	(None, 64)	0
Fourth_Layer (Dense)	(None, 64)	4160
dropout_3 (Dropout)	(None, 64)	0
Output_Layer (Dense)	(None, 6)	390

Total params: 53,382
Trainable params: 53,382
Non-trainable params: 0

Figure 4.6 Bangla Dataset and Saarth Dataset Trained Model Parameters Statistics

Algorithm	Bangla Music	Hindi Music
SVM	**0.7583**	0.5127
NB	0.667	0.6857
k-NN	0.6417	0.7171
Random Forest	0.7491	**0.8033**

Table 4.1 : Accuracy of Algorithm Implementation on Bangla Dataset and Hindi Dataset Genre wise Classification

The number of parameters needed for naive Bayes classifiers is linear in the number of variables (features/predictors) in a learning task, making them extremely scalable. One method of achieving maximum likelihood training is to evaluate a closed-form expression. [17]

An SVM training algorithm creates a non-probabilistic binary linear classifier by classifying new instances into one or the other category. SVM having ability to handle non-linear boundaries is beneficial in music classification, where the differences between genres can be complex and non-linear, since ***BanglaMusicStylo*** contains 29 features in input layer to handle with such huge complex and non-linear data SVM outperforms to find best hyperplane for classification, thus with comparison to Random Forest where classification lead by SVM provides a bit of good accuracy in accordance with the reason to find best separates classes by maximizing the margin

between them. Uses kernel functions to handle non-linearly separable data by transforming it into a higher-dimensional space. [16]

The K-NN method classifies new information based on similarity after storing all the relevant data. This indicates that the K-NN algorithm is able to categorize newly acquired data into a suitable category. [18]

A number of variables, including the use case, the model's structure, and the data's dimensions, affect how best to divide the test, validation, and train sets. Hence 4 analyses performed with different distribution using Random Forest Algorithm.

Training Testing Ratio

- 50% - 50%
- 70% - 30%
- 60% - 40%
- 80% - 20%

	Accuracy (Centroid)	Accuracy (MFCC)	Accuracy (ZCR)	Accuracy (Roll Off)	Accuracy
Party Songs	0.76	0.71	0.71	0.71	0.728
Remix Songs	0.80	0.87	0.87	0.87	0.859
Romantic Songs	0.80	0.81	0.81	0.81	0.816
Filmy	0.83	0.74	0.74	0.74	0.768
Dance Songs	0.98	0.72	0.6	0.72	0.755
Philosophical Songs	0.98	0.96	0.96	0.96	0.965
Rock Songs	0.84	0.66	0.78	0.66	0.735
Happy Songs	0.78	0.96	0.96	0.96	0.916
Pop Songs	0.7	0.68	0.82	0.68	0.72
Sad Songs	0.7	0.84	0.7	0.84	0.77
Training- Testing Ratio 50-50 Accuracy					**0.8033**

Table 4.2 Accuracy Comparison with Different Audio Features

	Accuracy (Centroid)	Accuracy (MFCC)	Accuracy (ZCR)	Accuracy (Roll Off)	Accuracy
Party Songs	0.843	0.857	0.864	0.886	0.862
Remix Songs	0.945	0.769	0.958	0.853	0.881
Romantic Songs	0.786	0.864	0.795	0.921	0.841
Filmy	0.828	0.928	0.985	0.885	0.907
Dance Songs	0.885	0.914	0.958	0.972	0.932
Philosophical Songs	0.945	0.886	0.685	0.914	0.858
Rock Songs	0.642	0.771	0.814	0.642	0.717
Happy Songs	0.886	0.885	0.985	0.928	0.921
Pop Songs	0.914	0.614	0.842	0.985	0.839
Sad Songs	0.972	0.693	0.886	0.833	0.846
Training- Testing Ratio 70-30 Accuracy					**0.8609**

Table 4.3 Accuracy Comparison with Different Audio Features

	Accuracy (Centroid)	Accuracy (MFCC)	Accuracy (ZCR)	Accuracy (Roll Off)	Accuracy
Party Songs	0.769	0.967	0.845	0.759	0.835
Remix Songs	0.810	0.705	0.769	0.882	0.792
Romantic Songs	0.95	0.866	0.85	0.967	0.908
Filmy	0.983	0.967	0.816	0.916	0.921
Dance Songs	0.766	0.933	0.816	0.683	0.800
Philosophical Songs	0.833	0.895	0.983	0.967	0.920
Rock Songs	0.8	0.85	0.733	0.683	0.766
Happy Songs	0.833	1.00	0.810	0.779	0.855
Pop Songs	0.967	0.816	0.966	0.983	0.933
Sad Songs	0.967	0.933	0.923	0.9	0.930
Training- Testing Ratio 60-40 Accuracy					**0.866**

Table 4.4 Accuracy Comparison with Different Audio Features

	Training-Testing Ratio			
Genre Name	**50-50**	**70-30**	**60-40**	**80-20**
Party Songs	0.728	0.863	0.835	32.5
Remix Songs	0.859	0.882	0.792	21.5625
Romantic Songs	0.816	0.842	0.909	35.625
Filmy	0.768	0.907	0.921	28.75
Dance Songs	0.755	0.933	0.8	25.9375
Philosophical Songs	0.965	0.858	0.92	19.375
Rock Songs	0.735	0.718	0.767	19.6875
Happy Songs	0.916	0.922	0.856	22.5
Pop Songs	0.72	0.839	0.933	22.5
Sad Songs	0.77	0.846	0.931	21.5625

Table 4.5 Genre wise Classification results of Training Testing

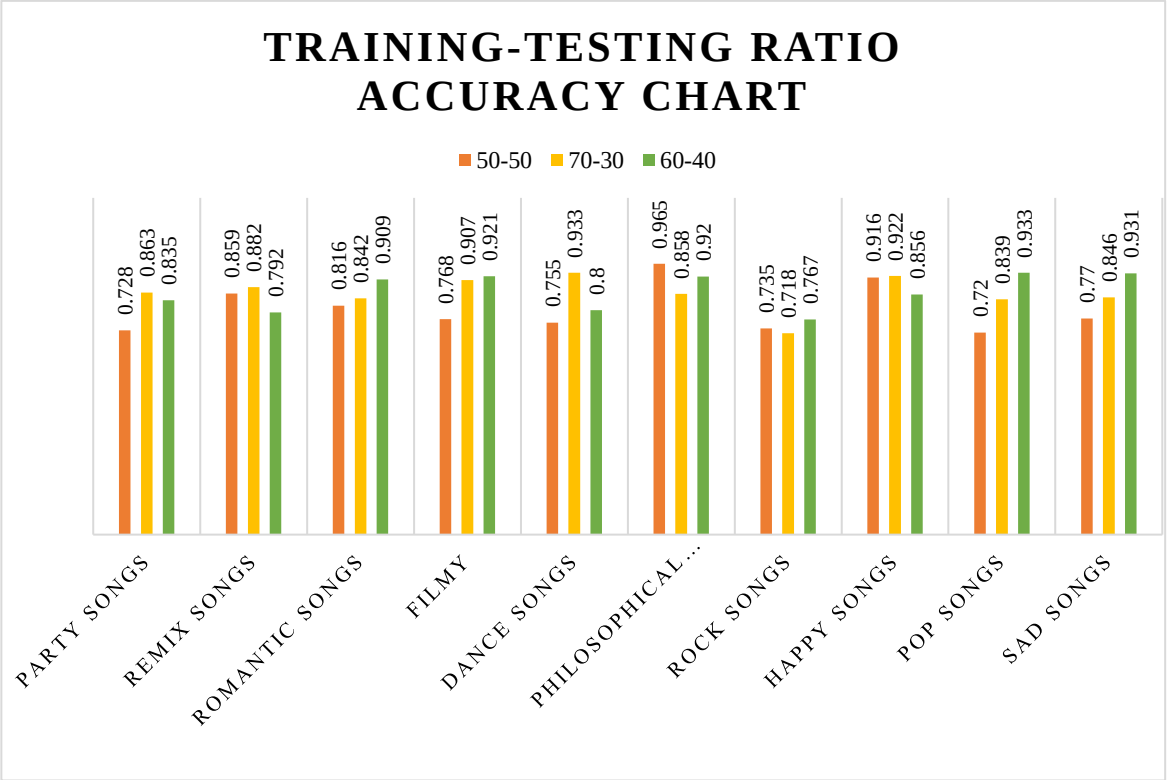


Figure 4.6 Accuracy Chart Representation of different training-testing ratios

SVM is good over kNN ,but in case of Random Forest provides increased precision as music beats are different energy level of linear in SVM is good in Bangla music, while as Hindi music contains majority of beat variation and energy level of a greater number of instruments. Hence, linear regression is not suitable of Hindi Music classification.

A 60/40 training/testing split can be a good choice as split allows for a substantial amount of data to be used for training while still reserving a significant portion for testing. Random Forest and Support Vector Machines where the choice of a 60/40 training/testing ratio (i.e., 60% of the data for training and 40% for testing) is one of many possible splits for evaluating machine learning models as per results discovered.

4.3 Stage 2 Lyrics Classification

The first step must be carried out, which is the preprocessing data in order for the system to easily computerize the text data and using text data is to reduce data that was less relevant or noise such as stop word. The stages of preprocessing data in this study include the following stages:

1. Case Folding

The song lyrics data were first preprocessed by converting all of the letters to lowercase. Non-letter characters were eliminated since they served as a separator.

2. Tokenization

At this point, case folding has been done on the song lyrics. The lyrics were fragmented into individual words (tokens).

3. Stop word Removal

Subsequently, the tokens generated in the earlier phases were chosen by eliminating terms deemed less significant (stop words), for instance, stop words in the Indonesian language such “thi”, “hau”, “ho”, “wo”, and “vahi”.

4. Data Cleaning

Next, as token had been generated, some of text content are having similar meaning in which phrases may comprise of same meaning in such situation those duplicate parts are required to be cleaned like ‘wahi wahi’ repeated terms are merged into a single ‘wahi’. Thus, Date cleaning plays essential part in lyrics tokenization phase. Such similar words are removed by using Similarity matching algorithm implemented by [24].

5. Stemming

In this last stage of pre-processing the data, the chosen words were changed into main/basic terms. Nazief and Adriani stemming algorithms were employed. The data is saved as a preprocessed dataset once all preparation steps have been completed.

6. Lyrics Translation phase

Label encoding also preceded the genre and artist features. The label encoding method involved the conversion of categorical data on genre and artist into numerical data.

18	Aatee Rahengee Bahare, Jatee Rahengee Bahare - हे आती रहे ` गी बहारे ` , जाती रहे ` गी बहारे `	Gulshan Bawra	hey aati rahengi bahare, jati rahengi bahare dil ki najar se duniya ko dekho duniya sada hi hasin hai ho aati rahengi bahare, jati rahengi bahare dil ki najar se duniya ko dekho duniya sada hi hasin hai ho aati rahengi bahare, jati rahengi bahare dil ki najar se duniya ko dekho duniya sada hi hasin hai aati rahengi bahare, jati rahengi bahare maine toh bas yahi mangi hain duwaye phulo ki tarah ham sada muskuraye maine toh bas yahi mangi hain duwaye phulo ki tarah ham sada muskuraye gate rahe ham khushiyo ke git yuhi jaye bit jindagi aati rahengi bahare, ho jati rahengi bahare dil ki najar se duniya ko dekho duniya sada hi hasin hai aati rahengi bahare, hey jati rahengi bahare tumse hain jab jivan me sahare jahan jaye najre wahi hain najare tumse hain jab jivan me sahare jahan jaye najre wahi hain najare leke aayegi har nayi bahar rang bhara pyar aur khushi aati rahengi bahare, hey jati rahengi bahare ha dil ki najar se duniya ko dekho duniya sada hi hasin hai aati rahengi bahare, jati rahengi bahare hey ham jo mile ho toh dil ko yakin hai dharati pe swarg jo hai toh yahi hai ham jo mile ho toh dil ko yakin hai dharati pe swarg jo hai toh yahi hai bhule se bhi gum aaye na waha, pyar hai jaha bandagi aati rahengi bahare, jati rahengi bahare dil ki najar se duniya ko dekho duniya sada hi hasin hai aati rahengi bahare, jati rahengi bahare	Philosophical Songs	Kishore Kumar, Amit Kumar, Asha Bhosle	Kasme Vaade	1978

Figure 4.8 Database row entry of Hindi Song

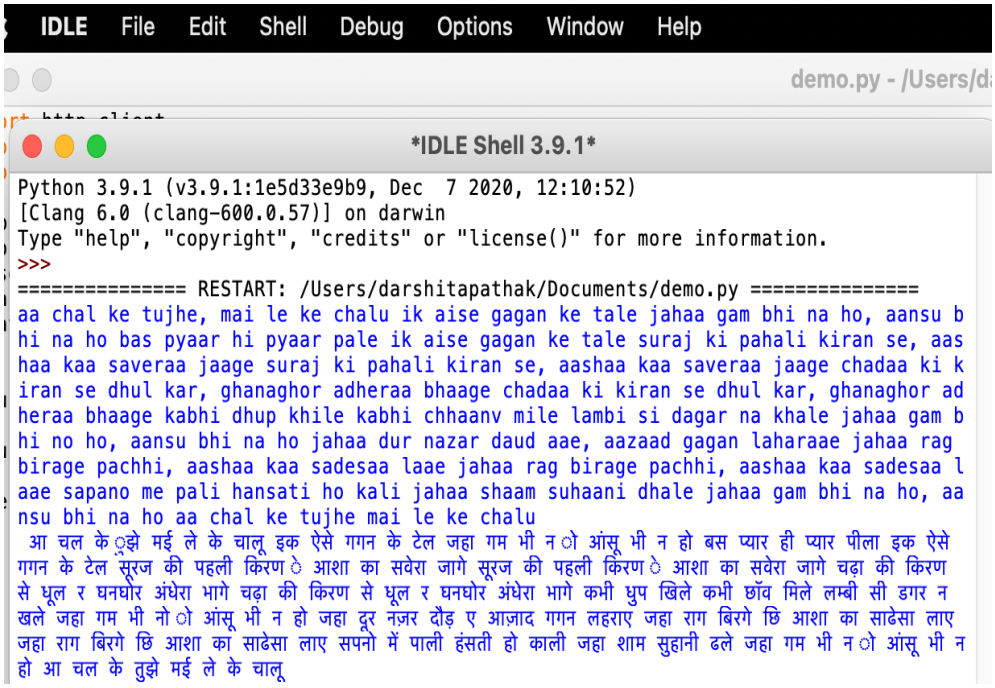


Figure 4.9 Translation of lyrics into Devnagari Hindi Language

```
In [1]: import matplotlib.pyplot as plt
from wordcloud import WordCloud, STOPWORDS
from nltk.tokenize import word_tokenize
import re
stopword=open("/Users/darshitapathak/Desktop/stopwords.txt",'r')
stop_words=[]
# pre-process stopword
for i in stopword:
    i = re.sub('[\n]', '', i)
    stop_words.append(i)
stopwords = set(stop_words)
text = "आ चल् के तुझे, मै ले के चलु इक् ऐसे गगन् के तले जहा गम् भि न हो, आन्सु भि न हो बस् प्यार् हि प्यार् पले इक् ऐसे गगन
new = ""
for word in text.split():
    if word not in stopwords:
        new = new +" "+word
        text = new
print(text)
```

चल् तुझे, ले चलु इक् गगन् तले जहा गम् हो, आन्सु बस् प्यार् प्यार् पले इक् गगन् तले सुरज् पहलि किरन् से, आशा सवेरा जागे सुरज् पहलि किरन् से, आशा सवेरा जागे चदा किरन् धुल् कर्, घनघोर् अधेरा भागे चदा किरन् धुल् कर्, घनघोर् अधेरा भागे कभि धुप् खिले कभि छान्व् मिले लम्बि सि दगर् खले जहा गम् नो हो, आन्सु जहा दुर् नज़र् दौद् आए, आज़ाद् गगन् लहराए जहा रग् बिरगे पछि, आशा सदेसा लाए जहा रग् बिरगे पछि, आशा सदेसा लाए सपनो पलि हन्सति कलि जहा शाम् सुहानि धले जहा गम् हो, आन्सु चल् तुझे ले चलु

Figure 4.10 Lyrics after stop words removal

Once content of lyrics is passed through all pre-requisite phases and trained dataset is generated improved random forest algorithm is applied with optimal split value. Estimators' of 1000 words are paired and generated for matching with depth of decision tree at level 8, which provides preferred resultant output with best fit random forest classifier.

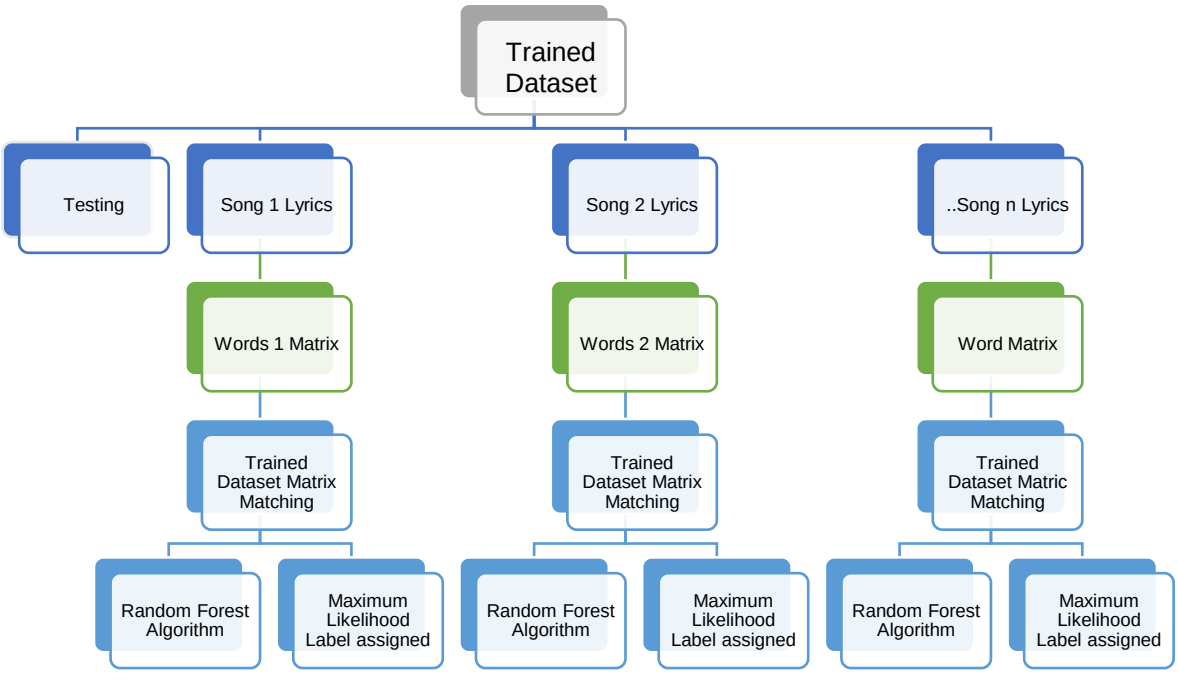


Figure 4.11 Random Forest Algorithm Matrix Matching Calculation

Pair wise similarity between multiple sentences, to generate Thematic feature among Lyrics words. Decision Tree classifier provides matching result among all possible similarities. Threshold value: 8 Depth Random Forest is improvised version of Decision Tree where a greater number of Decision Trees are made with all possible predictions.[3]

$$dk(x_i, x_j) = \begin{cases} 0, if\, l_l(x_i) = l_k(x_j) \\ 1, \quad otherwise \end{cases}$$

Random Forest each pair's similarity score of data instances is extracted. The similarity of two data instances is measured by the percentage of trees where the two data instances appear in the same leaf node.[3] Other algorithm implementation study shows the Improved Random Forest algorithm provides more accurate result over Naïve Bayes, SVM and kNN Algorithm.

पल एक पल - Jalebi(2018) and पल पल दिल के पास- Blackmail (1978)

	पल	पल	दिल	जीवन	मीठी	प्यास
पल	1	1	0	0	0	0
एक	0	0	0	0	0	0
पल	1	1	0	0	0	0
थम	0	0	0	0	0	0
हाथ	0	0	0	0	0	0
दायें	0	0	0	0	0	0

Figure 4.12 Random Forest Similarity Distance Matrix Calculation

t[2]:

	index	Song_name	type	artist	lyrics	नीले	अम्बर	चाँद	प्यार	बरस	...	बख्शी	अक्रिती	भसीन	मुच्छल	नीति	जहाँगीर	मेहताब	हिमेश	रेशमिया	कुमा
0	1	नीले नीले अम्बर पर	romantic	किशोर कुमार	नीले नीले अम्बर चाँद प्यार बरस तरस नीले नीले अ...	1	1	1	1	1	...	0	0	0	0	0	0	0	0	0	
1	2	अक्कड़ बक्कड़	party	बादशाह	अक्कड़ बक्कड़ बॉम्बे बो पुरे रात बज पौ अक्कड़ ...	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0	
2	3	अखियाँ	sad	पोपोन	थक गेया अँख जग्ग दिय अख माह ना लभद अँखियाँहाये...	0	0	0	1	0	...	0	0	0	0	0	0	0	0	0	
3	4	अंग से अंग लगाना	romantic	अलका याग्निक , बिनोद राठौड़ , सुदेश भोसले	आए आए चाहो बाँहो भर अंग अंग लग सजन रंग लग अंग ...	0	0	0	0	1	...	0	0	0	0	0	0	0	0	0	
4	5	अगर ज़िन्दगी हो	romantic	आशा भोसले	ज़िन्दगी संग ज़िन्दगी संग मौत मौत ज़िन्दगी संग...	0	0	0	1	0	...	0	0	0	0	0	0	0	0	0	

5 rows × 9593 columns

Figure 4.13 Numerical Label Assignment to each word

In [1]:
```
import pandas as pd
import numpy as np
```

In [2]:
```
#loading the dataset
df = pd.read_csv("/Users/darshitapathak/Data/songs_without_stopwords.csv", low_memory=False)
#df = pd.read_csv("binary.csv", low_memory=False)
df.head()
```

Out[2]:

	index	Song_name	type	artist	lyrics
0	1	नीले नीले अम्बर पर	romantic	किशोर कुमार	नीले नीले अम्बर चाँद प्यार बरस तरस नीले नीले अ...
1	2	अक्कड़ बक्कड़	party	बादशाह	अक्कड़ बक्कड़ बॉम्बे बो पुरे रात बज पौ अक्कड़ ...
2	3	अखियाँ	sad	पोपोन	थक गेया अँख जग्ग दिय अख माह ना लभद अँखियाँहाये...
3	4	अंग से अंग लगाना	romantic	अलका याग्निक, बिनोद राठौड़, सुदेश भोसले	आए आए चाहो बाँहो भर अंग अंग लग सजन रंग लग अंग ...
4	5	अगर ज़िन्दगी हो	romantic	आशा भोसले	ज़िन्दगी संग ज़िन्दगी संग मौत मौत ज़िन्दगी संग...

In [3]:
```
#define features andtags and split
from sklearn.model_selection import train_test_split
y = df.type
x=df.lyrics
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.2) # use random number generator
```

Figure 4.14 Training and Testing phase of Lyrics Classification

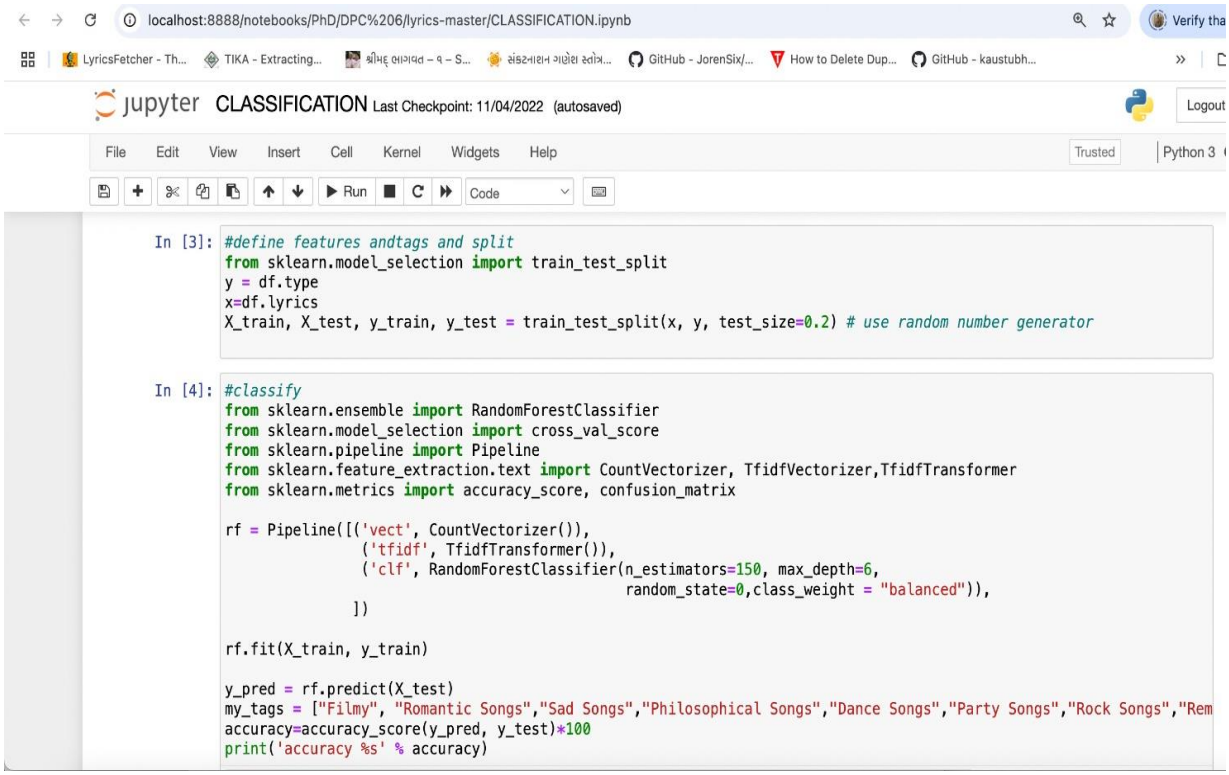


Figure 4.15 Random Forest Classifier Training and Testing for Lyrics Classification

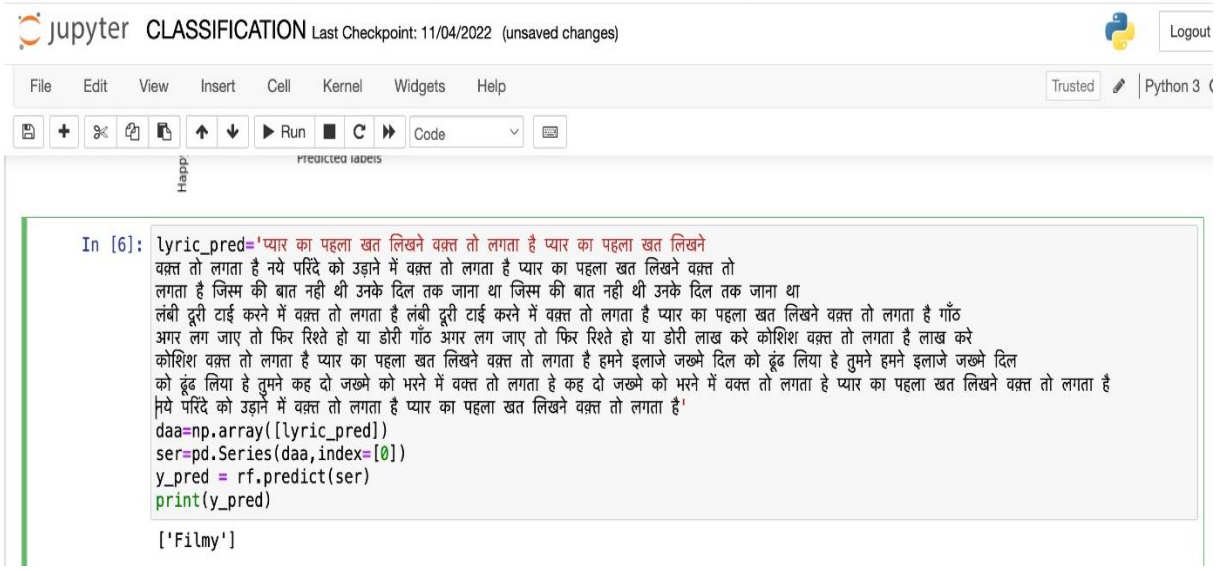


Figure 4.16 Input of Lyrics to RandomForestClassifier Technique

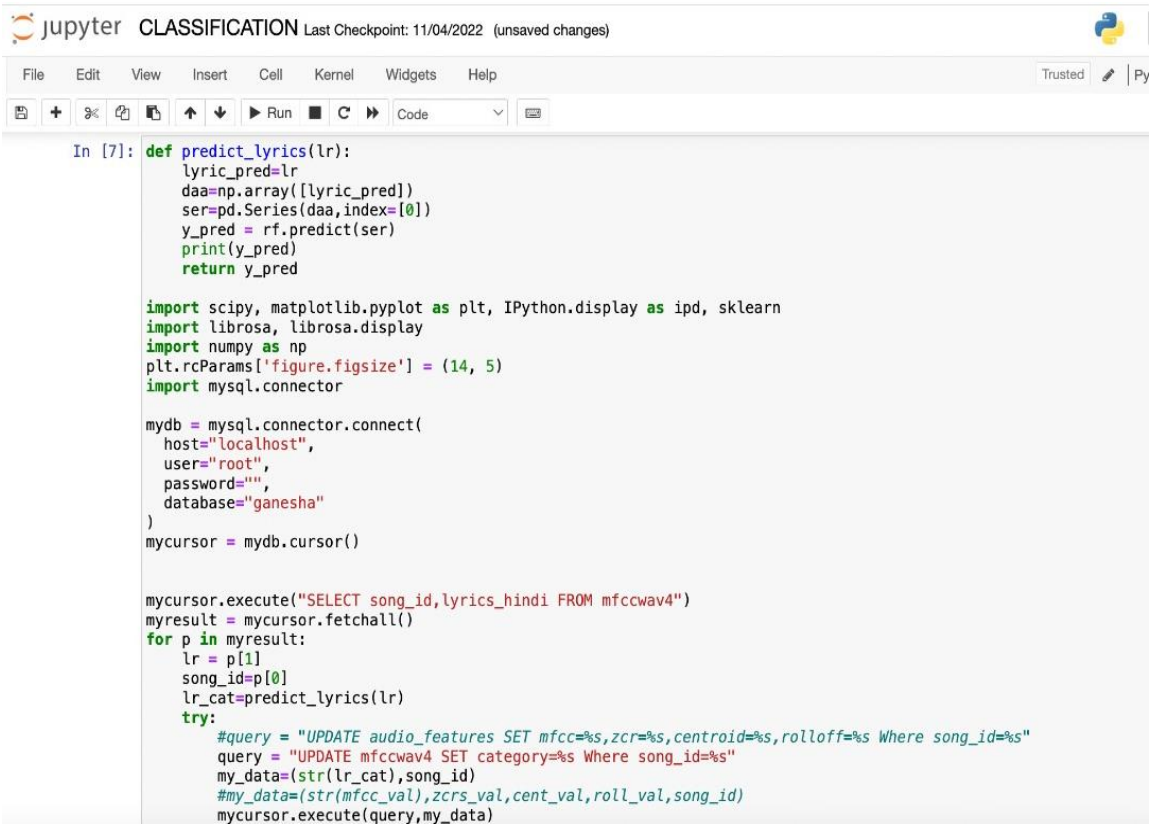


Figure 4.17 Lyrics Prediction Function Execution of Classification to Sanchay Dataset

RANDOM FOREST CLASSIFIER

In [19]:
```
#Import Random Forest Model
from sklearn.ensemble import RandomForestClassifier

#Create a Gaussian Classifier
clf=RandomForestClassifier(n_estimators=1000, max_depth=8,
                                                random_state=0,class_weight = "balanced")

#Train the model using the training sets y_pred=clf.predict(X_test)
clf.fit(X_train,y_train)

y_pred=clf.predict(X_test)

#Import scikit-learn metrics module for accuracy calculation
from sklearn import metrics
# Model Accuracy, how often is the classifier correct?
accuracy_rf=metrics.accuracy_score(y_test, y_pred)*100
print("Accuracy (in %):",accuracy_rf)
```

```
Accuracy (in %): 70.0
```

Figure 4.18 Random Forest Classifier Accuracy

```
In [7]: #print confusion matrix
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred)
ax= plt.subplot()
sns.heatmap(cm, annot=True, ax = ax); #annot=True to annotate cells

# labels, title and ticks
ax.set_xlabel('Predicted labels');ax.set_ylabel('True labels');
ax.set_title('Confusion Matrix');
ax.xaxis.set_ticklabels(my_tags); ax.yaxis.set_ticklabels(my_tags);
```

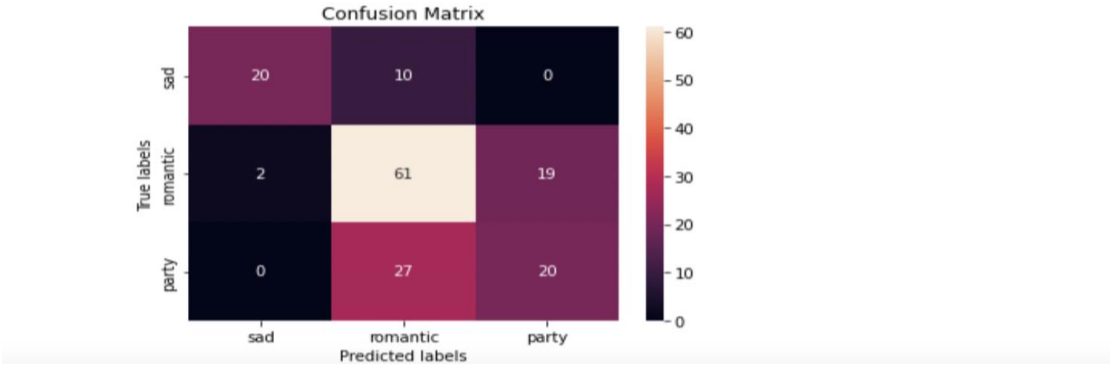


Figure 4.19 Confusion Matrix of Sample Three Class

In training phase 500 songs are given operational implementation, Time complexity to run this algorithm is 43.23 mins. Predicted and True labels values from confusion matrix indicated accuracy of improved random forest algorithm of 70%. After trained module is prepared, for testing perspective testing songs' lyrics is given as input and output will be predicted as class label of song which is shown in following screenshot.

```
    i += 1
print(f'{song} is present in songlist at indexes {matched_index}')



#fetching lyrics of that song
lyrics=l[matched_index]
#print(lyrics)

#creating series from string
list = [lyrics]
ser = pd.Series(list)
print(ser)

#predicting mood of that song
y_pred = rf.predict(ser)
print(y_pred)
```

```
Enter song name for which you want to classify  :मिले हो तुम हमको
मिले हो तुम हमको is present in songlist at indexes 585
0    मिले नसीब सेचुर क़िस्मत लकीर सेमिले नसीब सेचुर...
dtype: object
['romantic']
```

Figure 4.20 Label output based on Lyrics of songs

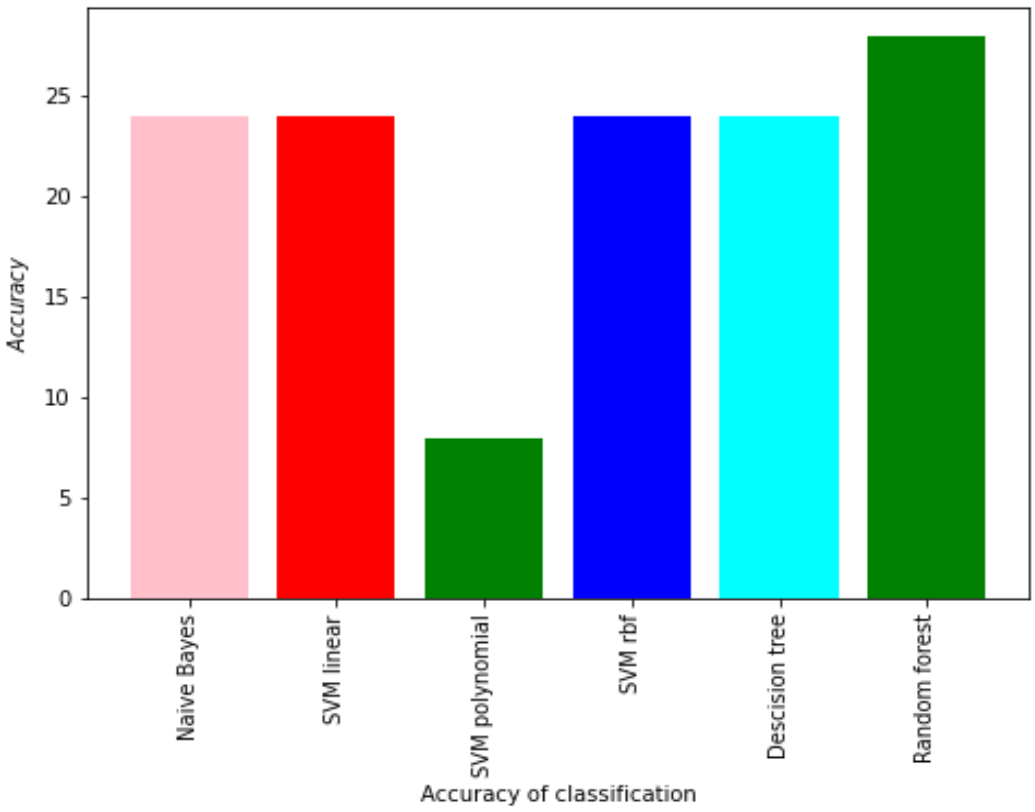


Figure 4.21 Accuracy of classification of various algorithmic results

4.4 Stage 3 Lyricist Classification

Unigram classification, also known as unigram language model classification, is a technique used in natural language processing (NLP) and text classification tasks. It involves representing documents or pieces of text using unigrams, which are single words occurring in the text, without considering their order or context. In context with lyricist wise classification label encoding between lyricist and words presented in lyrics are mapped with binary values. Further each uni-gram word mapping is processed with binary artist similarity value that classifies songs into one of the classes.

unigrams for artist

In [15]:

```
features=[]
newest_data_frame = new_data_frame
for ind in range(1,len(new_data_frame['artist'])):
    my_str_series = new_data_frame.loc[ind]['artist']
    new_data_frame["artist"].dropna(inplace=True)
    new_data_frame['artist'] = new_data_frame['artist'].astype(str)
    words_DF = nltk.tokenize.word_tokenize(my_str_series)
    #words_DF=new_data_frame['artist'].apply(nltk.tokenize.word_tokenize)
    strwithoutstopwords = ""
    stem_word=""
    for str_ in words_DF:
        #print(str_)..

        #words_DF = nltk.tokenize.word_tokenize(str_)
        #for wrd in words_DF:
        if str_ not in unique_word_list:
            #stem_word=generate_stem_words(str_)
            strwithoutstopwords=strwithoutstopwords+str_+" "
            features.append(str_)
    #without_stopword_songs.append(strwithoutstopwords)
    newest_data_frame.at[ind,'artist']=strwithoutstopwords

def unique(list1):

    # intilize a null list
    unique_list = []
```

Figure 4.22 Unigram for Artist

Out[6]:

	index	Song name	type	artist	lyrics	जब्	पल्	कभि	मयुसियो	घदि	...	Phani	Neelesh	Misra	Adheesh	Verma	Shabbir
0	1	Jab Aaye Aise Pal Kabhi (Male) - जब आये ऐसे पल...	Filmy	Faiz Anwar	जब् पल् कभि मयुसियो घदि जब् पल् कभि मयुसियो घद...	1	1	1	1	1	...	0	0	0	0	0	0
1	2	Maula Maula (Harshdeep Kaur) - मेंडा इश्क भी त...	Party Songs	Baba Farid	मेन्द इश्क़ तु मेन्द यार् तु मेन्द इश्क़ तु मे...	0	0	0	0	0	...	0	0	0	0	0	0
2	3	Main Kesar Kasturi Mere Surat	Happy Songs	Rajinder Krishan	केसर् कस्तुरि मेरि सुरत्	0	0	0	0	0	...	0	0	0	0	0	0

Figure 4.23 Lyricist wise Numerical label assignment

```
In [27]: score, acc = model.evaluate(X_test, y_test)
         print(score, acc)

         11/11 [==============================] - 0s 1ms/step - loss: 1.0886 - accuracy: 0.7163
         1.0886452198028564 0.7163323760032654

In [28]: # Summarize history for accuracy
         plt.plot(history.history['accuracy'])
         plt.plot(history.history['val_accuracy'])
         plt.title('Train vs Validation Accuracy')
         plt.ylabel('accuracy')
         plt.xlabel('epoch')
         plt.legend(['training accuracy', 'validation accuracy'], loc='lower right')
         plt.show()
```

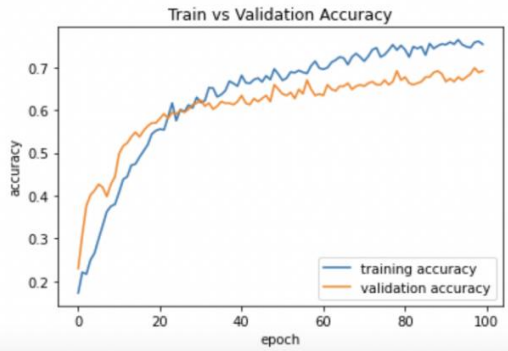


Figure 4.24 Genre wise Training vs Validation Accuracy

As depicted in the figure above, the accuracy curves demonstrate the precision of the model's performance on the input data. However, if the curve shows significant and abrupt fluctuations, it may suggest a decline in overall training effectiveness. A smooth learning curve indicates that the models' performance evolves gradually and steadily during training, reflecting consistent improvement over time. Here smooth learning curve as per figure suggests the model is learning in stable and consistent manner.

CHAPTER-5

Calculation Result

Human Survey Prediction Stage

Result & Analysis

CHAPTER 5

Results & Analysis

This chapter of Results and Analysis is an important component of research project as it involves textual descriptions, tables, charts, graphs, or any other relevant visuals to convey the data collected during research. It involves explaining what the data means in the context of research questions or hypotheses. It includes with findings with existing literature in your field and highlight areas of agreement or divergence between your results and previous studies.

5.1 Calculation Result

In the realm of proposed thesis theory of three stage classification based on proposed implementation framework, following table includes critical analysis of songs' lyricist, genre and lyrics wise class label result. Here in table few of songs are listed in which some of songs' result is having all class as same while other are giving different class label. Based on different class label predicted further research study shows to include human survey analysis which deals with Psychology of Human Emotion with Music.

ID	TITLE	Class	Lyricist	Genre	Lyrics
255	Zindagi Ke Safar Mein Gujar Jate Hai Jo Makaam	Philosophical Songs	Philosophical Songs	Romantic Songs	Happy Songs
762	Zindagi Kaisi Hai Paheli Haye	Philosophical Songs	Philosophical Songs	Sad Songs	Sad Songs
602	Yeh Hasin Vaadiya Yeh Khula Aasmaan - ये हसीन वादिया ये खुला आसमान	Tareef Songs	Happy Songs	Happy Songs	Happy Songs
760	Woh Shaam Kuchh Ajeeb Thi - वो शाम कुछ अजीब थी	Sad Songs	Sad Songs	Philosophical Songs	Sad Songs
101	Welcome Back (Title Track)	Title Song	Party Songs	Dance Songs	Pop Songs

7	- वेलकम एवरीबॉडी कम वेलकम				
207	Wada Raha Sanam Honge Juda Na Hum (Duet) - वादा रहा सनम हो जुदा ना हम	Romantic Songs	Romantic Songs	Romantic Songs	Rock Songs
98	Ude Dil Befikre - उड़े दिल बेफिक्	Romantic Songs	Happy Songs	Pop Songs	Romantic Songs
573	Tumse Achchha Kaun Hai, Dil Lo Jigar Lo Jaan Lo - तुम से अच् कौन है, दिल लो जिगर लो जान लो	Masti Bhare Geet	Party Songs	Party Songs	Sad Songs

Table 5.1 Three Stage wise classification results

5.2 Human Survey Prediction Stage

Human survey for music songs can provide insightful information on various aspects of music perception, preferences, and emotional responses. As predicted output by three stage yields into different category, for multi label classification algorithm final predicted output to which stage is to be provided must analyzed properly. As Music deals with Psychology Human surveys for music songs play an essential part of comprehending audience preferences, exploring emotional responses to music, informing business decisions in the music industry, and contributing to our broader understanding of music's role in society. Music has a profound impact on emotions. Surveys can aid researchers in comprehending how different songs evoke specific emotional responses in listeners.

For multi label classification algorithm in human survey popular songs are given as input and based on machine learning algorithm of three stages wise classification labels are provided as options to decide which song categorize into which class. More than 20 human responses were taken, and then based on posteriori probability final predicted class is assigned.

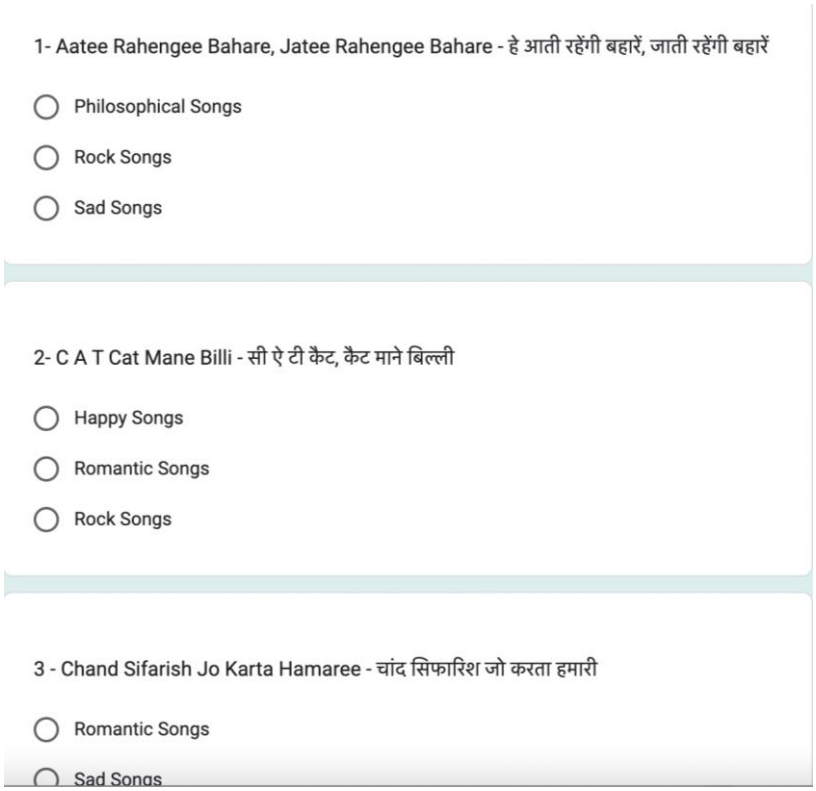


Figure 5.1 Sample Human Survey Google Form

	A	B	C	D	E	F	G	H	
1	Timestamp	Your Happy Name :)	1- Aatee Rahengee Bahe	2- C A T Cat Mane Billi -	3 - Chand Sifarish Jo Kar	4- Chhoti Si Umar Me La	5- Dance Pe Chance Ma	6- Din Dhal Jaye Haye R	7- Do L
2	07/07/2023 12:20:08	Rita	Philosophical Songs	Happy Songs	Romantic Songs	Rock Songs	Dance Songs	Sad Songs	Romar
3	07/07/2023 12:30:10	Chitz	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Dance Songs	Sad Songs	Romar
4	07/07/2023 12:44:59	Archana Kurde	Philosophical Songs	Rock Songs	Romantic Songs	Romantic Songs	Dance Songs	Sad Songs	Romar
5	07/07/2023 12:47:37	Riddhi kotak	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Rock Songs	Sad Songs	Romar
6	07/07/2023 12:56:10	Tejas Patalia	Philosophical Songs	Happy Songs	Romantic Songs	Rock Songs	Dance Songs	Sad Songs	Romar
7	07/07/2023 13:12:45	Kartik Trambadia	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Dance Songs	Sad Songs	Romar
8	08/07/2023 12:33:34	Devanshi Pathak	Philosophical Songs	Happy Songs	Romantic Songs	Rock Songs	Dance Songs	Sad Songs	Sad Sc
9	08/07/2023 19:29:31	SK	Sad Songs	Happy Songs	Romantic Songs	Romantic Songs	Dance Songs	Filmy	Romar
10	08/07/2023 19:51:08	Sandy	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Dance Songs	Sad Songs	Romar
11	08/07/2023 21:50:35	Madhuri	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Dance Songs	Filmy	Romar
12	09/07/2023 10:46:04	Rutvi Desai	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Filmy	Filmy	Romar
13	11/07/2023 08:00:23	Dinky	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Dance Songs	Sad Songs	Romar
14	13/07/2023 08:36:03	Mihir	Philosophical Songs	Happy Songs	Romantic Songs	Happy Songs	Rock Songs	Filmy	Romar
15	14/12/2023 16:59:23	Vandana Keshwala	Philosophical Songs	Happy Songs	Romantic Songs	Rock Songs	Dance Songs	Filmy	Sad Sc
16	14/12/2023 17:02:33	Ambasana Shweta Bhar	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Dance Songs	Filmy	Romar
17	14/12/2023 17:14:52	Prarthna Dhokiya	Philosophical Songs	Rock Songs	Romantic Songs	Romantic Songs	Rock Songs		Romar
18	14/12/2023 17:15:28	Bhakti Mehta	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Dance Songs	Sad Songs	Romar
19	14/12/2023 17:25:28	Vrunda Chavda	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Dance Songs	Sad Songs	Romar
20	14/12/2023 17:56:14	Mahek Desai	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Rock Songs	Sad Songs	Romar
21	14/12/2023 18:25:51	Kandarp Joshi	Sad Songs	Romantic Songs	Romantic Songs	Romantic Songs	Dance Songs	Filmy	Romar
22	14/12/2023 19:13:48	Rutvi Dekiwadia	Sad Songs	Happy Songs	Romantic Songs		Rock Songs		Sad Sc
23	07/07/2023 12:20:08	SGP	Philosophical Songs	Happy Songs	Romantic Songs	Romantic Songs	Dance Songs	Sad Songs	Romar

Form responses 1

Table 5.2 Human Survey Responses

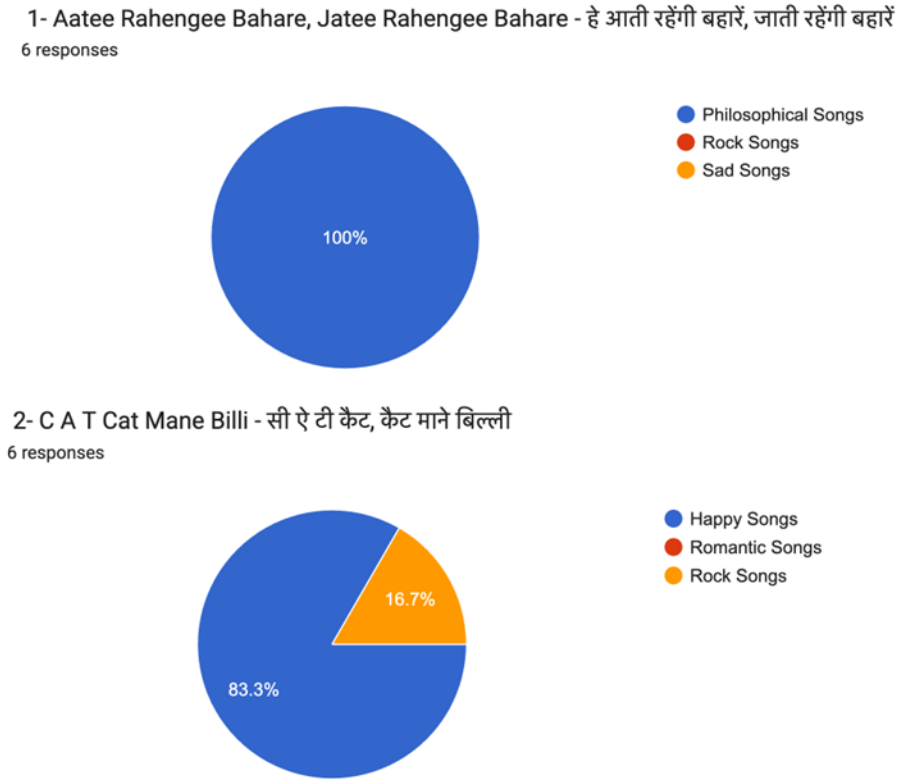
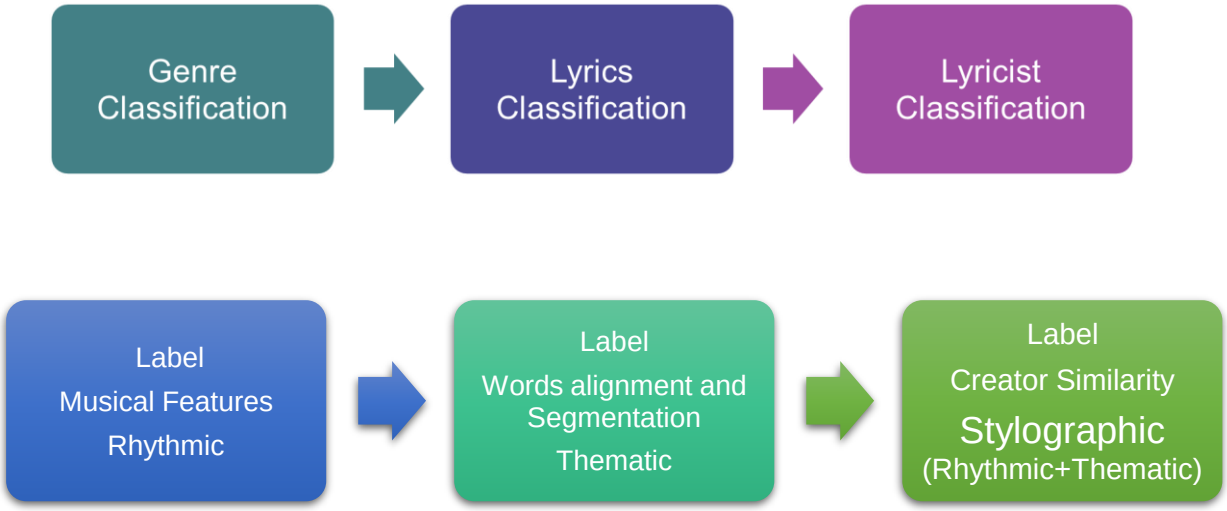


Figure 5.2 Sample Human Survey Response's Pie Chart representation of Songs

Since there plays psychology of human in songs classification, classification results also play vital role in classifying songs based on mood. Psychology plays A key component of recognizing human mood. Mood refers to a relatively long-lasting emotional state that influences how individuals perceive and interact with the world around them. Psychologists study mood through various theoretical frameworks and methodologies to understand its causes, effects, and potential interventions.

Hence while there occurs dissimilarity of results between genre, lyrics and lyricist classified label, human survey has been performed. between persons who listens songs on daily basis, based upon results it is seen that lyrics plays important role among lyricist and genre, thus for final stage of classification Creators Similarity which affects both Rhythmic and Thematic features both are combined for resultant output label.



(Romantic, Happy, Romantic) => Posteriori Probability=> Romantic

Figure 5.3 Final Predicted Output based on Posteriori Probability

Human prediction survey result shows classification majority focuses on Lyricist, Lyrics and then at last Genre. In Bayesian statistics, this is the updated or revised probability that an event will occur after additional information has been considered. Applying Bayes' method to update the prior probability yields the posterior probability. [60]

$$C_{map} = argmax_{c \in C} \left[\log \mathrm{P}(\mathrm{c}) + \sum_{1 \le \mathrm{lyric} < \mathrm{n}_d} \log \mathrm{P}(t_{lyric}|\mathrm{c}) + \log \mathrm{P}(t_{artist}|\mathrm{c}) + \log \mathrm{P}(t_{genre}|\mathrm{c}) \right]$$

Using data testing, the classification procedure is executed by computing posterior probability values for each class. Because of the prior classifier development, the posterior probability calculation method uses likelihood and prior probability values that are retrieved from the database. Each class's posterior probability is estimated using equation.[59]

```
data = pd.read_excel('/Users/darshitapathak/PhD/DPC 7/final_predict.xls')

# Use the multi-axes indexing funtion
for i in range(0,300):
    lyrics_label=data.loc[i,'Lyrics']
    lyricist_label=data.loc[i,'Lyricst']
    genre_label=data.loc[i,'Genre']
    if lyrics_label is lyricist_label:
        final_label=lyrics_label
    elif lyricist_label is genre_label:
        final_label=lyricist_label
    elif lyrics_label is genre_label:
        final_label=lyrics_label
    else:
        final_label=lyricist_label
    print(final_label)
    sheet.write(i, 0, final_label, style)
workbook.save("/Users/darshitapathak/PhD/DPC 7/sample.xls")
```

Figure 5.4 Multi Label Classification Implementation

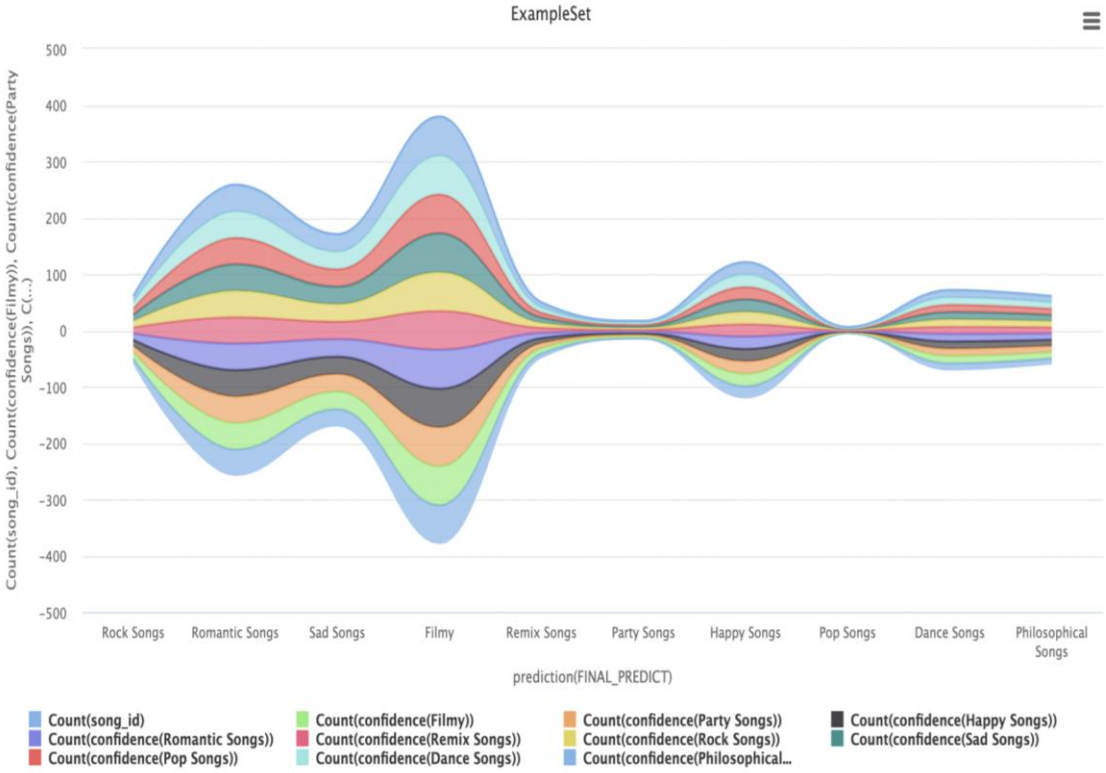
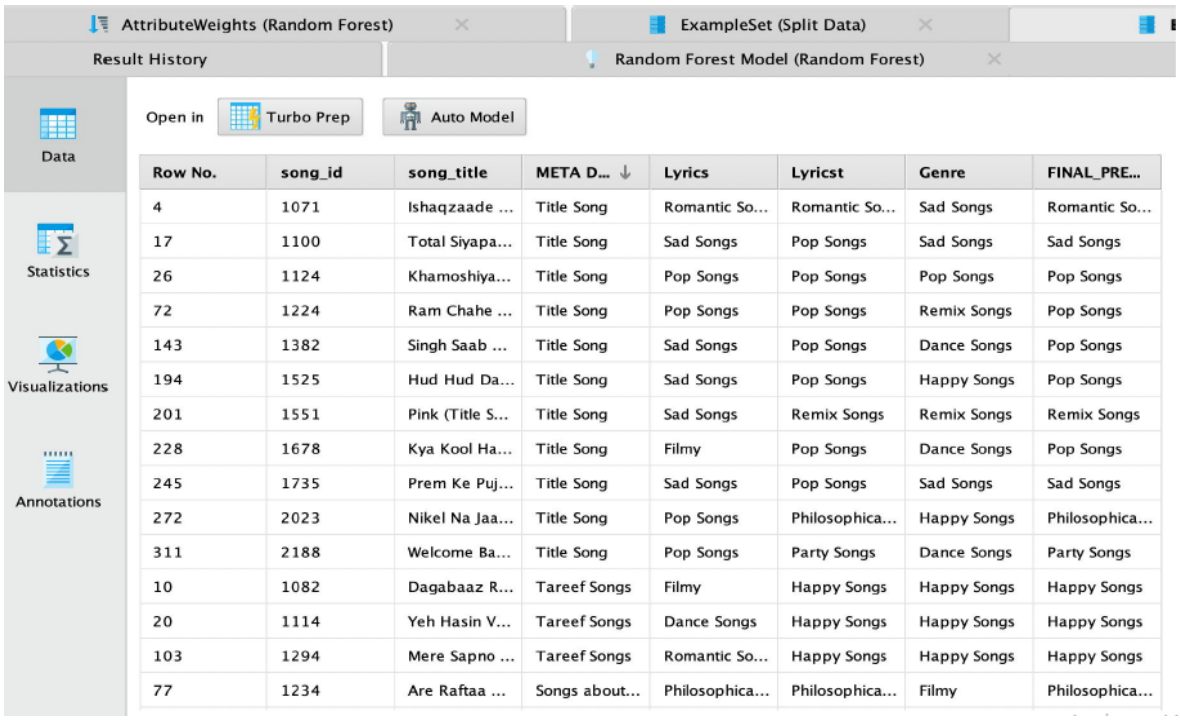


Figure 5.5 Multi Label Classification Example Set

song_id	song_title	META DATA GE	Lyrics	Lyricst	Genre	FINAL_PREDICT
1069	Maula Maula (Harshdeep Kaur) - मे ं डा इश् क भी तू , मे ं डा य	Qawwalis	Party Songs	Party Song	Party Songs	Party Songs
1070	Vah Re Vah Tu Hi Hai Mera Sara Jahan - वह रे वह तू ही है मे	Funny Songs	Romantic Son	Happy Son	Happy Songs	Happy Songs
1071	Ishaqzaade - इश् कजादे	Title Song	Romantic Son	Romantic S	Sad Songs	Romantic Songs
1072	Son Of Sardaar (Remix)	Remix Songs	Filmy	Remix Son	Remix Songs	Remix Songs
1077	I Love My India, Jis Desh Me Mere Piya (Part 2) - आइ लव माय	Patriotic Songs	Sad Songs	Happy Son	Party Songs	Happy Songs
1080	Baby Doll Main Sone Di - ओ बे बी डोल मै ं सोने दी	Rock Songs	Rock Songs	Rock Song	Filmy	Rock Songs
1082	Dagabaaz Re, Tore Naina Bade Dagabaaz Re - दगाबाज़ रे हा	Tareef Songs	Filmy	Happy Son	Happy Songs	Happy Songs
1083	Shayad Meri Shaadi Ka Khayaal, Dil Me Aaya Hai - शायद मे री	Masti Bhare Geet	Rock Songs	Rock Song	Romantic Sor	Rock Songs
1084	Do Lafzon Ki Hai Dil Ki Kahani - दो लफ् ज़ों की है , दलि की कहा	Romantic Songs	Romantic Son	Romantic S	Party Songs	Romantic Songs
1085	Hum Tum Tum Hum Ek Nadi Ke Hai Do Kinare - हम तू म तू म	Romantic Songs	Romantic Son	Romantic S	Happy Songs	Romantic Songs
1093	Dil Dhadak Dhadak Nain Phadak Phadak - दलि धड़क धड़क नै न	Romantic Songs	Filmy	Romantic S	Filmy	Filmy
1094	Give Me Love - गवि मी लव	Disco Songs	Romantic Son	Party Song	Dance Songs	Party Songs
1095	Mere Sathiya Mere Sath Chal Aisa Na Ho Jaye - मे रे साथयि	Romantic Songs	Romantic Son	Romantic S	Filmy	Romantic Songs
1100	Total Siyapaa - Title Song - टोटल सयिाप् पा	Title Song	Sad Songs	Pop Songs	Sad Songs	Sad Songs
1108	Aa Gale Lag Ja, Mere Sapne, Mere Apne, Mere Paas Aa - आ	Romantic Songs	Romantic Son	Romantic S	Romantic Sor	Romantic Songs
1113	Jab Jab Phool Khile Tujhe Yaad Kiya Humne - जब जब फ ू ल ख	Sad Songs	Sad Songs	Sad Songs	Filmy	Sad Songs
1114	Yeh Hasin Vaadiya Yeh Khula Aasmaan - ये हसीन वादयिा ये	Tareef Songs	Dance Songs	Happy Son	Happy Songs	Happy Songs
1119	Tere Bin Ek Pal Dil Nahi Lagda - ते रे बनि एक पल दलि नइयो लग	Sad Songs	Sad Songs	Sad Songs	Filmy	Sad Songs
1120	Zaraa Saa Jhum Lun Main - ज़रा सा झू म लू मै ं	Masti Bhare Geet	Pop Songs	Pop Songs	Filmy	Pop Songs
1121	Maine Jise Abhi Abhi Dekha Hai - मै ं ने जसि अभी अभी दे ख	Ched Chad Songs	Rock Songs	Rock Song	Romantic Sor	Rock Songs
1124	Khamoshiyan (Title Song) - खामोशयिां	Title Song	Pop Songs	Pop Songs	Pop Songs	Pop Songs
1125	Dhoom Machale Dhoom Machale Dhoom - धू म मचाले धू म	Dance Songs	Pop Songs	Dance Son	Remix Songs	Dance Songs
1131	O Sathi Re, Tere Binaa Bhi Kya Jeena - ओ साथी रे , ते रे बनि	Sad Songs	Sad Songs	Sad Songs	Romantic Sor	Sad Songs
1132	Abhi Abhi Solah Baras Ki Hui अभी अभी सोलह बरस की ह ई	Dance Songs	Dance Songs	Dance Son	Filmy	Dance Songs

Table 5.3 Final predicted output based on multilabel classification



Row No.	song_id	song_title	META D...	Lyrics	Lyricst	Genre	FINAL_PRE...
4	1071	Ishaqzaade ...	Title Song	Romantic So...	Romantic So...	Sad Songs	Romantic So...
17	1100	Total Siyapa...	Title Song	Sad Songs	Pop Songs	Sad Songs	Sad Songs
26	1124	Khamoshiya...	Title Song	Pop Songs	Pop Songs	Pop Songs	Pop Songs
72	1224	Ram Chahe ...	Title Song	Pop Songs	Pop Songs	Remix Songs	Pop Songs
143	1382	Singh Saab ...	Title Song	Sad Songs	Pop Songs	Dance Songs	Pop Songs
194	1525	Hud Hud Da...	Title Song	Sad Songs	Pop Songs	Happy Songs	Pop Songs
201	1551	Pink (Title S...	Title Song	Sad Songs	Remix Songs	Remix Songs	Remix Songs
228	1678	Kya Kool Ha...	Title Song	Filmy	Pop Songs	Dance Songs	Pop Songs
245	1735	Prem Ke Puj...	Title Song	Sad Songs	Pop Songs	Sad Songs	Sad Songs
272	2023	Nikel Na Jaa...	Title Song	Pop Songs	Philosophica...	Happy Songs	Philosophica...
311	2188	Welcome Ba...	Title Song	Pop Songs	Party Songs	Dance Songs	Party Songs
10	1082	Dagabaaz R...	Tareef Songs	Filmy	Happy Songs	Happy Songs	Happy Songs
20	1114	Yeh Hasin V...	Tareef Songs	Dance Songs	Happy Songs	Happy Songs	Happy Songs
103	1294	Mere Sapno ...	Tareef Songs	Romantic So...	Happy Songs	Happy Songs	Happy Songs
77	1234	Are Raftaa ...	Songs about...	Philosophica...	Philosophica...	Filmy	Philosophica...

Table 5.4 Different Audio Labels

5.3 Results & Analysis

It has been demonstrated by the results that adding artist and genre criteria can enhance performance. The system was able to classify songs more accurately as a result of the addition of genre and artist information. It was also demonstrated that genre and artist were both essential in identifying a subject's song.

The classification of Hindi songs by various algorithms pursued by only a handful of researchers. In contrast, other classification techniques have predominantly focused on English, Bengali, and Telugu songs. This disparity highlights a significant gap in the research landscape, suggesting an opportunity for further exploration and development in the classification of Hindi music.

Algorithm	**Language**	**F1-Score**	**No. of Classes**	**No. of Songs**
NB+TF-IDF [14]	Bengali	57%	3	2152
BLSTM + GloVE [14]	Bengali	59%	3	2152
BERT [14]	Bengali	**65%**	3	2152
Quadrant Classification [15]	Telgu	62.58%	4	481
LibSVM + NG [16]	**Hindi**	**68.30%**	**2(Positive-Negative)**	**461**
SARTH - Proposed Algorithm	**Hindi**	**63.46 %**	**10**	**1000**

Table 5.5 Comparative Analysis of F1-Score of various Algorithms

Table 5.5 shows proposed SARTH Algorithm using improved random forest algorithm for classifying Hindi songs based on lyricist, genre, and lyrics demonstrates superior performance when analyzing a larger dataset encompassing more songs and categories which is of 10 categories of 1000 songs for training and testing model, that further can be examined on Sanchay dataset having songs more than 5000. This approach outperforms to existing datasets that have limited categories and fewer songs, providing a more robust framework for understanding the diverse landscape of Hindi music.

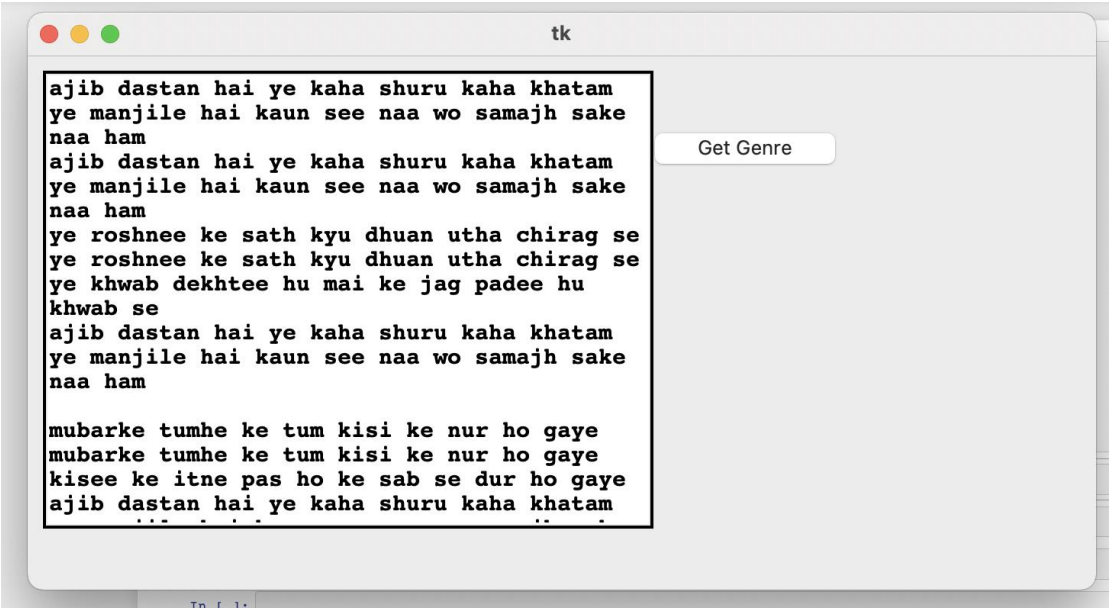


Figure 5.6 GUI Layout of Genre Prediction (Romanized Lyrics)

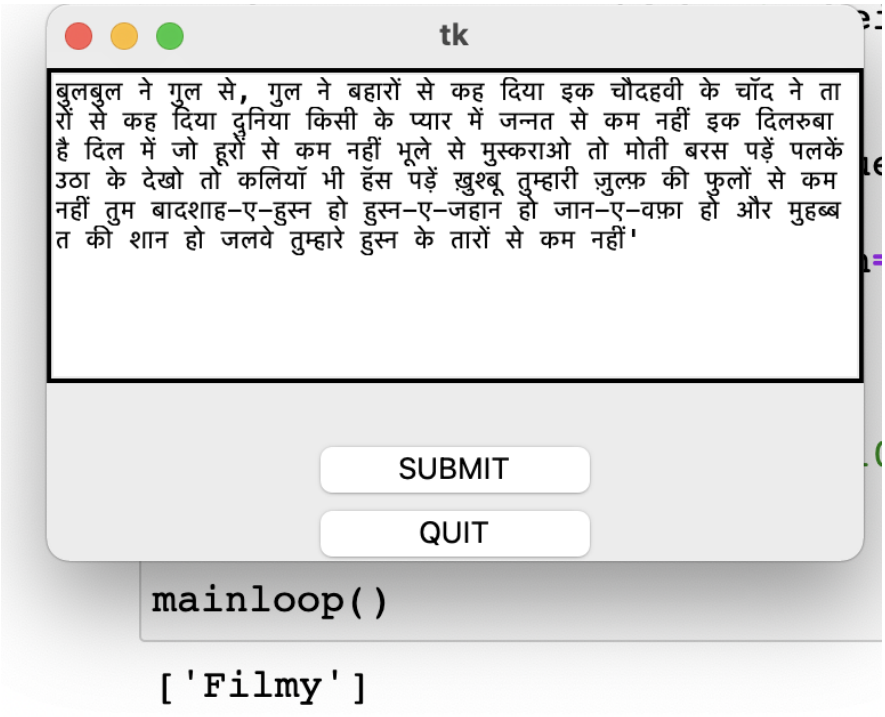


Figure 5.7 GUI of final predicted (Hindi Lyrics) Output

The proposed implementation framework for three-stage classification offers a comprehensive analysis of songs by examining the lyricist, genre, and lyrical content. Results elaborate this framework systematically categorizes songs for a detailed understanding of their characteristics and providing insights into the connections between lyrical themes and musical styles.

CHAPTER-6

Conclusion

Future Work

Chapter 6
Conclusion & Future Scope

6.1 Conclusion

Innovative approach for Lyrics Extraction by Metadata extraction and crawling is useful to build any music information retrieval related application. This music crawler will surely be unique in the field of search engine for music information retrieval and also beneficial if implemented as a commercial application. Genre wise classification Simulation by combination of MFCC and other Timbre based features has been observed and with combination of all features classification has improved the similarity metric for Random Forest pair wise Lyrics classification compared with Naive, SVM classifiers. The proposed system in this Thesis will be useful to fulfill its objective to make searching of song by its genre efficient and also to classify track of the song. The classification of music genres in this work is based on four spectral mode and genre correlations and the results are shown by leveraging technique. This indicates that it is important to classify audio tracks based on genre in order to distinguish human emotion.

Comparative analysis between Bangla Database and Hindi Database signifies enhanced Random Forest method for Hindi database provides accurate results with more features involved. Comparative analysis with Human prediction survey shows classification focuses more on Lyrics & Lyricist. Algorithm of three stage classification with 10 categories of Hindi songs provides optimum results with proposed algorithm approach. On the '*Sanchay*' dataset, the Improved Random Forest algorithm significantly outperforms the outcomes of other algorithms and produces state-of-the-art results.

Hierarchical classification and multiple labels are not supported by this research. Since certain songs contain more than one subject class, such as songs about friendship and love, this research should be able to classify more than one class. A further example of a song that belongs to the family class (only the parent class) features a class child and dad. Certain songs are solely classified into the primary

class and do not have any subclasses as determined by this study. This topic can also be pursued in more advanced studies where the datasets are audio files (mp3, wav) rather than text.

6.2 Future Scope

Overall system works best and fulfills its objectives but for more real time application and to extend this research further, it can be converted into a commercial application, with more user interaction, a system could be developed that shall contain more user-friendly GUI, various OS capability and an application be created so that variety of users may put it to use. With the advancements in artificial intelligence (AI) and machine learning (ML), the future of music classification holds great promise. AI algorithms can be trained on vast amounts of music data to develop more sophisticated classification models.

As with any AI application, ensuring fairness, transparency, and ethical use of music classification systems will be critical. Future research could focus on developing methods to mitigate biases and ensure equitable representation of various artists and genres. Future systems could be capable of analyzing music in real-time, allowing for dynamic classification and adaptation to changes in the music or user preferences. This could enhance user experiences in live events, streaming platforms, and interactive applications.

Overall, the future of music classification holds immense potential to enhance music discovery, understanding and enjoyment for both listeners and creators. As technology continues to advance, we can expect emergence of increasingly sophisticated and versatile music classification systems.

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List of Publications

1) Indian Film Songs Genre wise Classification based on Spectral Features Authors : Miss Darshita S. Pathak, Dr. Tejas P. Patalia ,Mr. Chitrang K. Vyas SJR : 0.32 Tuijin Jishu/Journal of Propulsion Technology ISSN : 10014055 H-Index : 24 (Scopus Index)

2) Sanchay : A Literary Dataset of Indian Filmy Songs Authors : Miss Darshita S. Pathak, Dr. Tejas P. Patalia Scopus Cite Score 2022 : 2.1 SJR : 0.285 Neuroscience: Cognitive Neuroscience Publisher : Anka Publishers, Turkey ISSN:1303-5150 (Scopus Index)

Sanchay: A Literary Dataset of Indian Filmy Songs

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Abstract.

With the rapid growth of Indian Film music industry huge volume of Hindi songs are produced every day. Numerous of lyricists, singers and artists are involved in production of songs from different genres. With evolved research in field of Music Information Retrieval analysis on Hindi songs emerged nowadays, but open source dataset availability is less. Thus, proposed technique to pre-pare dataset by crawling lyrics based on song title from the webpage will be extracted by using classical naïve based string matching algorithm. Metadata extraction procedure will be useful to extract various Meta information of song paired with music, which will be useful to prepared repository with essential song metadata. The prepared dataset given Hindi transliterate name "*Sanchay*" means a repository will helpful in area of mu-sic segmentation, digital signal processing, song feature extraction, emotion classification, lyrics mining, playlist generation, artist-genre paired clustering with commercial applications too.

Keywords: music information retrieval, lyrics mining, signal processing, metadata, genre, text transliterate

DOINumber:10.14704/nq.2022.20.10.NQ55859 **NeuroQuantology 2022; 20(10): 8772-8779**

1 Introduction

Songs are an integral part of movies. Songs represent, enhance, and complement an emotion in the movie. In Indian Film industry which is widely accepted as word Bollywood exemplify movie songs. Indian Film Industry music one of such huge part of human lives in India and with rapid growth of Industrial collaboration over world, Globalization has huge impact in Bollywood music which recognized as one of popular amongst all music industry in world many peoples from different stages of society are involved in this industry. Lyricists, producers, singers and artists are involved with this entertainment industry.Whether person is on car, auto-rickshaw or in radio, with wedding, sangeet ceremony or in funeral ceremony songs plays vital role.

There are large datasets of royalty-free music such as Jamendo (often used [14,15]) or others found in the DB Tunes link directory focus on the ones that cover commercial popular music and we will see that few propose metadata on cultural aspects, lyrics and audio altogether. However these all dataset contains popular English based songs, Apart from that some of Hindi Bollywood dataset available on Kaggle, UCI contains song's lyrics with minimum numbers. For commercial based application and research songs more than 2500 requires for training purpose in artificial neural network. MusicBrainz a large set of cultural metadata but nothing about lyrics, for example. The Last.fm dataset contains tags that were used by some researchers for computing moods and emotions [16, 17].

Apart from rhythm, tune, fusion, singer or genre; lyrics is the most vital element of songs, as it has direct impact on the listeners' choice and mood. In this paper, to the best of our knowledge, we presented the very first Sanchay – Hindi word of synonym as collection of data. Dataset having 5076 Songs lyrics of 257+ lyricists. In this paper refer author or lyricist having same meaning. The rest of the paper is divided into six sections. Literature review is illustrated in section II, the attributes of the dataset are described in section III. Data collection procedure, statistical analysis of the dataset and applications of the dataset are demonstrated in further section of paper. Finally, conclusion statements are in section VII.

2 Related Work

Music information retrieval (MIR) is the interdisciplinary science of retrieving information

Tuijin Jishu/Journal of Propulsion Technology
ISSN: 1001-4055
Vol. 44 No. 2 (2023)

Indian Film Songs Genre Wise Classification Based on Spectral Features

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***Abstract*:** The classification of musical genres has received a lot of attention in recent years due to the several uses it can serve in music information retrieval. Additionally, since genres frequently co-occur in music tracks in Indian Film Songs, it is desirable to record and model the genre correlations in order to enhance the effectiveness of multi-label music genre classification. Present a novel multi-modal strategy that uses a music spectral feature-based classification method to address these problems. In addition, a genre correlations extraction module is offered to identify and model probable genre correlations based on the characteristics of the proposed approach.

Index Terms: *Indian Music, Genre classification, spectral features, Indian filmy songs*

1. Introduction

Songs are an integral part of movies. Songs represent, enhance, and complement an emotion in the movie. In Indian Film industry which is widely accepted as word Bollywood exemplify movie songs. Indian Film Industry music one of such huge part of human lives in India and with rapid growth of Industrial collaboration over world .The versatility of musicality in the Hindi music industry. From dance music to orchestrated musicals, the industry serves you the multiple genre music at its epitome. Lyricists, producers, singers and artists are involved with this entertainment industry. Whether person is on car, auto-rickshaw or in radio, with wedding, sangeet ceremony or in funeral ceremony songs plays vital role as play by different genres. Classification of songs into different parts based on music spectral features involve music information retrieval algorithm for automatic categorization of songs.

2. Related Work

The interdisciplinary discipline of extracting information from music is known as music information retrieval (MIR). MIR is a young, expanding field of study with numerous practical applications. MIR practitioners may have training in signal processing, psychology, academic music research, musicology, machine learning, or a mix of these fields. [1]. Music signal feature extraction includes sound quality feature, melody feature and rhythm feature. “Genre” "kind" or "sort" is the term for any category of literature or other forms of art or entertainment.